

程序設計介紹

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Outline

- Introduction to Process Design
- Flow-sheeting
- Process Design Concept
 - Onion model
 - Hierarchical structure
 - Attentions for process design
- Optimization
 - Process
 - Equipment (Tower、Heat exchanger & Vessel)

• Introduction to Process Design

➤ 何謂程序設計

所謂「程序」就是組合各種化學反應及物理變化的單元操作，將原料變為附加價值較高的產品之軟體及硬體，而「程序設計」即是設計這些單元操作流程的一件工作。

單元程序：加氫脫硫、烷化、氯化

單元操作：蒸餾、萃取、吸收、加熱、冷卻、流體輸送

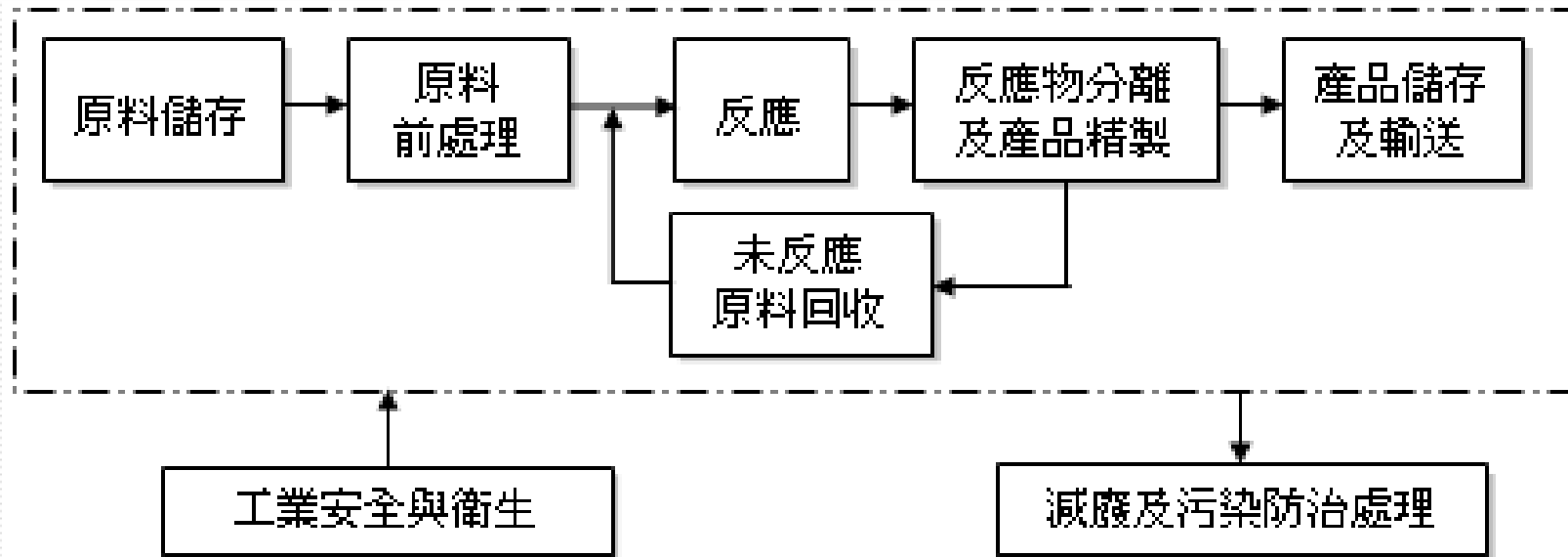
軟體：操作條件、配方

硬體：蒸餾塔、熱交換器、貯槽、泵浦

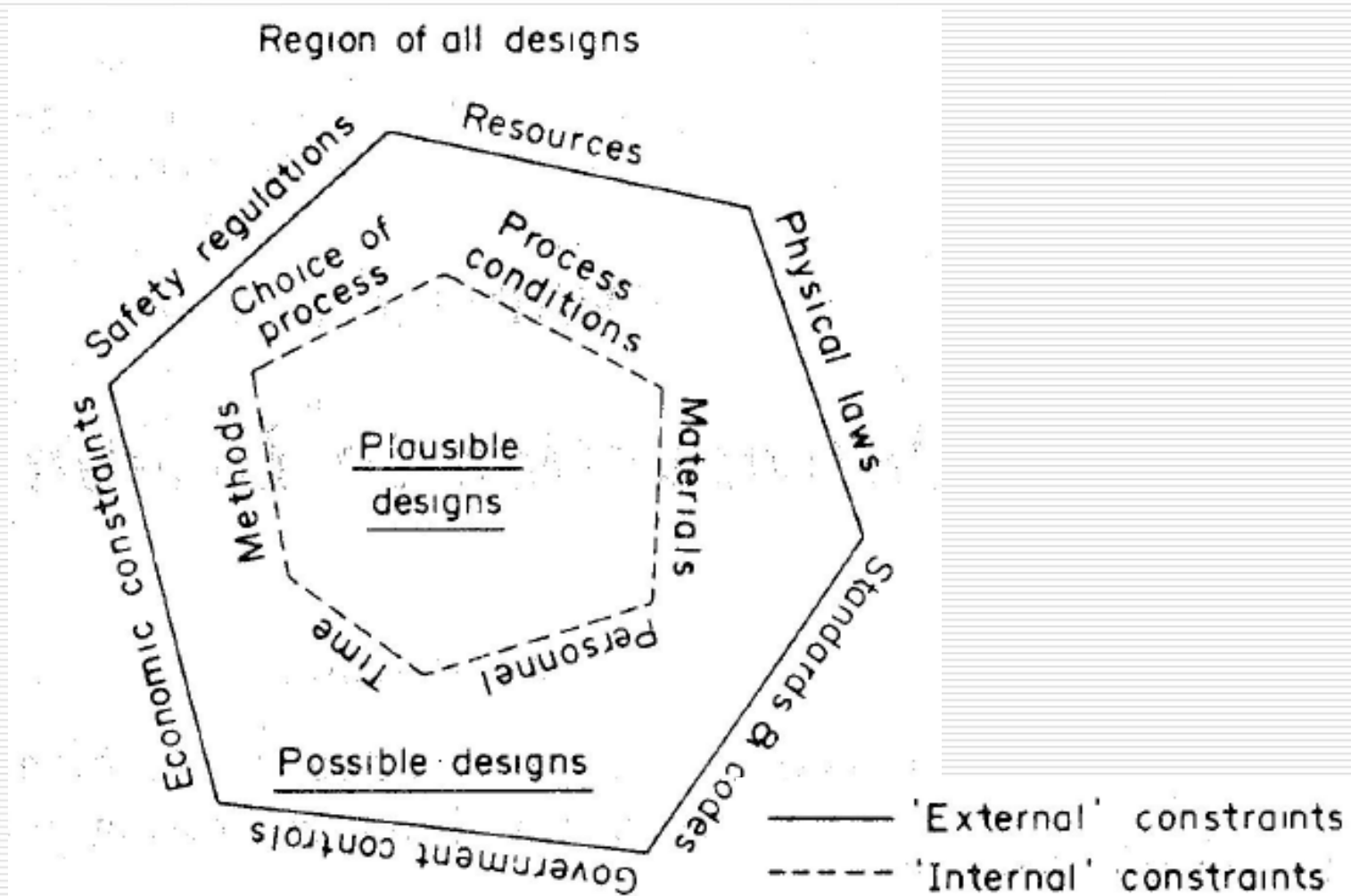
➤ 程序設計整合下列課程

- 質能平衡
- 反應工程
- 化工熱力學
- 單元操作
- 程序控制

➤ Anatomy of a Chemical Process



➤ Design Constraints



➤ Design Base

a. Standards & Codes

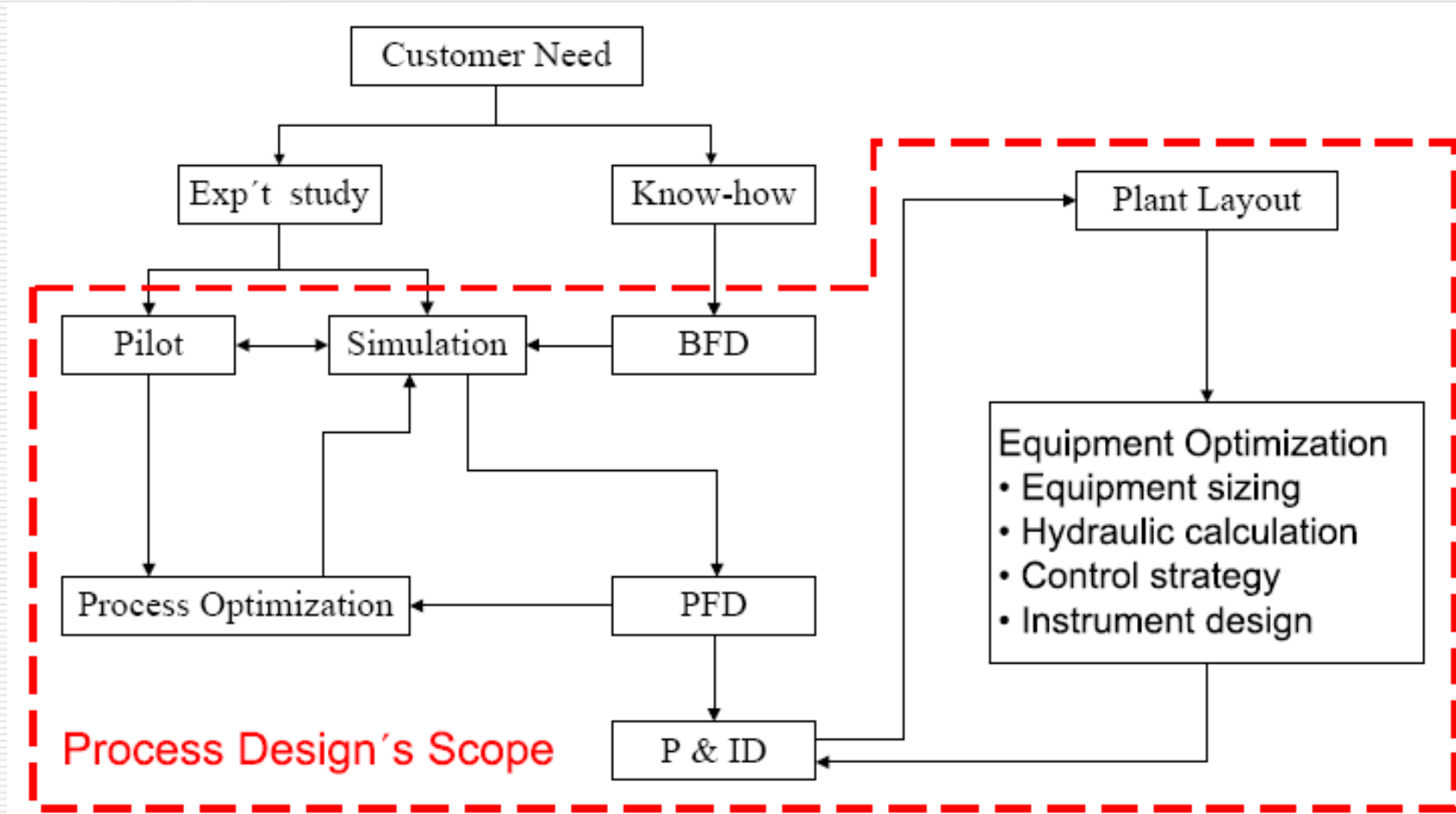
- ASME Standard (塔槽及泵浦)
- TEMA Standard (熱交換器)
- ANSI Standard (管件)
- API 650 (貯槽及大桶)
- API 617 (離心式壓縮機)

➤ Design Base (cont'd)

b. Basic Engineering Design Data

- Climatic data
- Topographic data
- Transportation
- Utility
- Pollution and environmental control
- Raw material & product specifications
- Equipment design guide (from customer)

• Flow-sheeting



Why Use Process Simulation

- Design a new plant or process
- Review an exist plant or process
- Obtain basic information and data for equipment design
- Improve operating condition or reduce energy
 - Optimization
 - Scale up
 - Improve product specification

製程模擬軟體之功能

➤ 提供各種單元操作模組，包括：

穩態單元操作：蒸餾、吸收、氣提、萃取、驟沸、熱交換器、壓縮機、泵、加熱爐、Turbine、Expander、管線模擬器、容器、氣液分離槽、氣液液分離槽、控制器...

動態單元操作：動態蒸餾塔、動態反應器、動態緩衝槽、動態三相分相槽、換熱器、混合器、分流器、控制閥、PID控制器、...

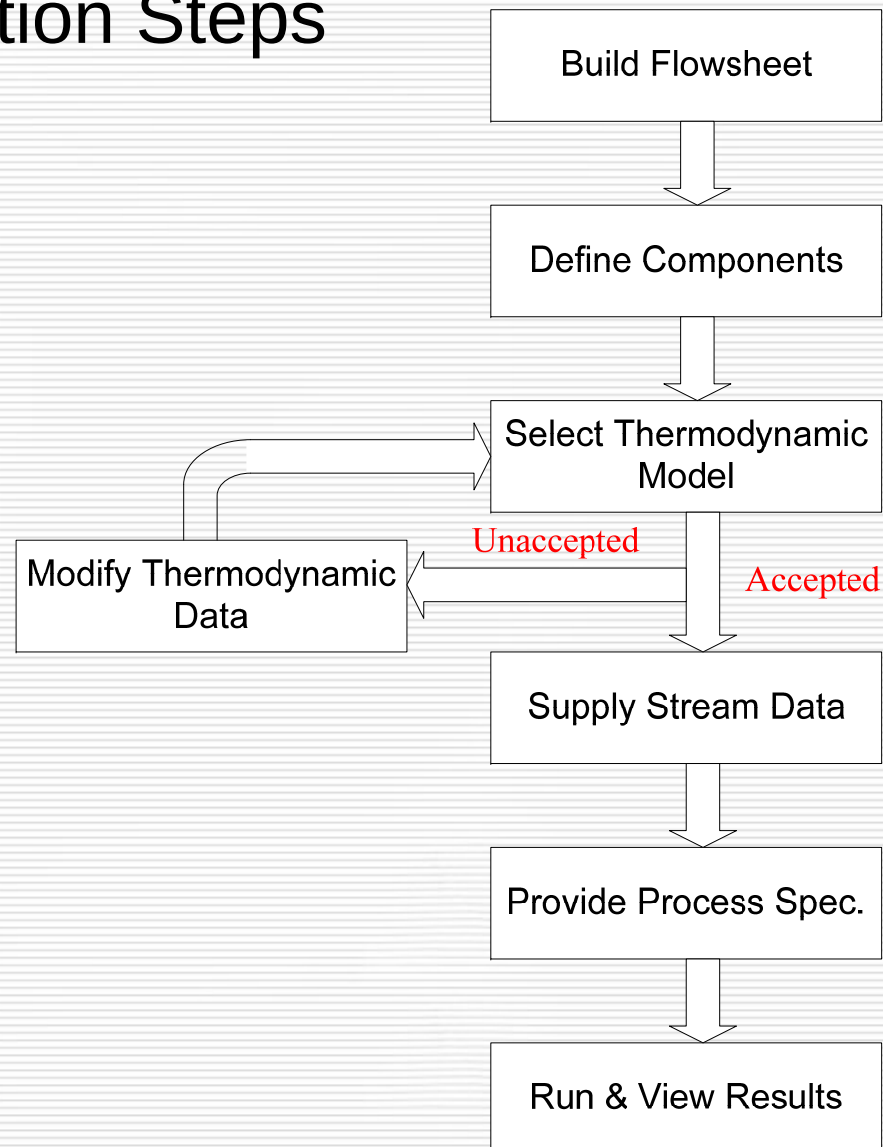
➤ 提供物性資料庫及熱力學模式：以 AICHE 之 DIPPR 為基礎，加上電解質資料庫、相平衡模式、和熱焓模式和物性計算

➤ 進行製程單元或整廠整合模擬、操作變數敏感度分析及最適化

➤ 提供工程設計所需數據：蒸餾塔(板塔、填充塔)、吸收塔、熱交換器、容器、轉動機械、管線、控制閥及緊急排放閥....等設備 Sizing

☆ 選擇『適合的』工具軟體，不一定要最好的

Simulation Steps



流程圖是利用各種符號（主要為設備、管線及儀控等三類）把流程具體化、圖像化的圖件，依照設計時間的先後順序為：

a. Block Flow Digram (方塊流程圖)

從原料進入至產品產出的製程中，主要的單元程序依照製程的順序表示的圖件

b. Process Flow Digram (程序流程圖)

除了將單元程序的使用設備(含編號)表示出來外，圖中亦包括操作條件、流體流量及物性等資料

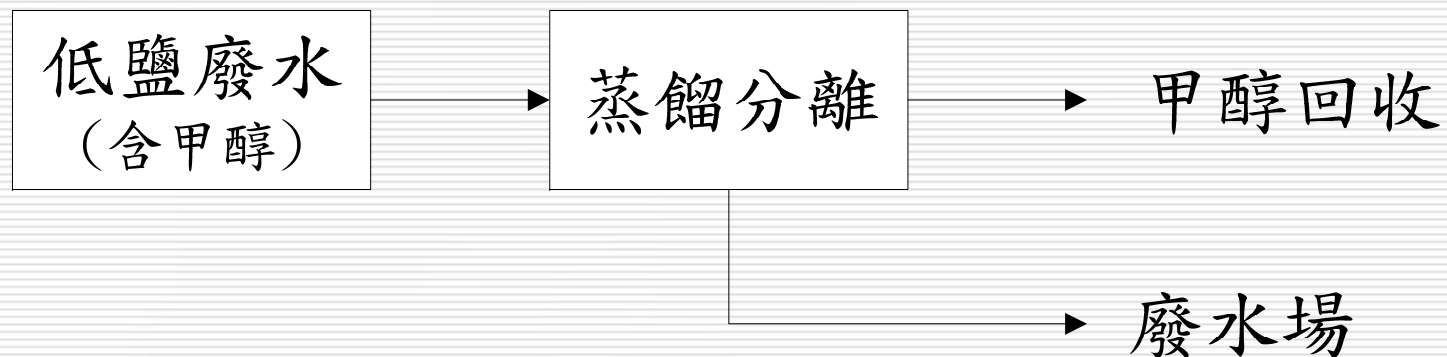
c. Process Control Flow Diagram (控制流程圖)

PFD中再增加流程控制迴路

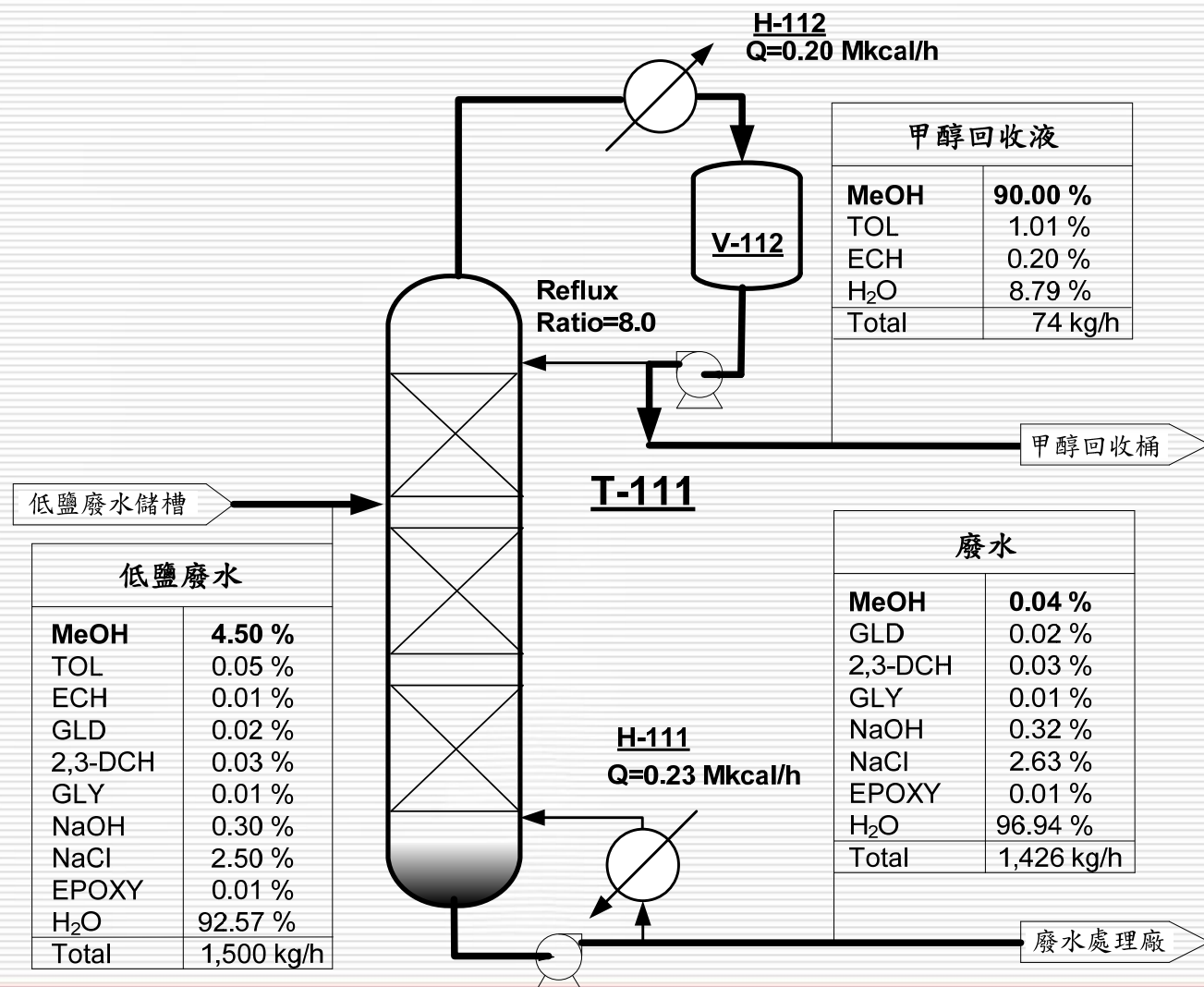
d. Piping & Instrument Diagram (管線及儀錶流程圖)

考慮實際操作 (包括開停車、正常運轉及緊急停車)、維修以及工安、環保等需求下的所有相關設備、管線和儀錶

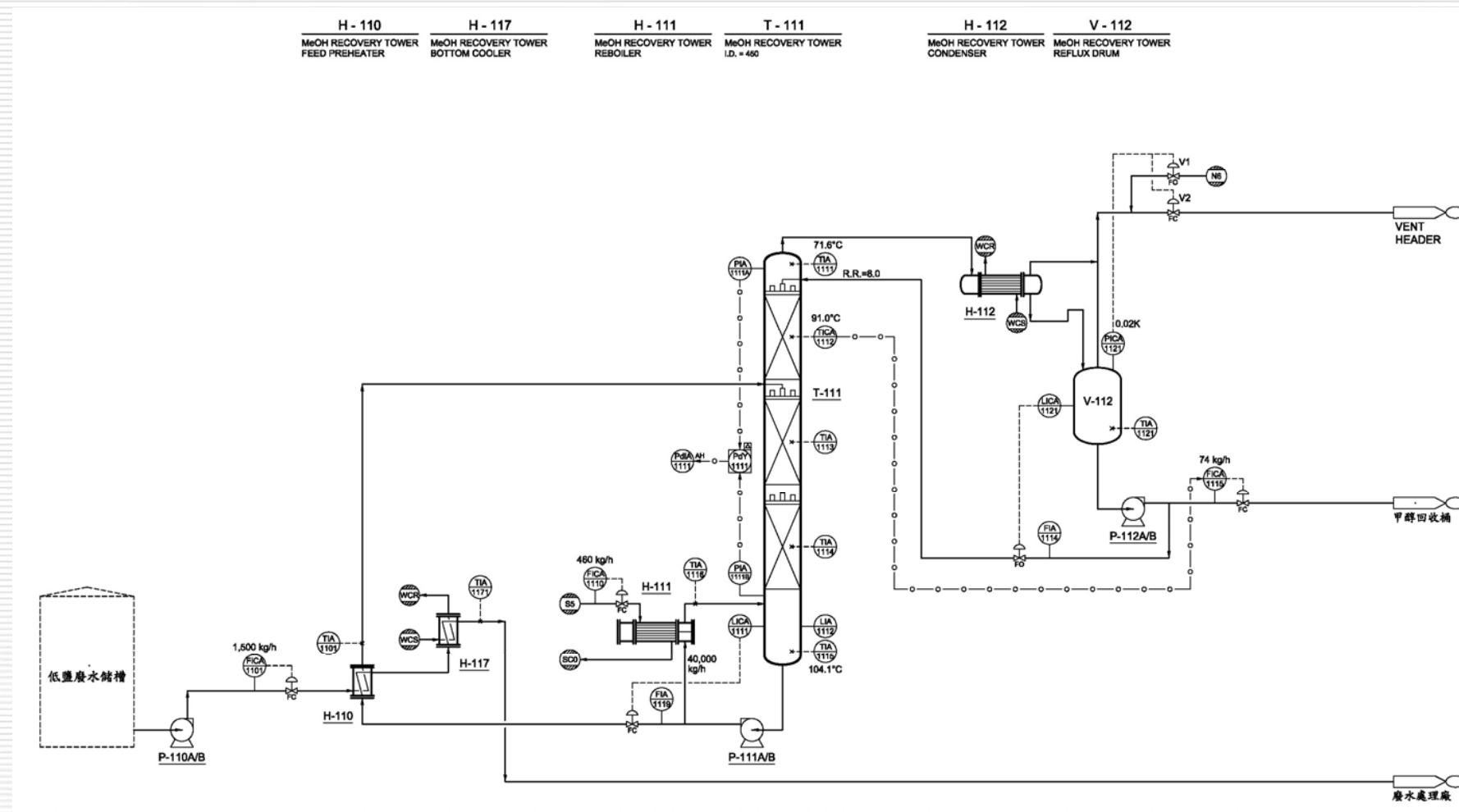
BFD (方塊流程圖)



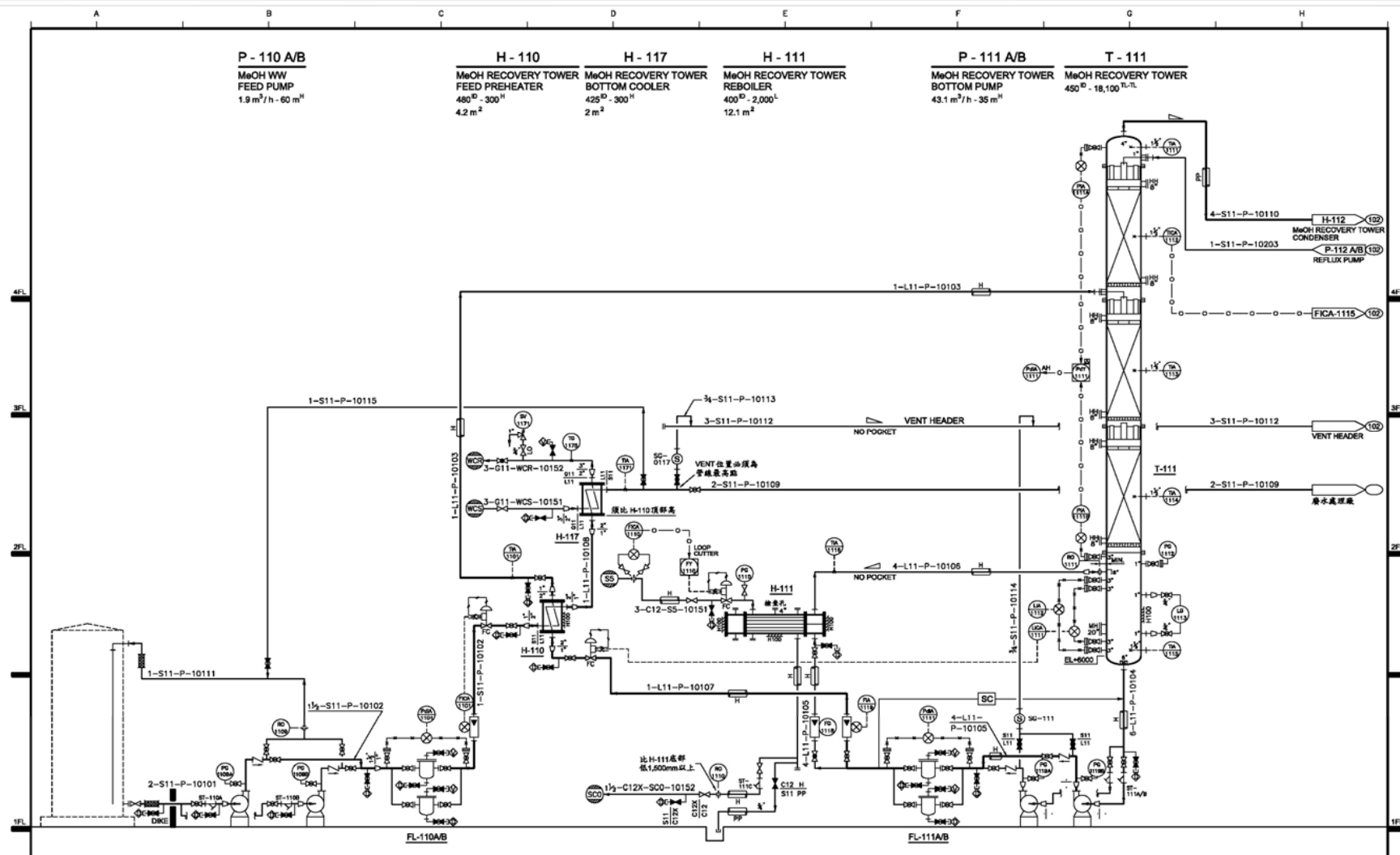
PFD (程序流程圖)



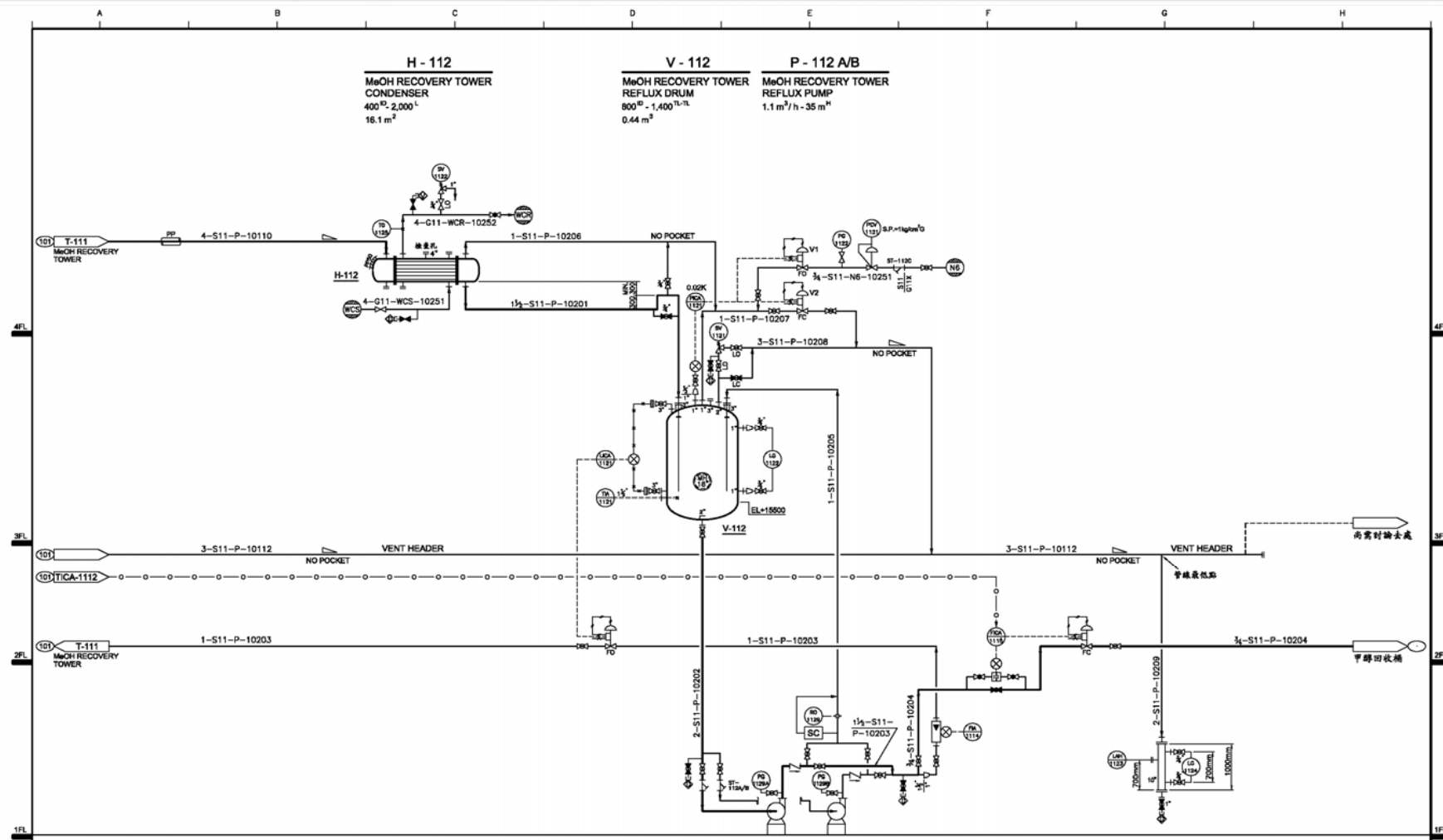
PCF (控制流程圖)



P & ID (管線及儀錶流程圖)- Sheet 1



P & ID (管線及儀錶流程圖)- Sheet 2

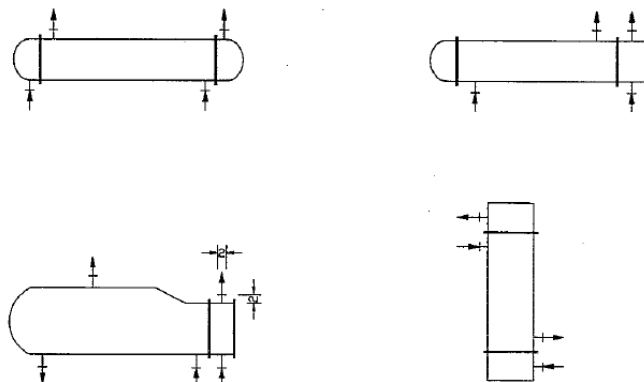


P & ID的內容主要分為設備、管線及儀錶三部份，
常用的各種符號如下：

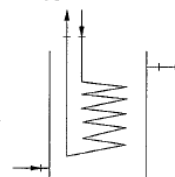
a. 設備部分

(1) Heat exchanger

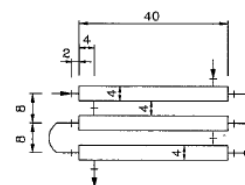
a. Shell & tube type



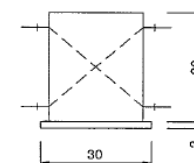
b. Coil type



c. Double pipe type

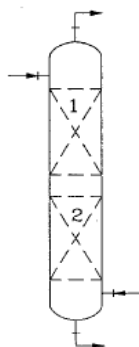


d. Plate type



(2) Tower

a. Packed Column



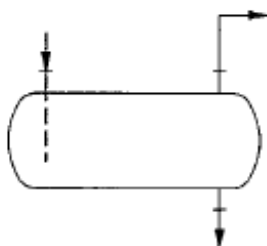
b. Plate tower



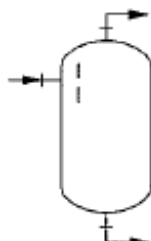
a. 設備部分 (續)

(3) Vessel

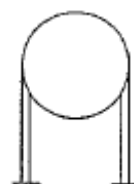
a. Horizontal type



b. Vertical type



c. Spherical type

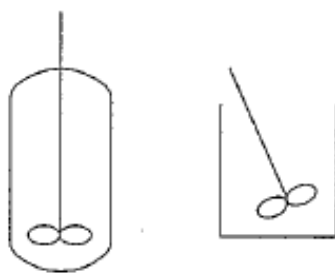


(4) Vessel with agitator

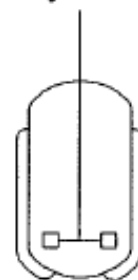
a. Turbine type



b. Propeller type

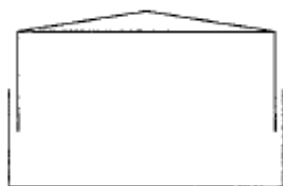


c. Turbine type (with jacket)

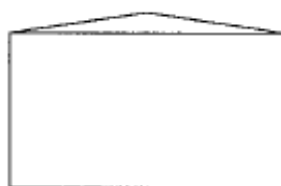


(5) Tank

a. Wet gas holder



b. Cone roof type



c. Flat roof type



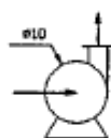
d. Floating roof type



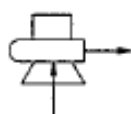
a. 設備部分 (續)

(6) Pump

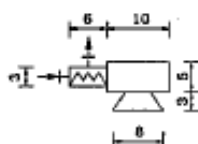
a. Centrifugal pump



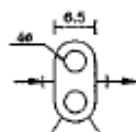
e. Vertical Centrifugal pump



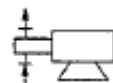
b. Screw pump



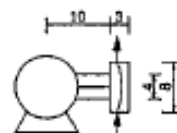
f. Gear pump



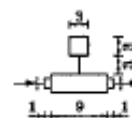
c. Plunger pump



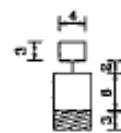
g. Diaphragm pump



d. Line pump



h. Under water pump

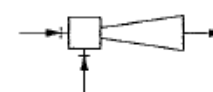


(8) Cyclone & Ejector(Injector)

a. Cyclone

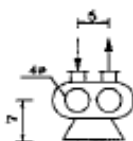


b. Ejector(Injector)

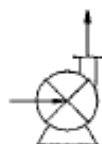


(7) Blower, Fan & Compressor

a. Roots blower



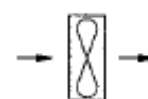
b. Turbo blower



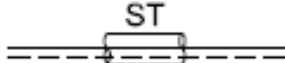
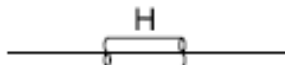
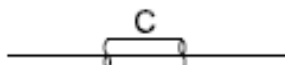
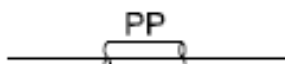
c. NASH blower



d. Axial Fan



b. 管線部分

	PRIMARY PIPING
	SECONDARY PIPING
	PIPING AND EQUIPMENT SHOWN ON OTHER P AND I DIAGRAMS
	FUTURE PIPING AND EQUIPMENT
	ELECTRIC TRACE AND INSULATE TO MAINTAIN TEMPERATURE AS INDICATED IN DEGREES CENTIGRADE
	STEAM TRACE AND INSULATE
	HOT INSULATE
	COLD INSULATE
	PERSONNEL PROTECTION

b. 管線部分 (續)

	PLUG VALVE		INTERMITTENT BLOWDOWN (BLOW-OFF) VALVE
	GLOBE VALVE		NEEDLE VALVE
	GATE VALVE		ANGLE VALVE
	BALL VALVE		SELF CONTAINED REGULATOR
	BUTTERFLY VALVE		SOLENOID VALVE
	CHECK VALVE		R MECHANICAL MANUAL RESET WHEN INDICATED DE DE-ENERGIZED STATE
	CONTINUOUS BLOWDOWN VALVE		STOP CHECK (NON-RETURN) VALVE
	FLOW CONTROL VALVE		THREE WAY VALVE
	FOUR WAY VALVE		ROTAMETER
	GATE VALVE WITH BODY BLEED		PURGE METER
	CHEMICAL SEAL		INTEGRAL ORIFICE FLOW METER
	STEAM TRAP		VORTEX FLOW METER
	IN-LINE Y STRAINER		FLEXIBLE HOSE W/ QUICK CONNECT
	PRESSURE RELIEF VALVE		INSTRUMENT TRANSMITTER
	RUPTURE DISK		HEADER (FLOW DIAGRAM FOR DISTRIBUTION OR COLLECTION TO BE REFERRER TO)
	BLIND FLANGE WITH BLIND PLATE		TEMPORARY STRAINER
	BLIND FLANGE		FLEXIBLE PIPE
	FOAM CHAMBER		SILENCER
	FLAME ARRESTER		BREATHER VALVE

C. 儀錶部分

1) Instrument Line Symbol

Lead or Impulse Piping



Pneumatic Signal



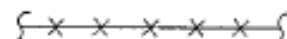
Hydraulic Signal



Electrical Signal



Capillary Tubing



Radiation Signal



2) Balloon symbols

Instrument

LOCAL	LOCAL PANEL or SUB- PANEL	CONTROL PANEL	DCS

Transmitter

Computing Relay

Pilot Lamp

CONTROL VALVE SYMBOLS



CONTROL VALVE W/ POSITIONER



90° CYLINDER ACTUATED SPRING LOADED

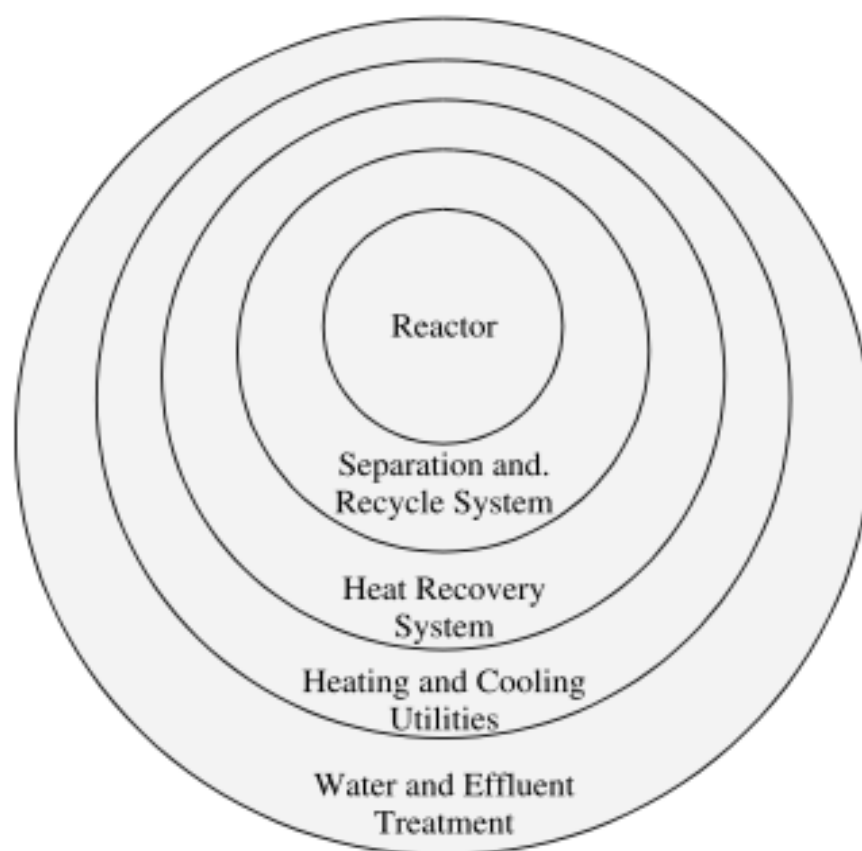


AXIAL CYLINDER ACTUATED

Process Unit

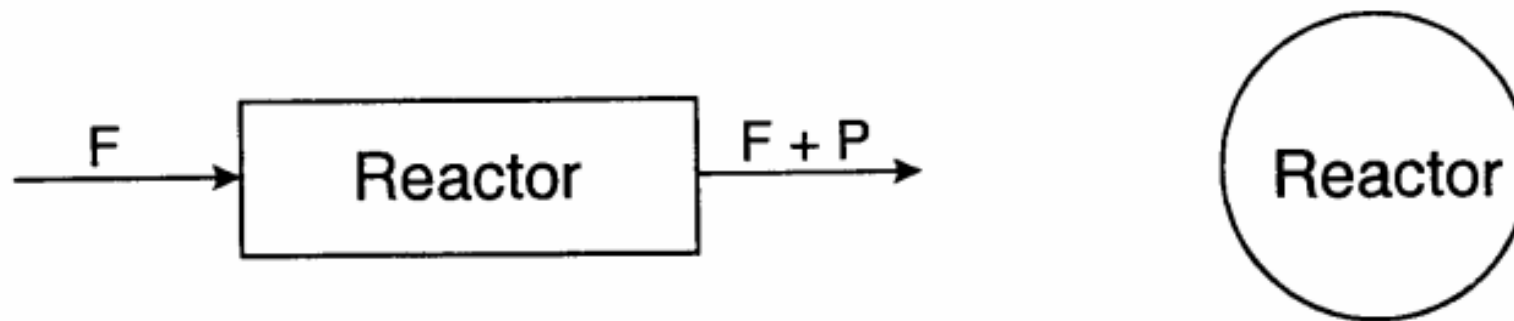
- Reactor : CSTR 、 PFR 、 Tubular
- Separator
 - Vapor-Liquid : Distillation 、 Evaporation
 - Liquid-Liquid : Extraction 、 Decanter
 - Liquid-Solid : Crystallization 、 Adsorption 、 Filter
- Heat exchanger : Shell & Tube 、 Plate 、 Screw
- Fluid transfer unit : Compressor 、 Blower 、 Pump 、 Pipe
- Vent gas treatment : Scrubber 、 Flare (for emergency)

- The onion model of process design



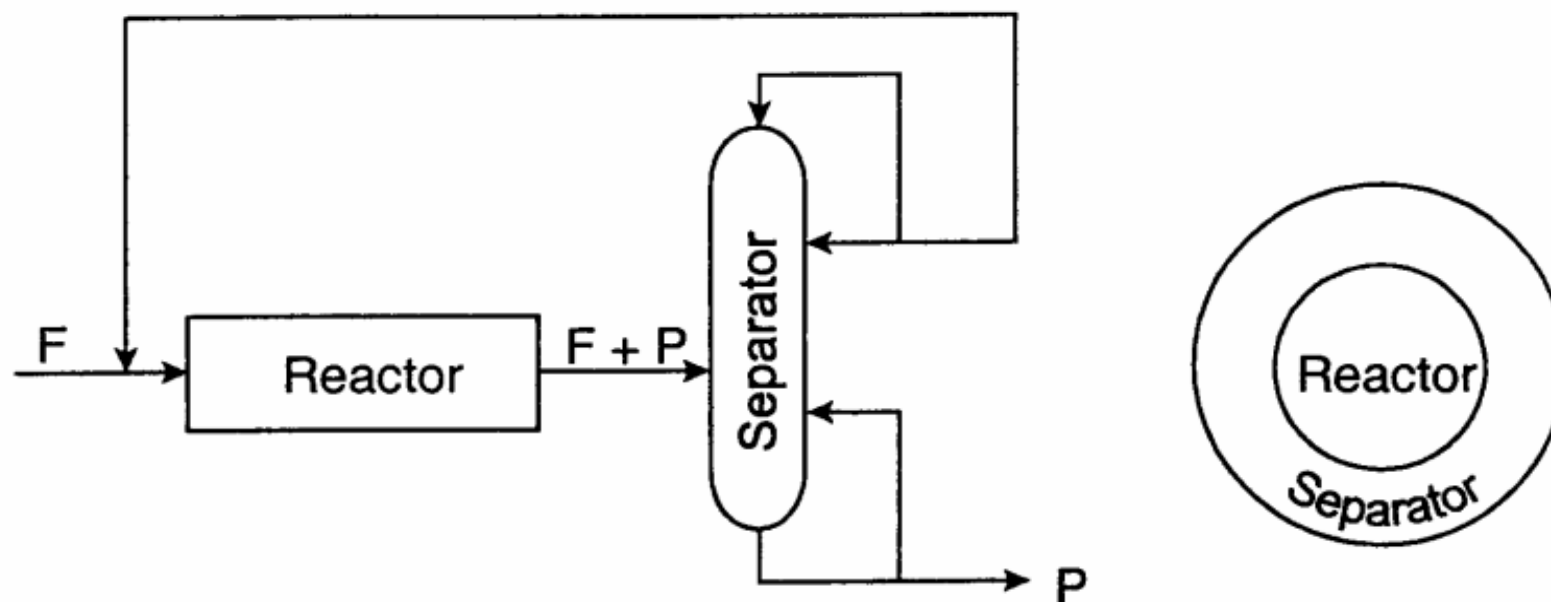
A reactor is needed before the separation and recycle system can be designed and so on

1st Layer: Reactor



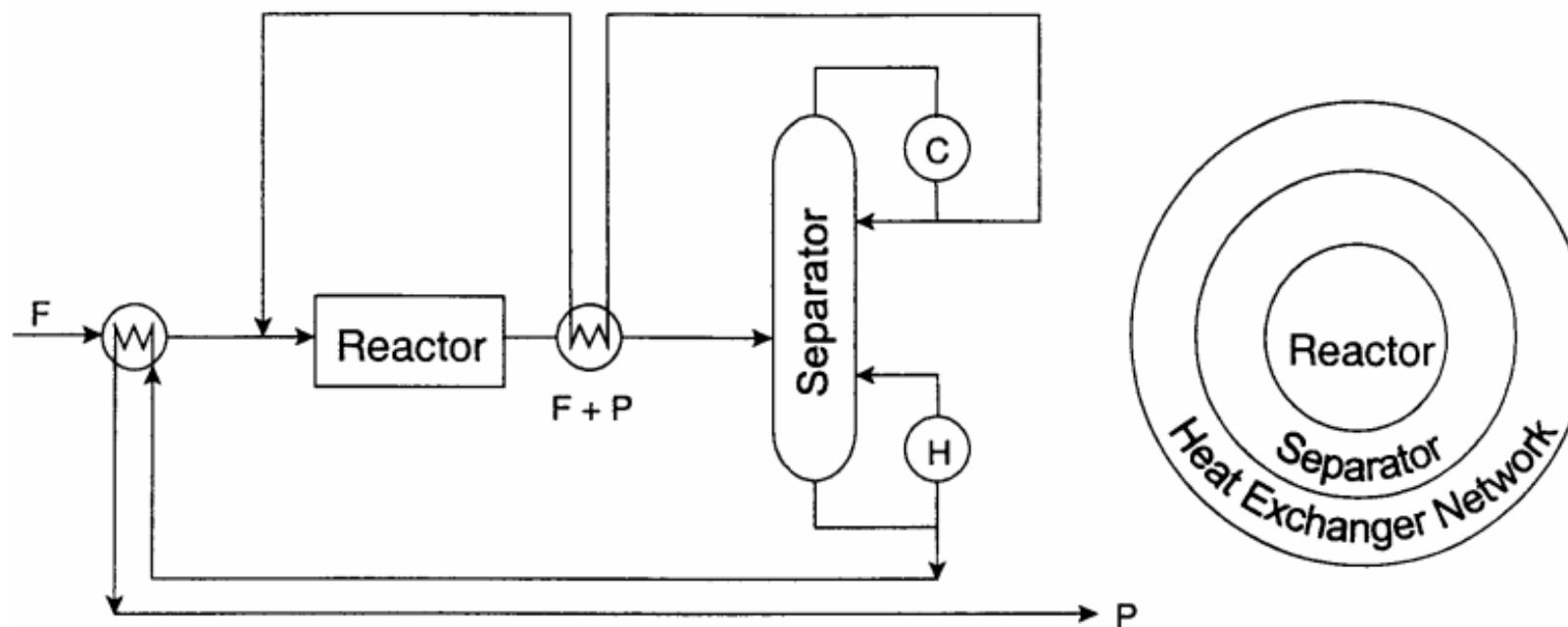
Process design starts with the reactor

2nd Layer: Separation and recycle system



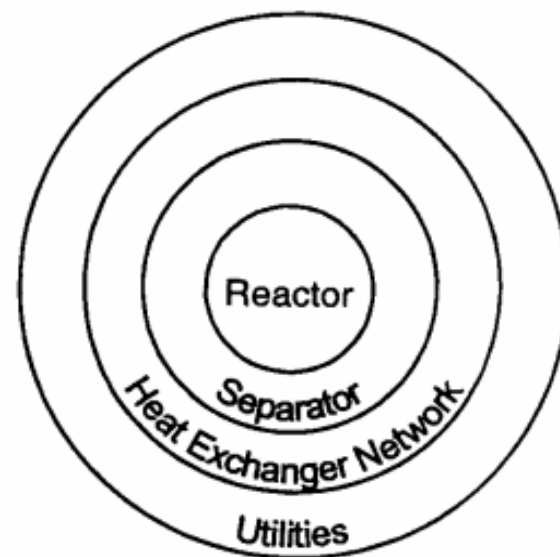
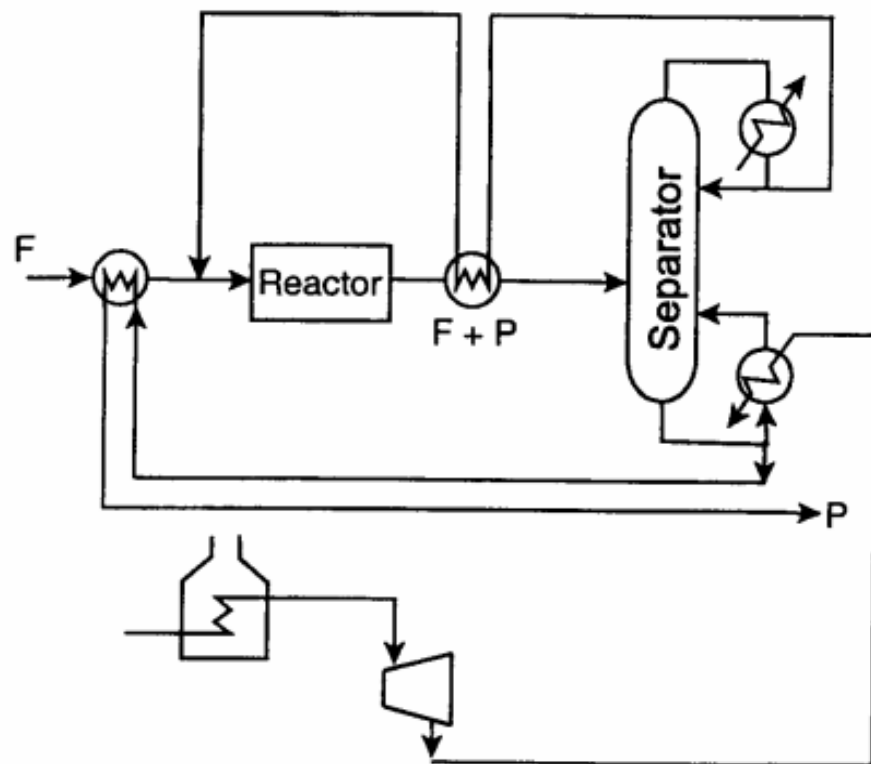
Once we know the reactor design,
we need to specify the separation system

3rd Layer: Heat recovery system



Once we know reactor and separator design, we know material and energy balance $\Rightarrow$ Heat Exchanger Network

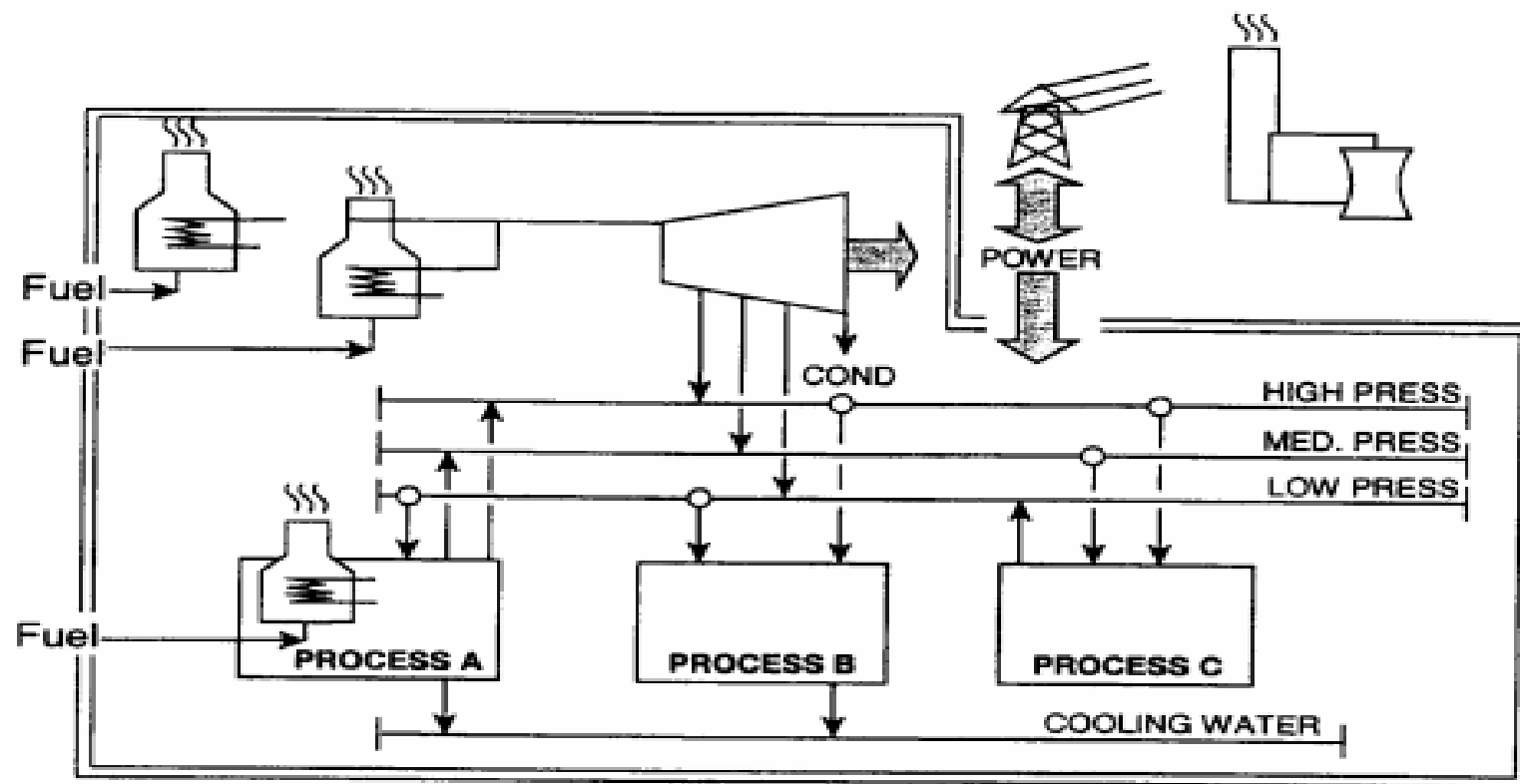
4th Layer: Heating and cooling utilities



Once we have the process and HEN design
⇒ Utilities

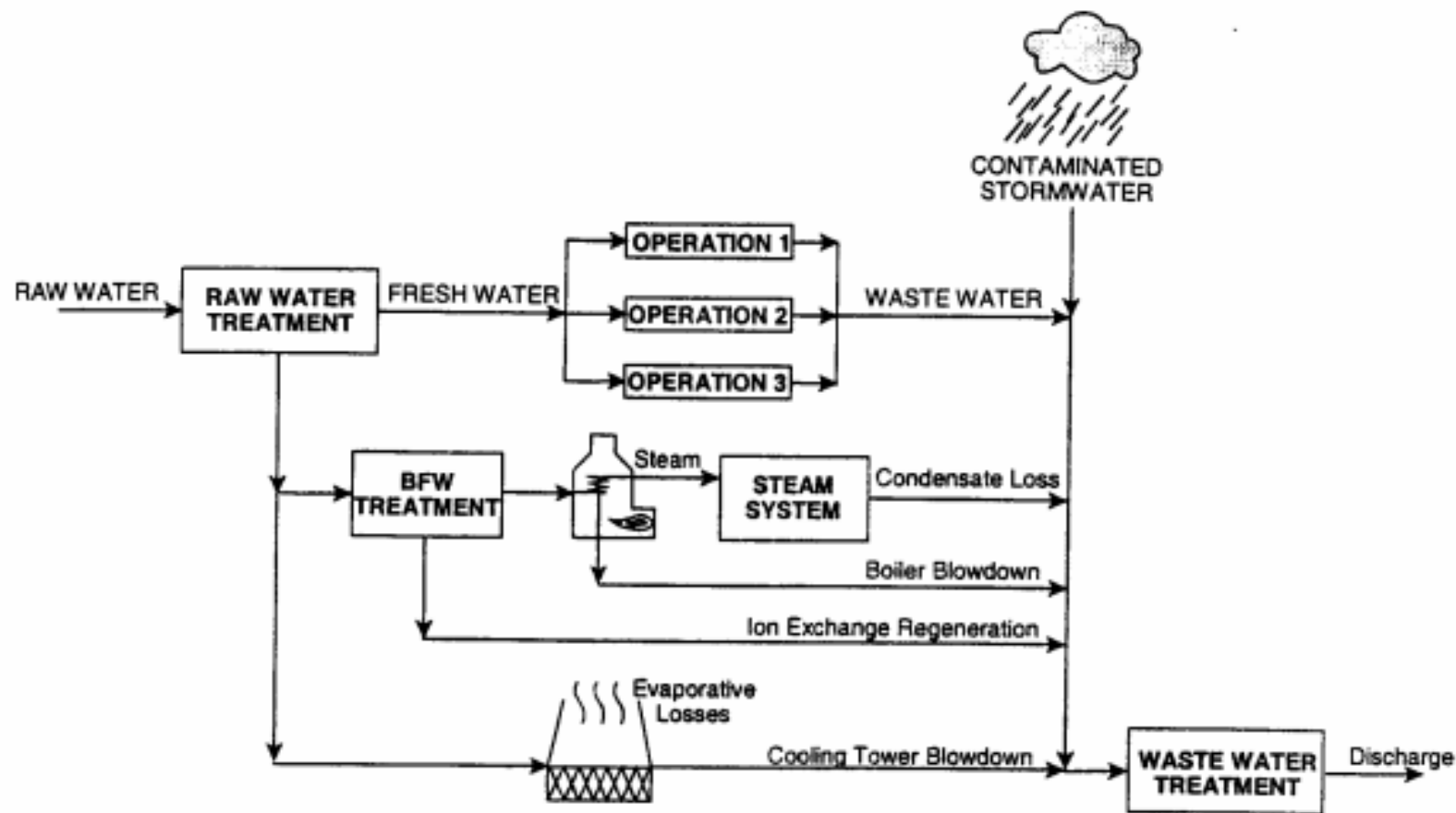
4th Layer: Heating and cooling utilities

The process requires external heating and cooling utilities



5th Layer: Water and effluent treatment

Water and effluent treatment system



- Hierarchical Structure (compare w/ Onion model)

Hierarchy of decisions

- | | |
|---|-----------------------------|
| 1. Batch versus continuous | Reactor |
| 2. Input-output structure of the flowsheet | |
| 3. Recycle structure of the flowsheet | Separation & Recycle System |
| 4. General structure of the separation system <ul style="list-style-type: none">a. Vapor recover systemb. Liquid recovery system | |
| 5. Heat-exchanger network | |

Advantages: Estimate the cost as proceed through the levels in the hierarchy

Objective: Find an optimum design through a systematic analysis

- Batch vs Continuous

Distinction between batch and continuous process is “Fuzzy”

(a) Production rate

- continuous process $> 10^7$ lb/year
- batch process $< 10^6$ lb/year

(b) Market forces

(c) Operation problems

(d) Multiple operations in a single equipment

Level 0: Input information

TABLE 4.1-1
Input information

1. The reactions and reaction conditions
2. The desired production rate
3. The desired product purity, or some information about price versus purity
4. The raw materials and/or some information about price versus purity
5. Information about the rate of the reaction and the rate of catalyst deactivation
6. Any processing constraints
7. Other plant and site data
8. Physical properties of all components
9. Information concerning the safety, toxicity, and environmental impact of the materials involved in the process
10. Cost data for by-products, equipment, and utilities

Level 0: Reaction information

TABLE 4.1-2
Reaction information

1. The stoichiometry of all reactions that take place
2. The range of temperatures and pressures for the reactions
3. The phase(s) of the reaction system
4. Some information on the product distribution versus conversion (and possibly reactor temperature, molar ratio of reactants, and/or pressure)
5. Some information about conversion versus space velocity or residence time
6. If a catalyst is used, some information about the state of the catalyst (homogeneous, slurry, packed bed, powder, etc.), some information about the deactivation rate and some idea of the regenerability of the catalyst as well as the method of regeneration (coke burn, solvent wash, etc.)

Level 0: Reaction information

Max. yield condition is not a optimum condition

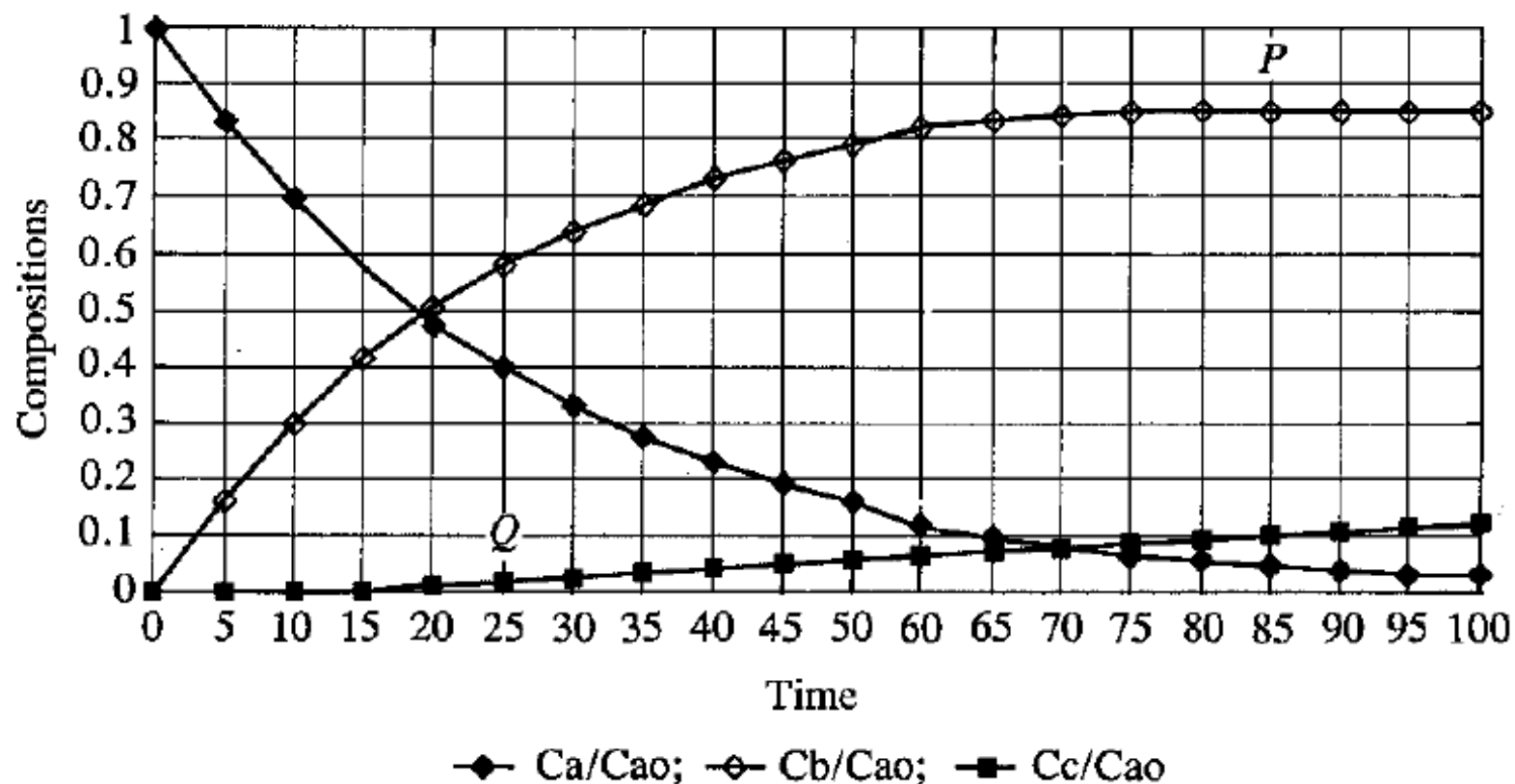


FIGURE 4.1-1
Batch composition profiles.

Level 0: Plant and site data

TABLE 4.1-3
Plant and site data

- 1. Utilities**
 - a. Fuel supply*
 - b. Levels of steam pressure*
 - c. Cooling-water inlet and outlet temperatures*
 - d. Refrigeration levels*
 - e. Electric power*
- 2. Waste disposal facilities**

The new plant should be compatible with the facilities that are already exist on the site.

Level 0: HDA process (example)

TABLE 4.1-4

Input information for the hydrodealkylation of toluene to produce benzene

1. Reaction information

a. Reactions:



b. Reaction inlet temperature $> 1150^\circ\text{C}$ (to get a reasonable reaction rate); reactor pressure = 500 psia.

c.
$$\text{Selectivity} = \frac{\text{Moles Benzene at Reactor Outlet}}{\text{Moles Toluene Converted}} = S$$

$$\text{Conversion} = \frac{\text{Moles Toluene Converted in Reactor}}{\text{Moles Toluene Fed to Reactor}} = x$$

$$S = 1 - \frac{0.0036}{(1 - x)^{1.544}} \quad x < 0.97 \quad (4.1-4)$$

d. Gas phase

e. No catalyst

Level 0: HDA process (example)

2. Production rate of benzene: 265 mol/hr
3. Product purity of benzene: $x_D = 0.9997$
4. Raw materials: Pure toluene at ambient conditions; H_2 stream containing 95% H_2 , 5% CH_4 at 550 psia, 100°F
5. Constraints: $H_2/\text{aromatic} \geq 5$ at the reactor inlet (to prevent coking); reactor outlet temperature < 1300°F (to prevent hydrocracking); rapidly quench reactor effluent to 1150°F (to prevent coking); $x < 0.97$ for the product distribution correlation
6. Other plant and site data to be given later

Level 1: Batch versus continuous

Design of Batch versus Continuous Processes

To develop a conceptual design for a continuous process, we must do the following:

1. Select the process units needed.
2. Choose the interconnections among these units.
3. Identify the process alternatives that need to be considered.
4. List the dominant design variables.
5. Estimate the optimum processing conditions.
6. Determine the best process alternative.

For a batch process we must make exactly the same decisions. However, we must also make the following decisions:

7. Which units in the flowsheet should be batch and which should be continuous?
8. What processing steps should be carried out in a single vessel versus using individual vessels for each processing step?
9. When is it advantageous to use parallel batch units to improve the scheduling of the plant?
10. How much intermediate storage is required, and where should it be located?

Level 1: Batch versus continuous

The factors that favor batch operations are

1. Production rate

- sometimes batch $< 10^7$ lb/year, usually $< 10^6$ lb/year
- multi-product

2. Market forces

- seasonal production
- short product lifetime

3. Scale up problems

- very long reaction time
- handling slurries at low flow rate
- rapidly fouling materials

Level 2: Input output structure

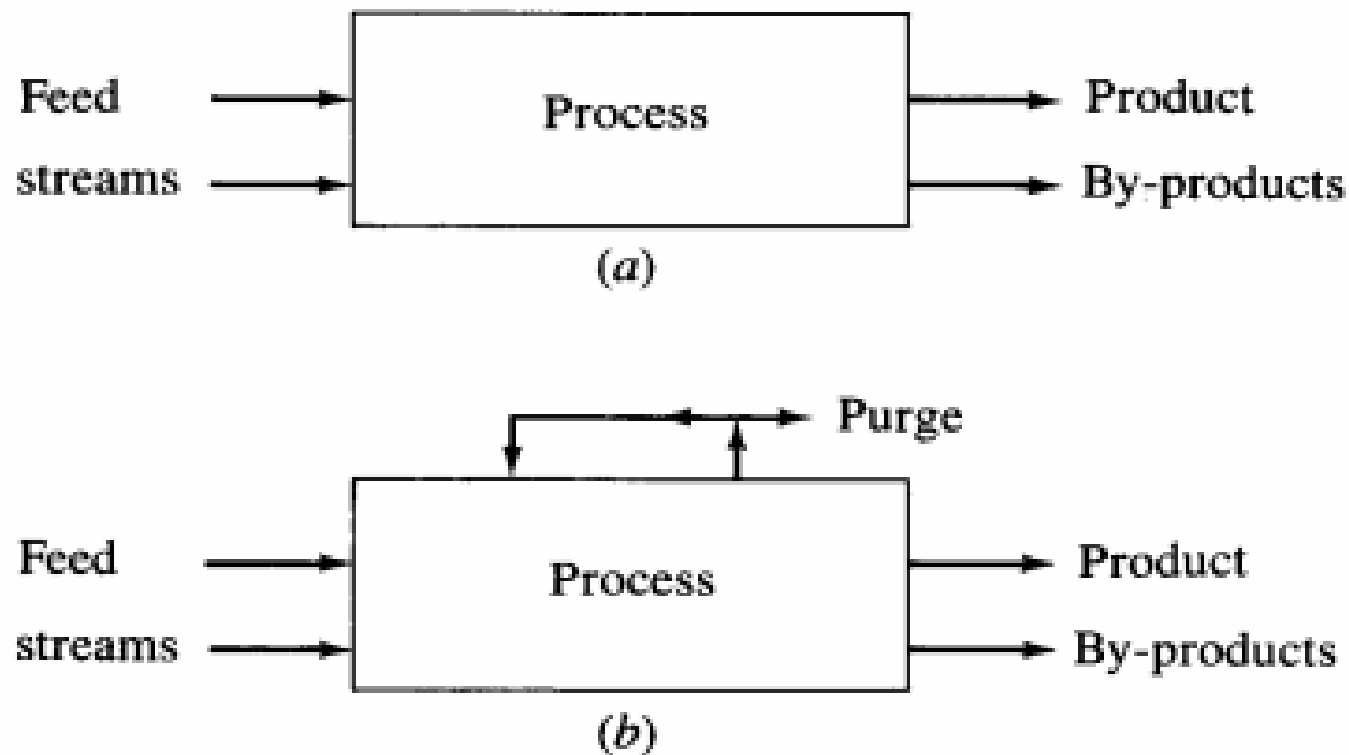


FIGURE 5.1-1

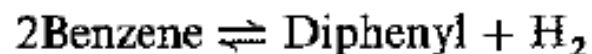
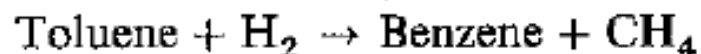
Input-output structure of flowsheet. [From J. M. Douglas, *AIChE J.*, 31: 353 (1985).]

Level 2: Decisions

TABLE 5.1-2
Level-2 decisions

1. Should we purify the feed streams before they enter the process?
2. Should we remove or recycle a reversible by-product
3. Should we use a gas recycle and purge stream?
4. Should we not bother to recover and recycle some reactants?
5. How many product streams will there be?
6. What are the design variables for the input-output structure, and what economic trade-offs are associated with these variables?

Example for pt. 2



remove CH_4 would increase raw material cost

recycle CH_4 would oversize all equipments in the loop

Level 2: Decisions (cont'd)

Some design guides for purification of feeds :

1. If feed impurity is
 - a. not inert and is present in large amounts
 - b. catalyst poison

→ **remove it**

2. If feed impurity is
 - c. present in a gas feed
 - d. a product or by product components
 - e. form an azeotrope with reactant
 - f. easier to separate from the product than the feed

→ **it is better to feed to process**

Level 2: Number of product streams

TABLE 5.1-3
Destination codes and component classifications

Destination code	Component classification
1. Vent	Gaseous by-products and feed impurities
2. Recycle and purge	Gaseous reactants plus inert gases and/or gaseous by-products
3. Recycle	Reactants Reaction intermediates Azeotropes with reactants (sometimes) Reversible by-products (sometimes)
4. None	Reactants—if complete conversion or unstable reaction intermedia
5. Excess—vent	Gaseous reactant not recovered and recycled
6. Excess—waste	Liquid reactant not recovered or recycled
7. Primary product	Primary product
8. Valuable by-product (<i>I</i>)	Separate destination for different by-products
9. Fuel	By-products to fuel
10. Waste	By-products to waste treatment

Level 2: Overall material balances

TABLE 5.2-2
Procedures for developing overall material balances

1. Start with the specified production rate.
2. From the stoichiometry (and, for complex reactions, the correlation for product distribution) find the by-product flows and the reactant requirements (in terms of the design variables).
3. Calculate the impurity inlet and outlet flows for the feed streams where the reactants are completely recovered and recycled.
4. Calculate the outlet flows of reactants in terms of a specified amount of excess (above the reaction requirements) for streams where the reactants are not recovered and recycled (recycle and purge or air or water).
5. Calculate the inlet and outlet flows for the impurities entering with the reactant streams in step 4.

Level 2: Overall material balances (cont'd)

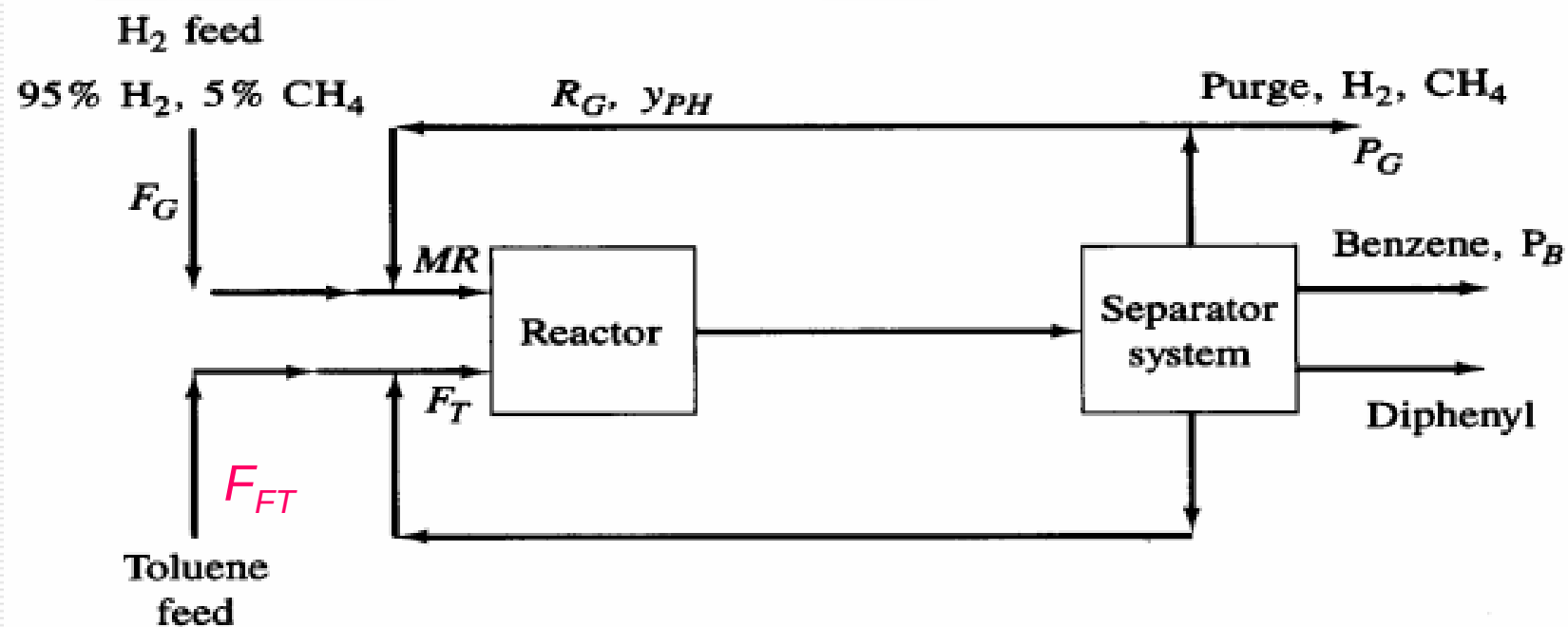


FIGURE 6.2-4
HDA, gas recycle.

Level 2: Overall material balances (cont'd)

Develop expressions in terms of design variables without consider any recycle flows

According M/B & stoichiometry, we could obtain the equations below :

$$F_{FT} = \frac{P_B}{S}$$

$$P_D = F_{FT} \frac{1-S}{2} = \frac{P_B}{S} \frac{1-S}{2}$$

$$F_G = \frac{P_B \left[1 - (1 - y_{PH}) \frac{1-S}{2} \right]}{S(y_{FH} - y_{PH})}$$

$$P_G = F_G + \frac{P_B}{S} \frac{1-S}{2}$$

Given design variables
 P_B , S , y_{PH} to calculate overall
input & output flow rates

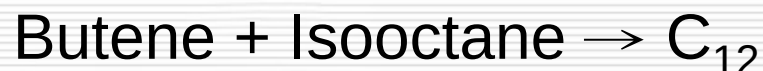
Level 3: Recycle structure

TABLE 6.1-1
Decisions for the recycle structure

1. How many reactor systems are required? Is there any separation between the reactor systems?
2. How many recycle streams are required?
3. Do we want to use an excess of one reactant at the reactor inlet?
4. Is a gas compressor required? What are the costs?
5. Should the reactor be operated adiabatically, with direct heating or cooling, or is a diluent or heat carrier required?
6. Do we want to shift the equilibrium conversion? How?
7. How do the reactor costs affect the economic potential?

Do not separate two components and remix them at a reactor inlet!

Example for pt. 3.



Use excess isobutane leads to an improved selectivity to produce isobutane, but the larger cost to recover and recycle isobutane.

Level 3: Recycle structure

Decisions on Reactor Heat Effects

- a. Estimate the heat load and adiabatic temp. change first, and judge adiabatic reaction if feasible.
- b. If operation adiabatically is unreasonable → use direct heating or cooling first
- c. For endothermic process,
Heat load $< (6\sim8) \times 10^6$ Btu/h → isothermal operation use direct heating
For larger head load → a diluent or heat carrier is used
- d. For exothermic process,
Adiabatic temp. rise is less than 10~15% inlet temp. → use adiabatic reactor, and for larger heat load, use direct cooling or add a diluent.

Level 3: HDA process (example)

Example 6.1-2 HDA process. The components and their destination for the HDA process are as follows:

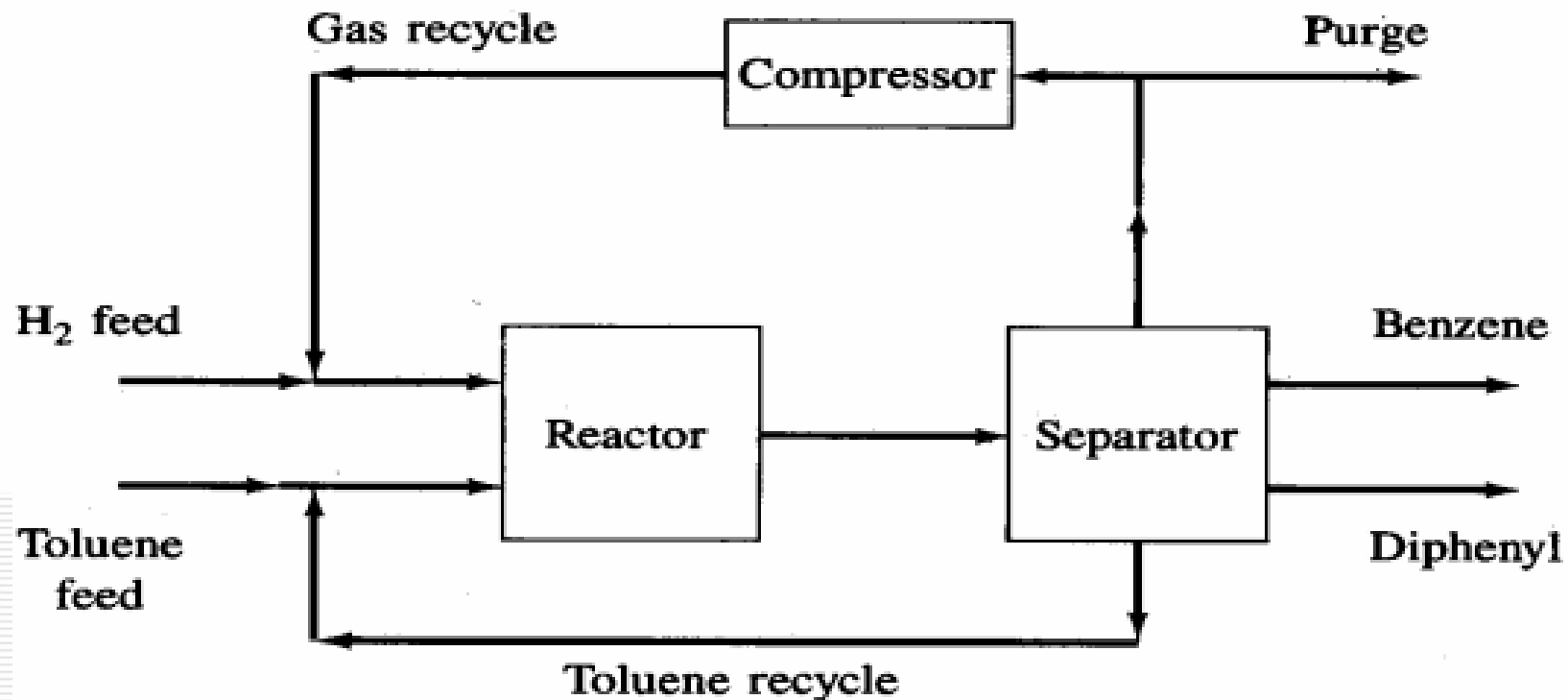


FIGURE 6.1-2

HDA recycle structure. [From J. M. Douglas, *AIChE J.*, 31: 353 (1985).]

Level 3: HDA process (example)

Example 6.1-2 HDA process.

Component	NBP, °C	Destination
H ₂	-253	Recycle + purge — gas
CH ₄	-161	Recycle + purge — gas
Benzene	80	Primary product
Toluene	111	Recycle — liquid
Diphenyl	255	Fuel by-product

Thus, there are three product streams—purge, benzene, and diphenyl—and two recycle streams, H₂ + CH₄ and toluene, where the first is a gas and the second is a liquid. A recycle flowsheet is given in Fig. 6.1-2, and it shows the reactor and the recycle gas compressor.

Level 4: Separation system

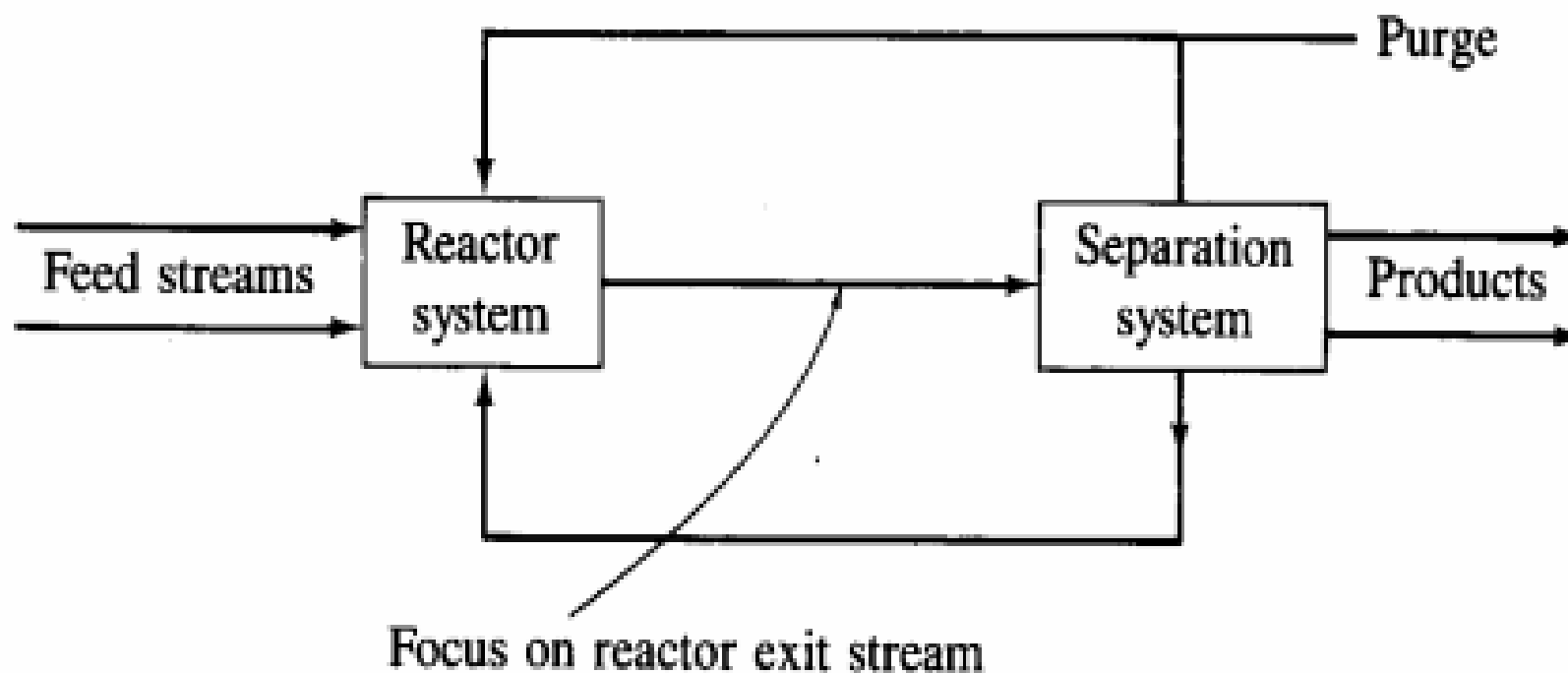


FIGURE 7.1-1

Phase of the reactor effluent stream.

Case 1 : general structure

Case 2 : vapor recovery system

Case 3 : liquid separation system

Level 4: Reactor exit is liquid

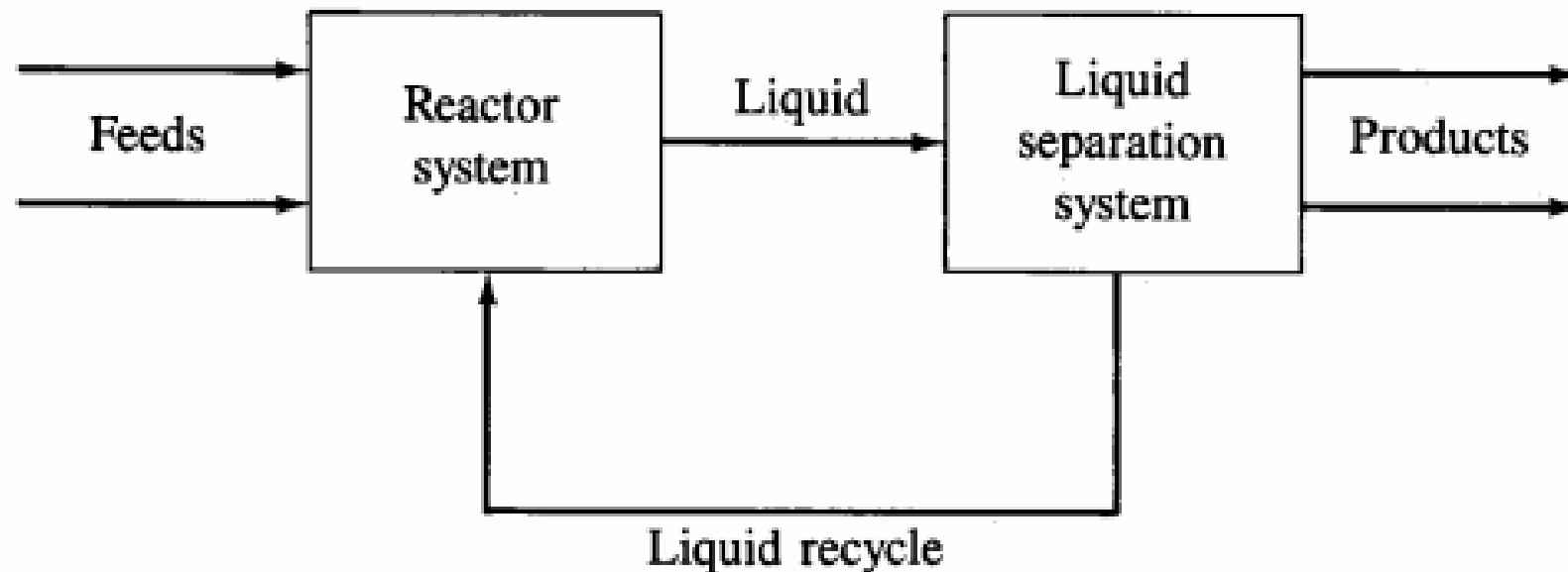


FIGURE 7.1-2

Reactor exit is liquid. [From J. M. Douglas, *AIChE J.*, **31**: 353 (1985).]

Liquid separation system: column, extraction, azeotropic distillation etc.

Level 4: Reactor exit is vapor and liquid

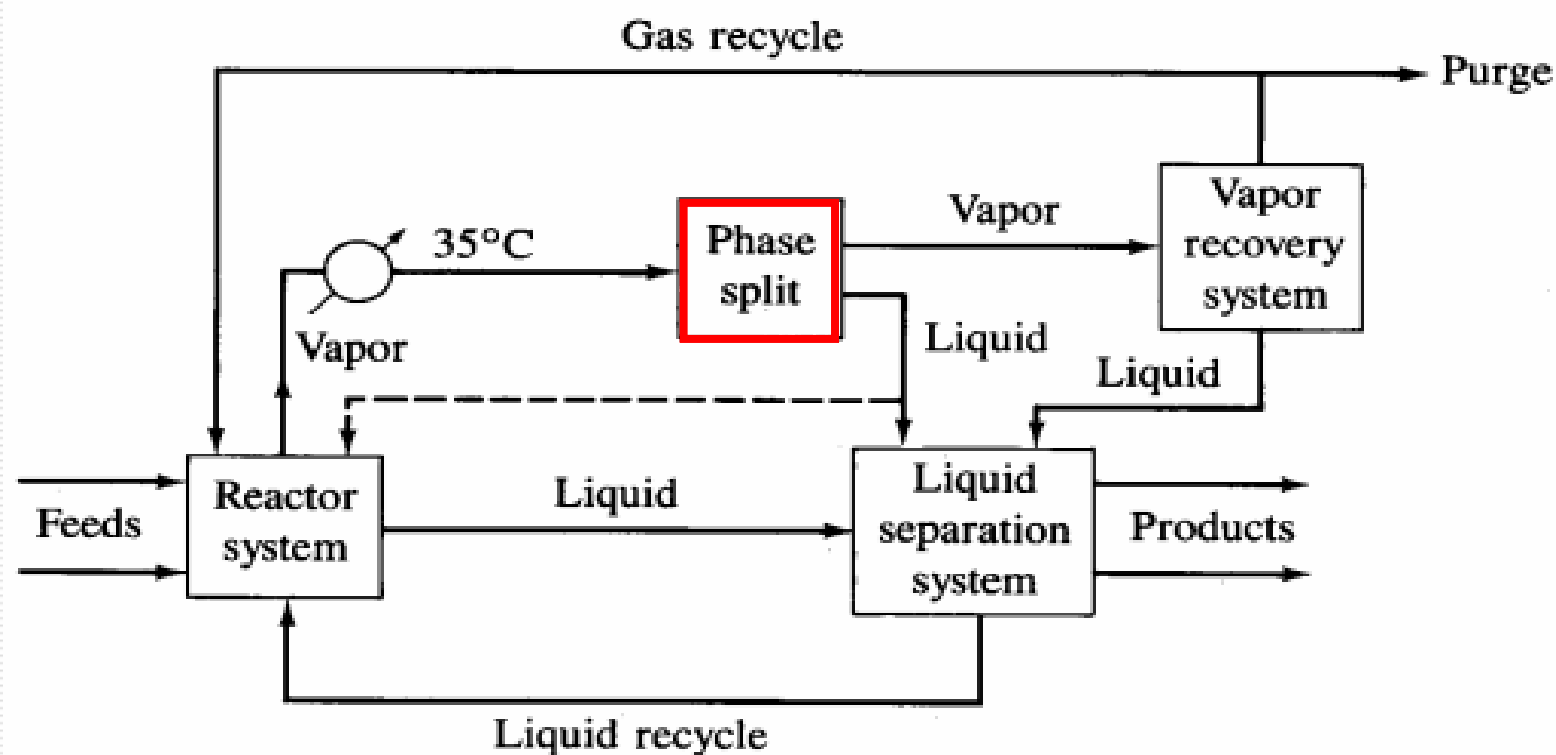


FIGURE 7.1-3

Reactor exit is vapor and liquid. [From J. M. Douglas, *AIChE J.*, 31: 353 (1985).]

Two phase: use reactor as a phase splitter or cool down to 35 C.

Level 4: Reactor exit is vapor

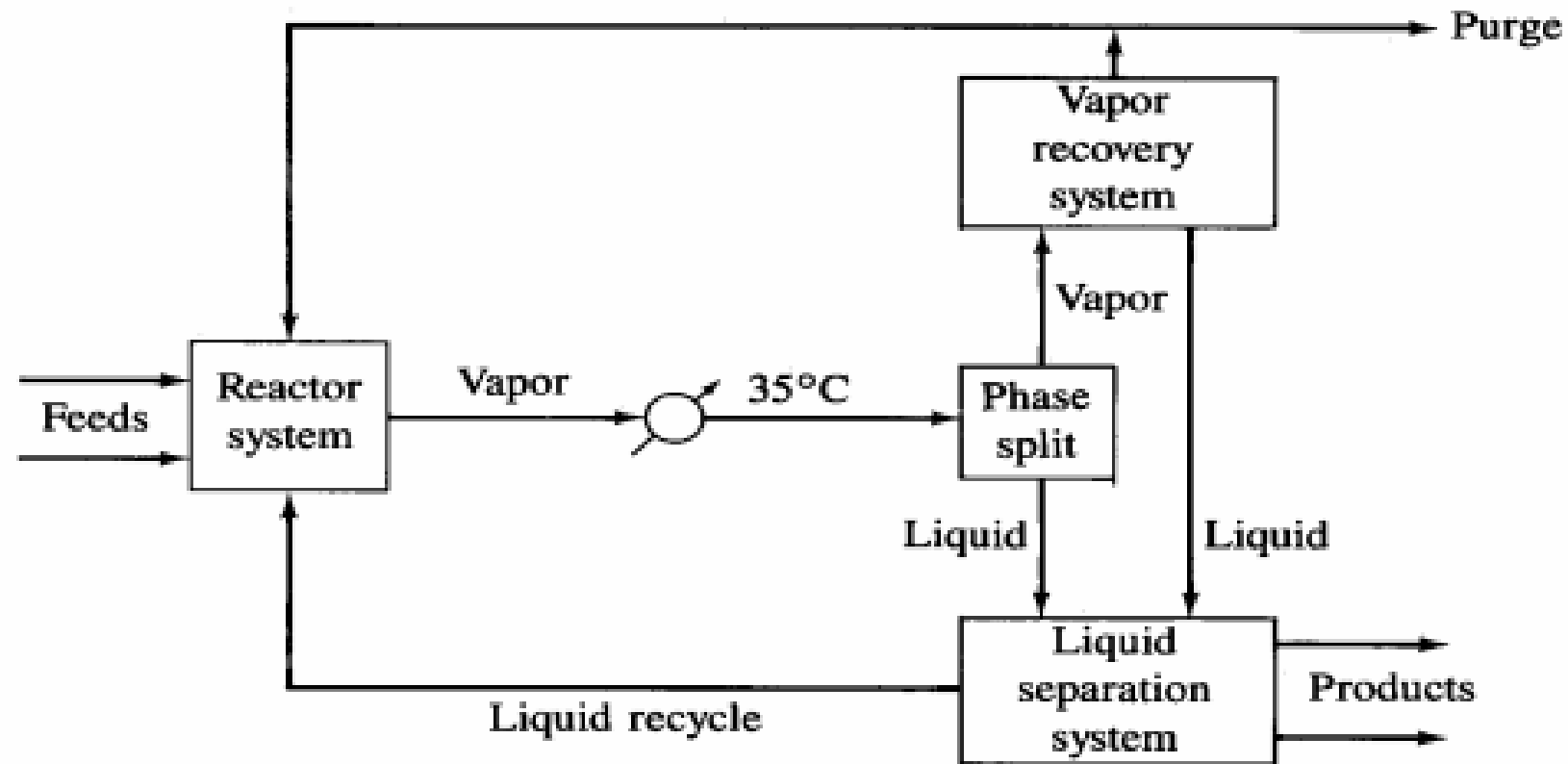
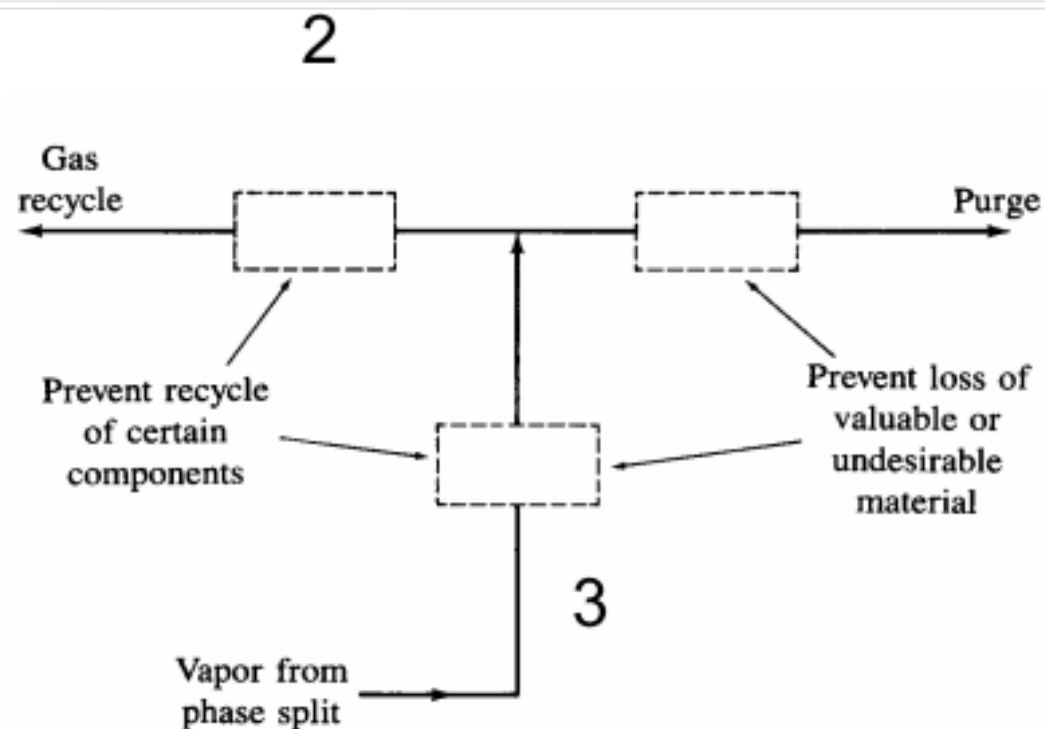


FIGURE 7.1-4

Reactor exit is vapor. [From J. M. Douglas, *AIChE J.*, **31**: 353 (1985).]

phase split is cheap.

Level 4: Vapor recovery system



Type: condensation
absorption
adsorption
membrane
reaction system
(removing CO₂, H₂S)

FIGURE 7.2-1
Vapor recovery system location.

Level 4: Vapor recovery system (cont'd)

Location 1: If significant amount of valuable materials are being lost in the purge.

Location 2: If materials are deleterious to the reactor operation.

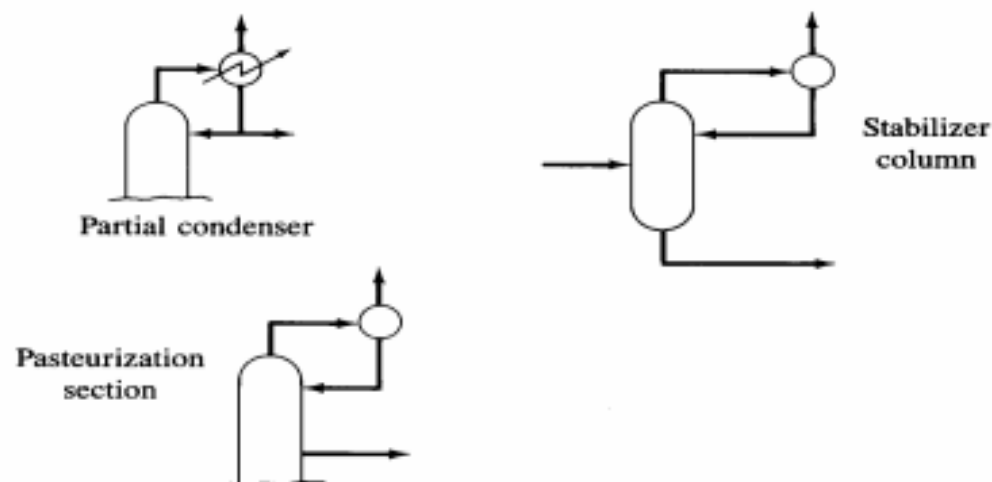
Location 3: Both factors are valid.

None: Both factors do not exist.

Level 4: Liquid separation system

• Light ends

If product quality is unacceptable, options are:



Note: Recycle vapor stream to vapor recovery system if possible.

FIGURE 7.3-1
Alternatives for removing light ends.

Method

- drop the pressure for removal
- partial condenser
- pasteurization column
- stabilizer column

Destination

- vent
- flare
- vapor recovery system

Level 4: Liquid separation system

• Azeotropes

- recycle the azeotropes
- split the azeotrope (need two columns for a binary azeotrope).

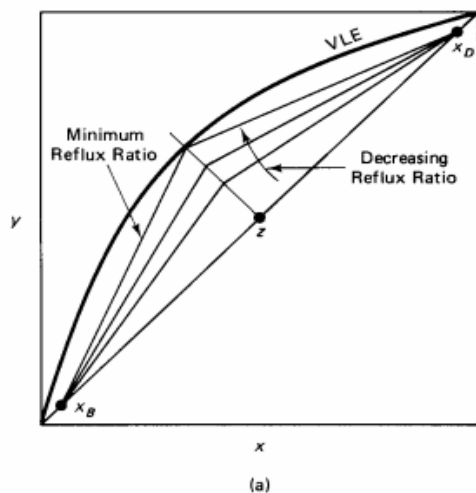
Why use distillation?

- ability to handle a wide range of throughput
- ability to handle a wide range of feed concentrations
- ability to produce high product purity

Separations not suited to distillation:

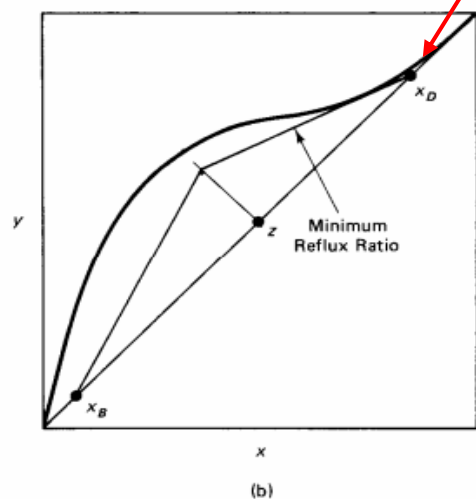
- low molecular weight materials
- high molecular weight heat sensitive materials
- separation of components with a low concentration
- separation of classes of components
- relative volatility < 1.1

Level 4: Liquid separation system



Understand the effects of changing reflux rate or boilup rate.

操作線與平衡線相切



定義：

q = heat need to convert 1 mol feed to sat'd vapor ÷
molar heat of vaporization

q line slope = $q / (q-1)$

Figure 8-9 Minimum reflux ratio.
(a) Pinch occurring at the feed plate
(b) Pinch occurring above the feed plate
(nonideal VLE).

Level 4: Liquid separation system

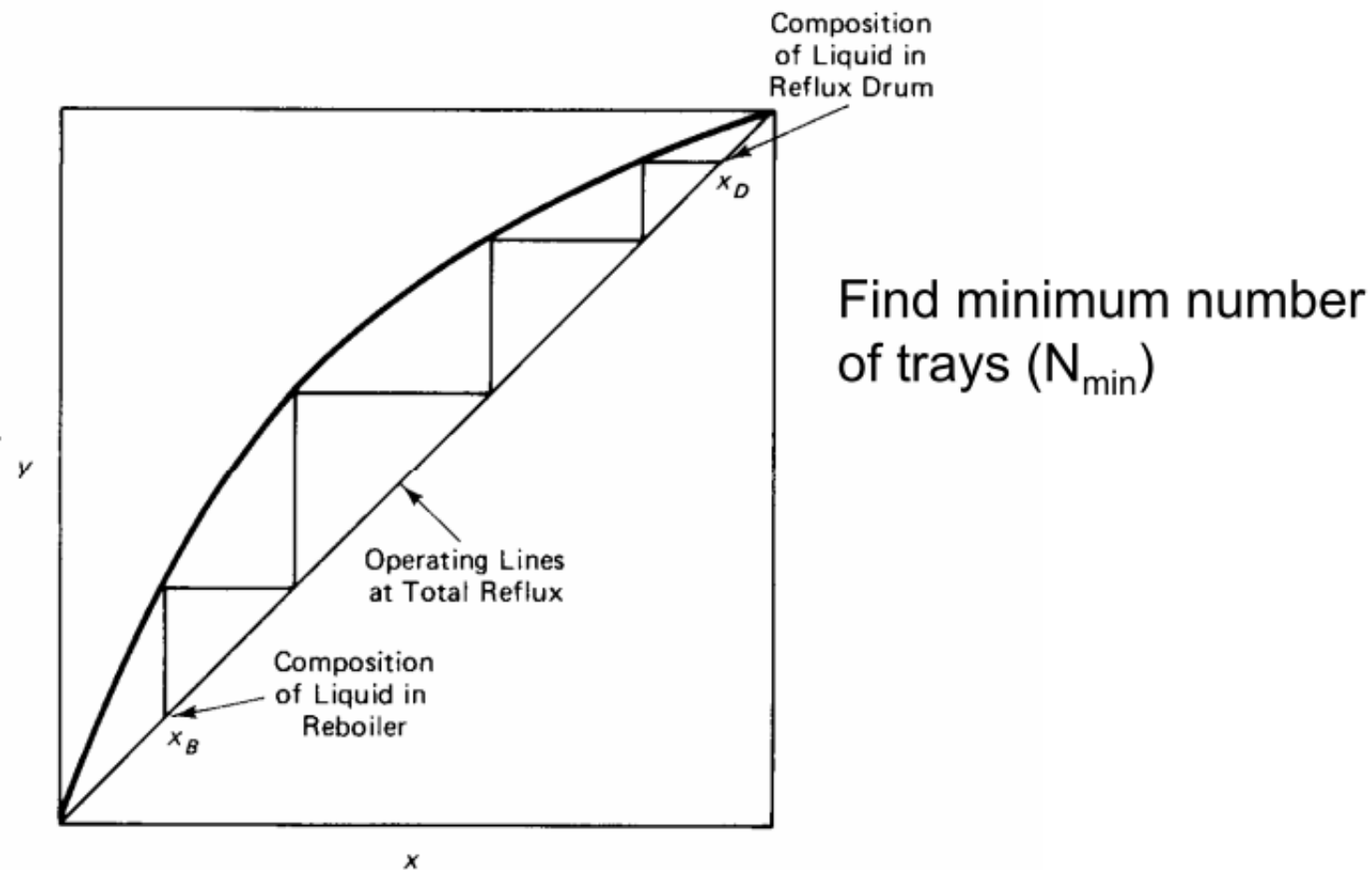


Figure 8-11 Total reflux on a McCabe-Thiele diagram.

Level 4: Liquid separation system

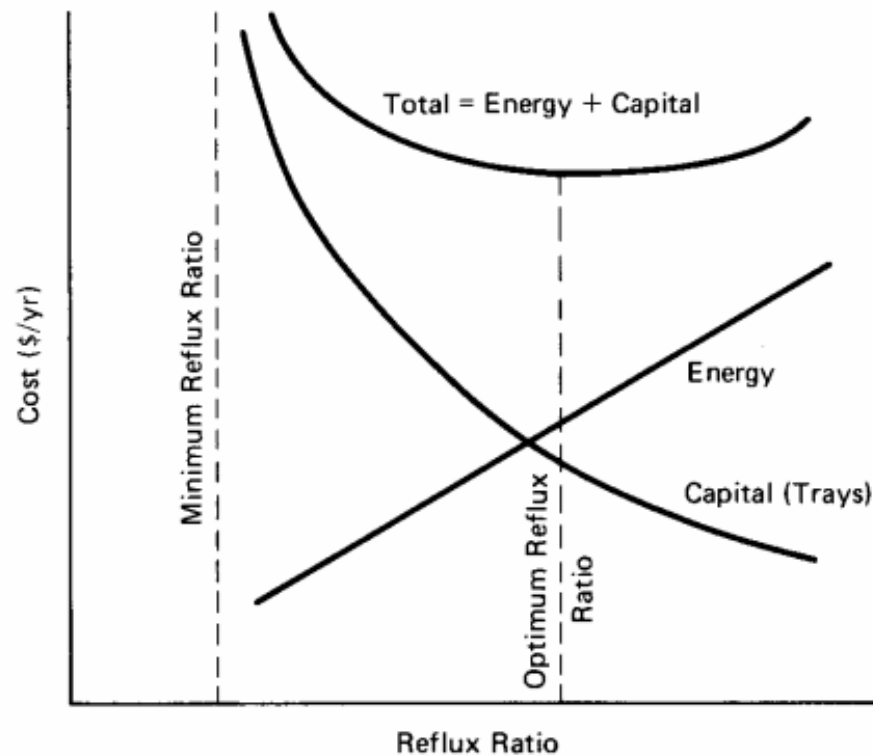
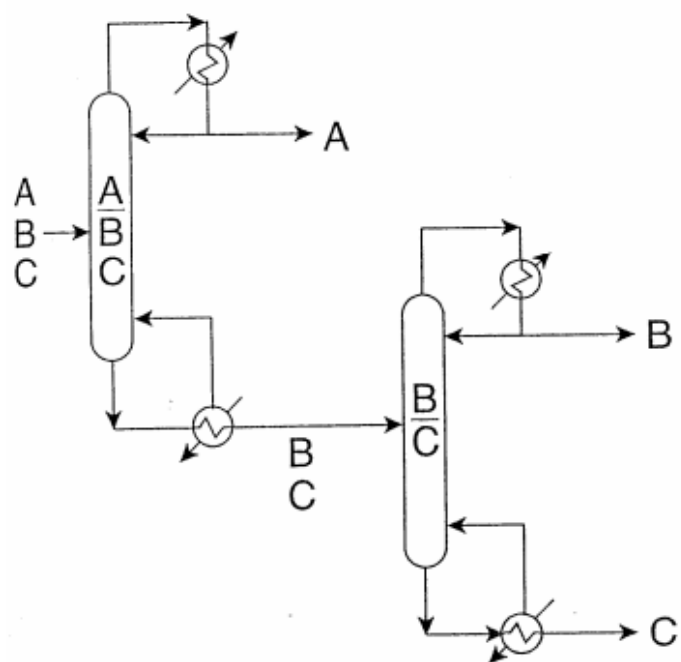


Figure 8-10 Effect of reflux ratio on costs.

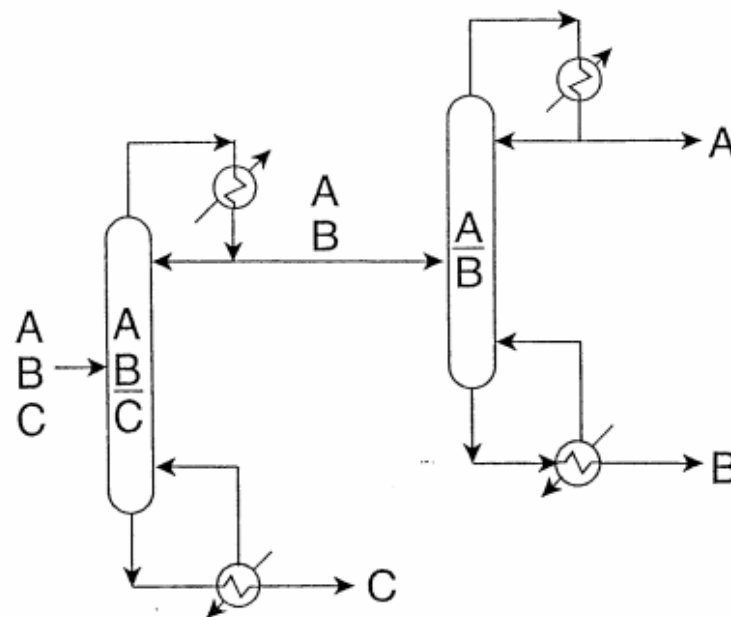
$$RR = 1.1 \sim 1.2 RR_{min}$$

Level 4: Liquid separation system



Direct Sequence

the lightest component is taken overhead in each column



Indirect Sequence

the heaviest component as bottom product in each column

Level 4: Liquid separation system

TABLE 7.3-3
General heuristics for column sequencing

1. Remove corrosive components as soon as possible.
2. Remove reactive components or monomers as soon as possible.
3. Remove products as distillates.
4. Remove recycle streams as distillates, particularly if they are recycled to a packed bed reactor.

TABLE 7.3-4
Heuristics for column sequencing

1. Most plentiful first.
2. Lightest first.
3. High-recovery separations last.
4. Difficult separations last.
5. Favor equimolar splits.
6. Next separation should be cheapest.

☆ *The best column sequence will change as we alter the design variables*

Level 4: Liquid separation system

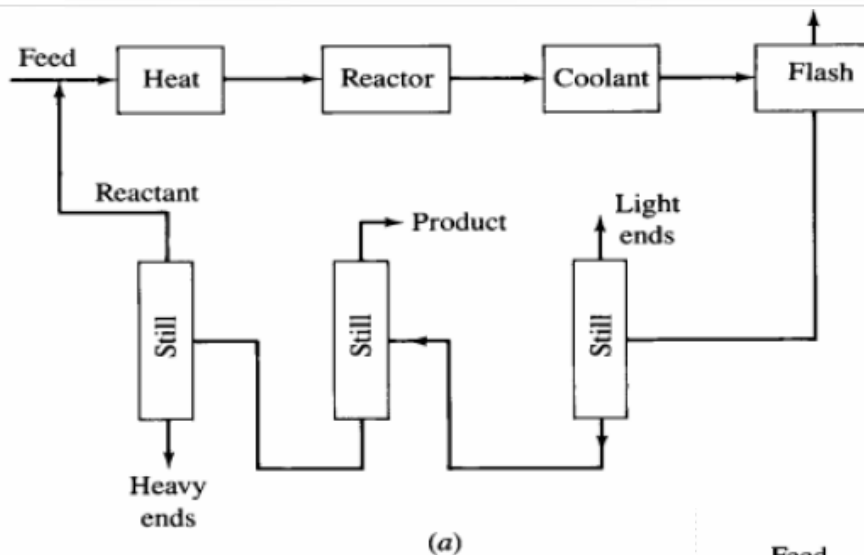
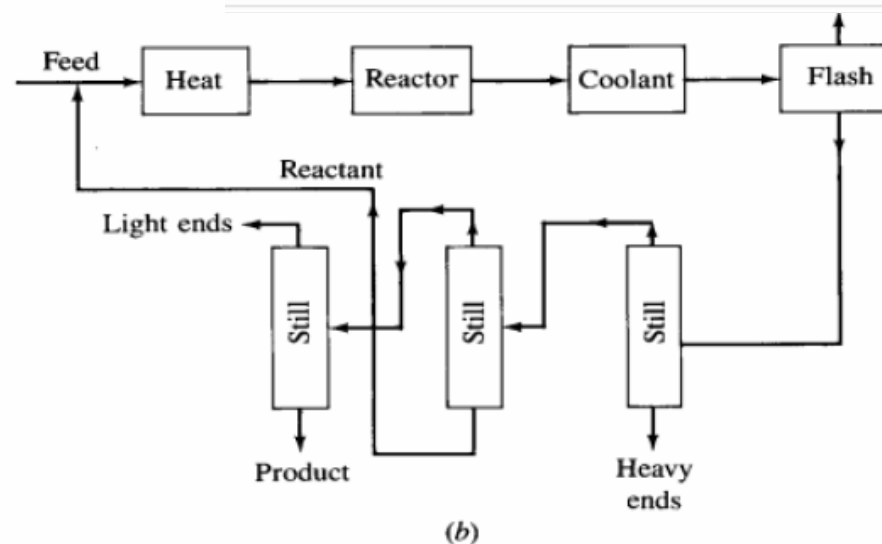


FIGURE 7.3-4

Sequence selection changes recycle costs.

light ends (H_2/CH_4)
product (Benzene)
reactant (Toluene)
heavy ends (Diphenyl)

★ *Select the sequence that minimizes the number of columns in a recycle loop*



Level 4: Liquid separation system

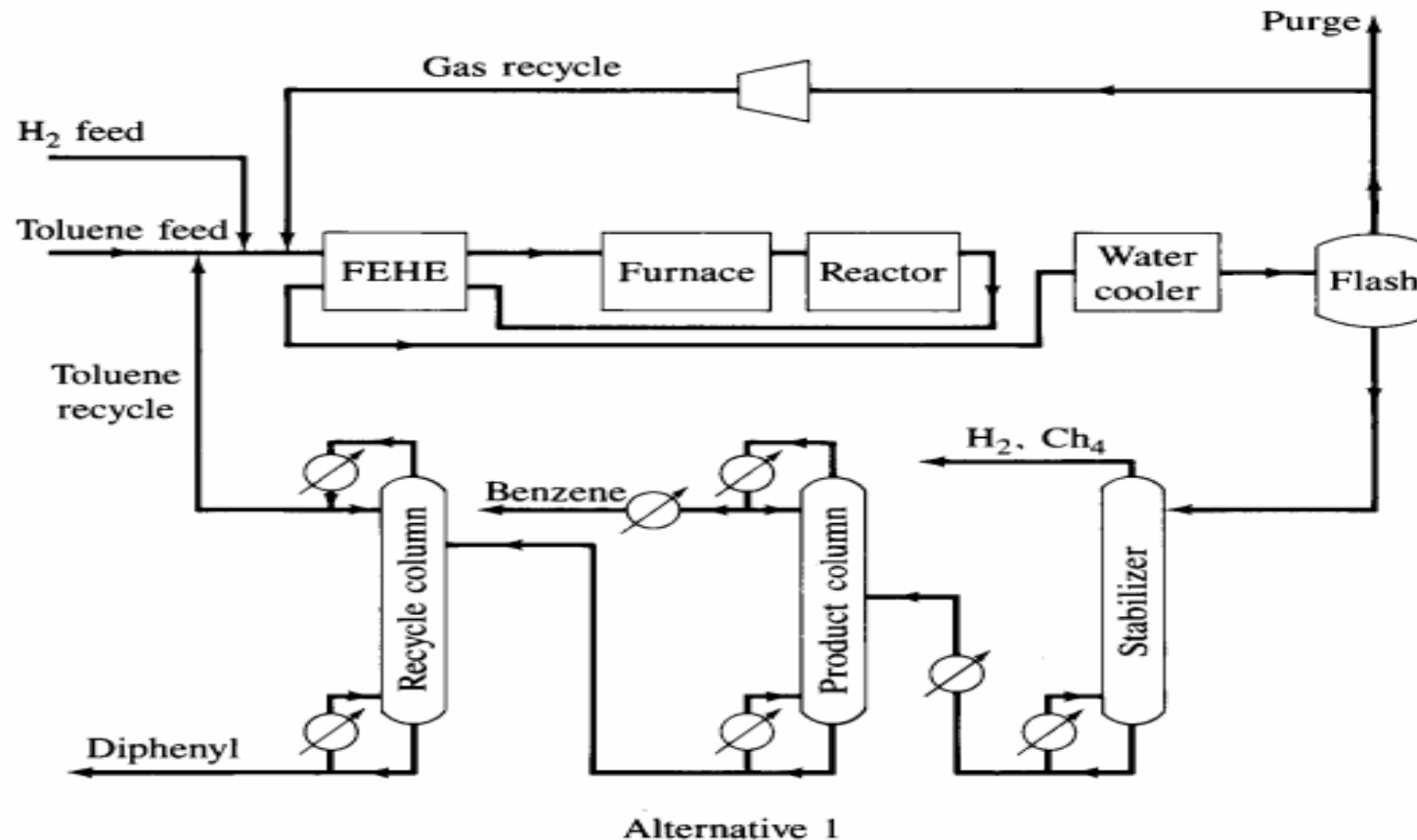


FIGURE 8.10-2

HDA process with a feed-effluent heat exchanger. (Note: The quench stream following the reactor is not shown on these graphs.) [From D. L. Terrill and J. M. Douglas, *I&EC Research*, **26**: 685 (1987), with permission of the American Chemical Society.]

Level 4: Liquid separation system

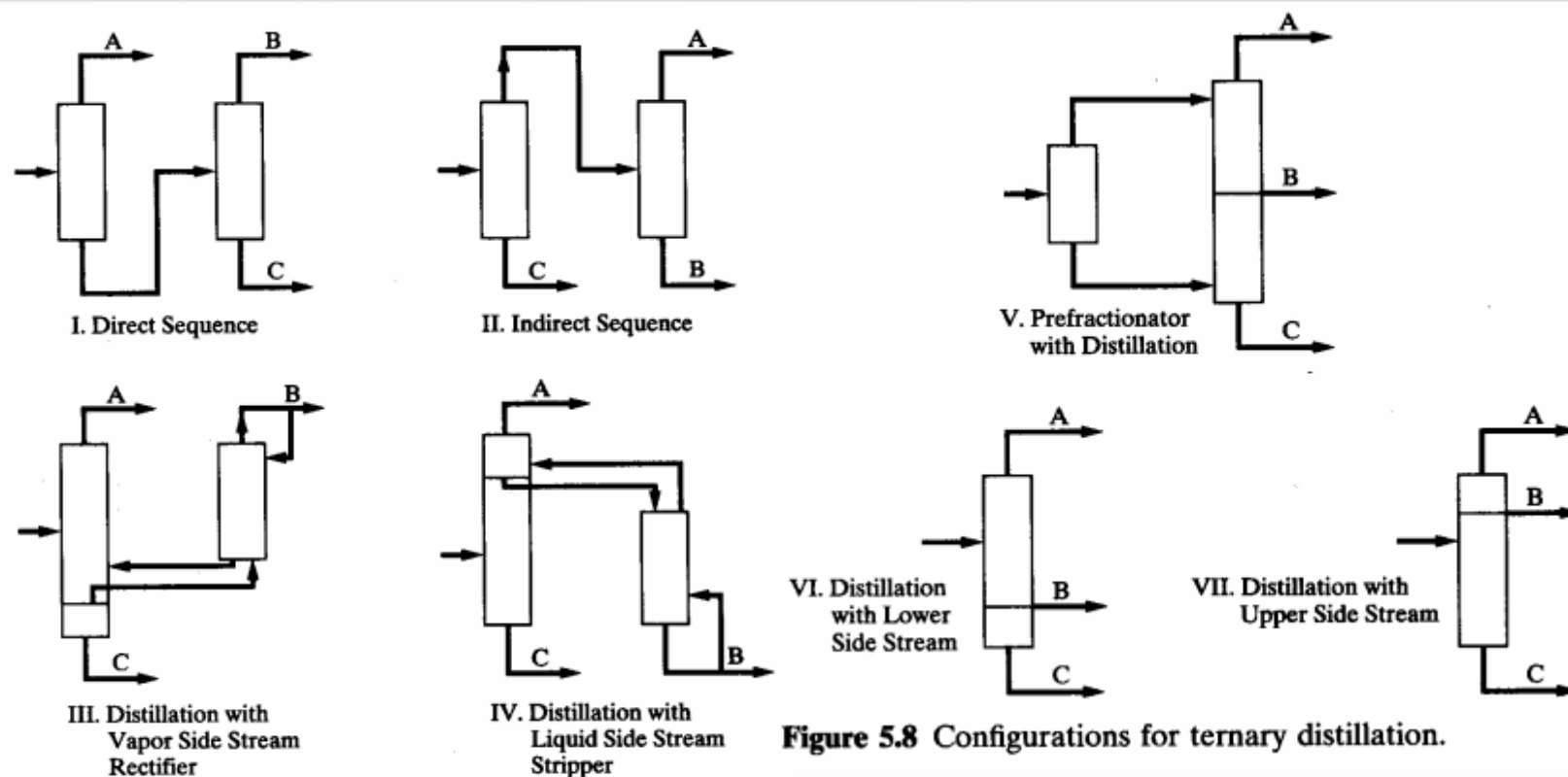


Figure 5.8 Configurations for ternary distillation.

Level 4: Liquid separation system

Minimum-boiling

repulsion

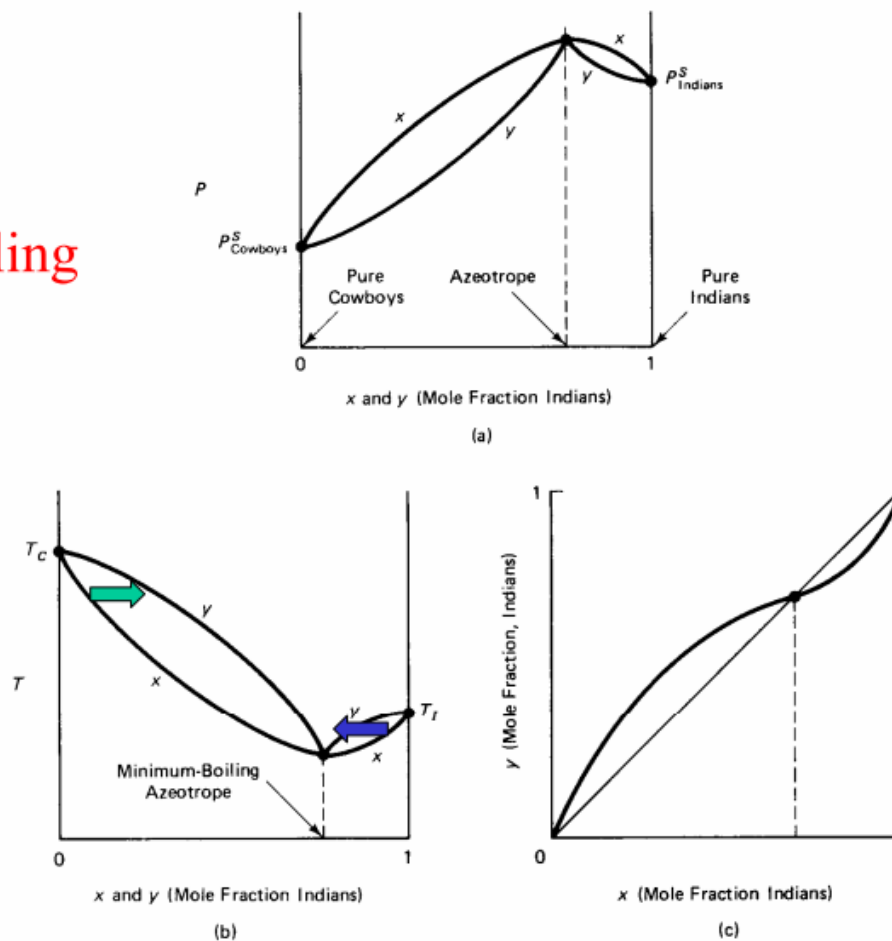


Figure 6-11 Homogeneous minimum-boiling azeotrope. (a) Pxy diagram (const. T) (b) Txy diagram (const. P) (c) xy diagram (either T or P const.).

Level 4: Liquid separation system

Maximum-boiling
attraction

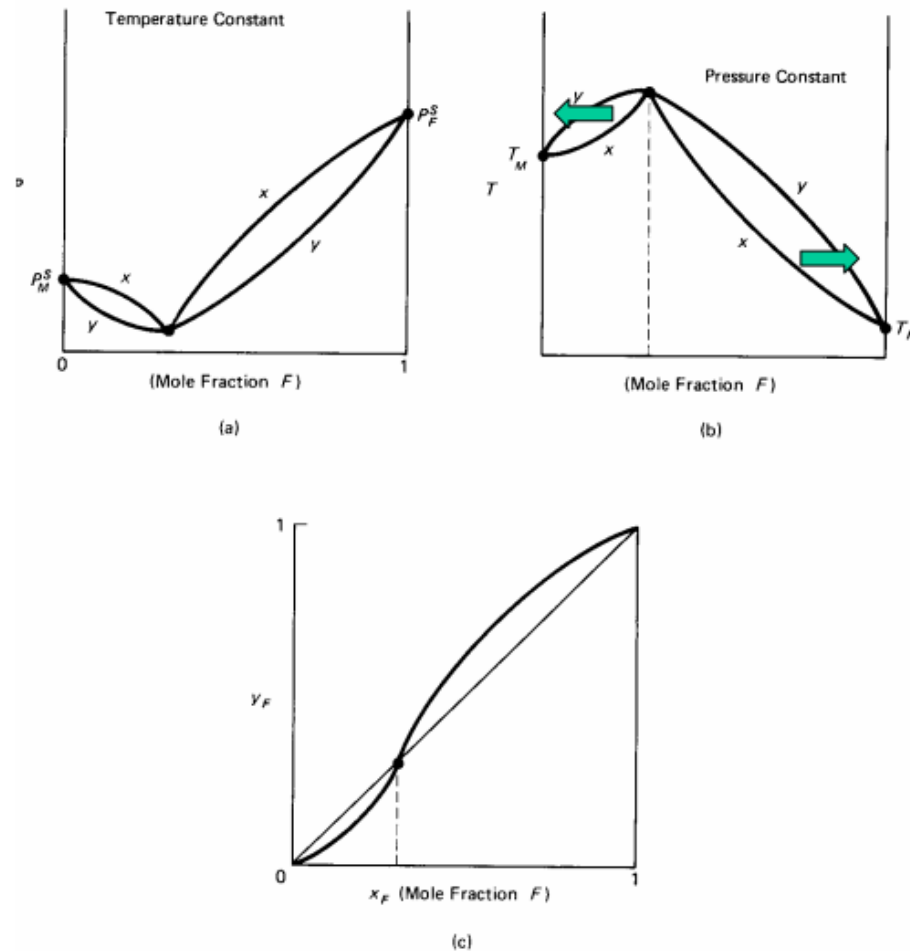


Figure 6-12 Maximum-boiling homogeneous azeotrope. (a) Pxy diagram (b) Txy diagram (c) xy diagram.

Level 4: Liquid separation system

Two-liquid phase

Repulsion force
is extremely large

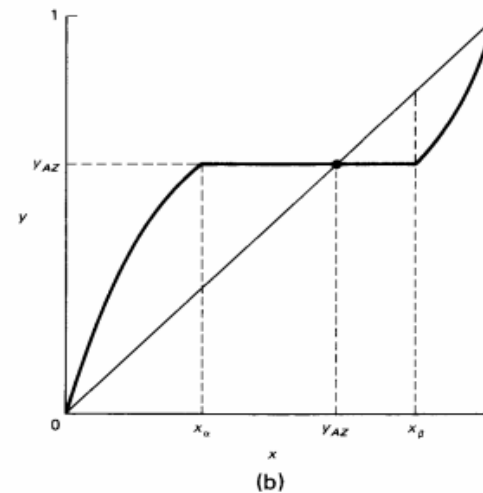
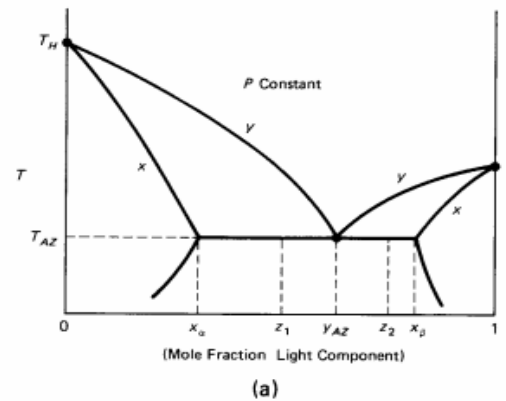


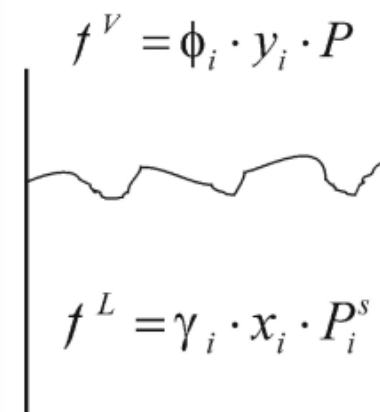
Figure 6-13 Heterogeneous minimum-boiling azeotrope. (a) Txy diagram
(b) xy diagram.

Level 4: Liquid separation system

Breaking Azeotrope

We have to change something to overcome azeotrope

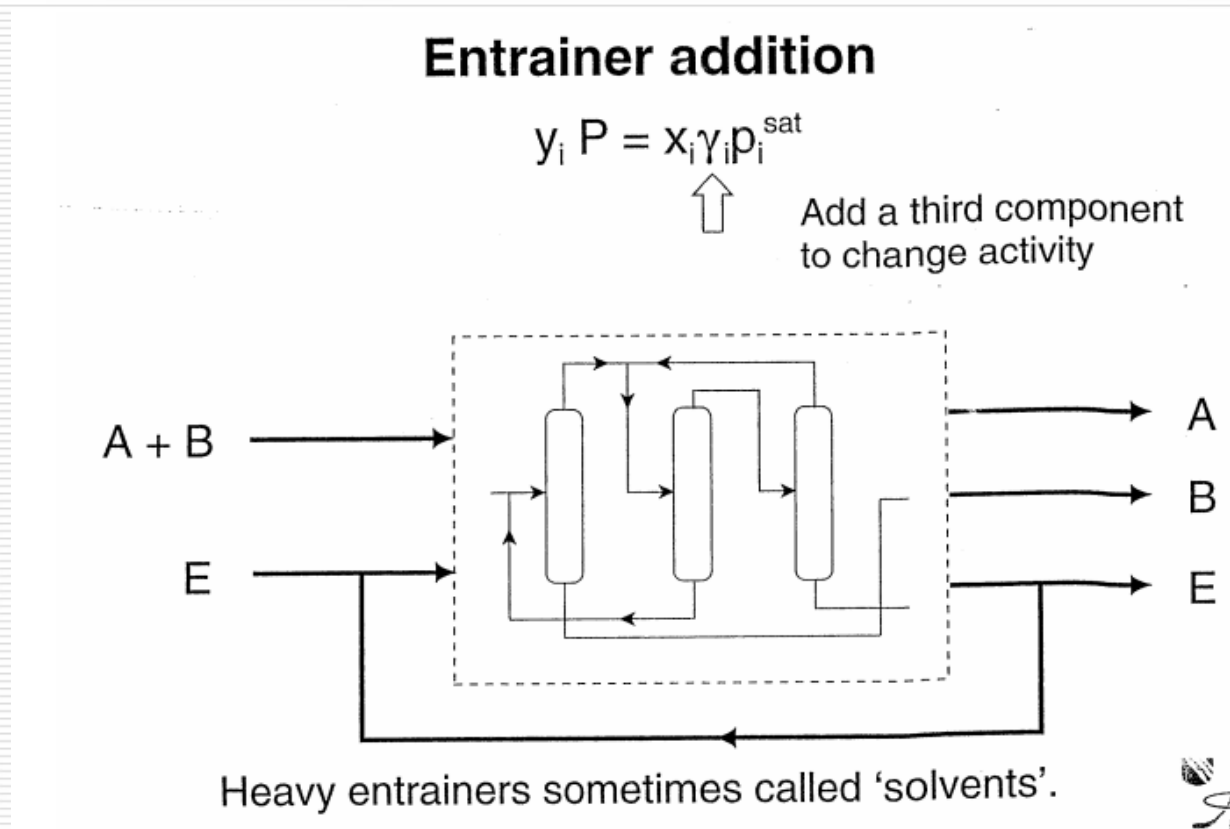
1. Change P- pressure swing
- Pressure swing distillation
2. Change gamma- add entrainer
- Azeotrope distillation & Extraction distillation
3. Change y- remove vapor across membrane
(pervaporization)



The diagram shows a vertical rectangular box representing a distillation column. Inside the box, there is a wavy horizontal line representing the liquid-vapor interface. Above the line, the equation $f^V = \phi_i \cdot y_i \cdot P$ is written. Below the line, the equation $f^L = \gamma_i \cdot x_i \cdot P_i^s$ is written.

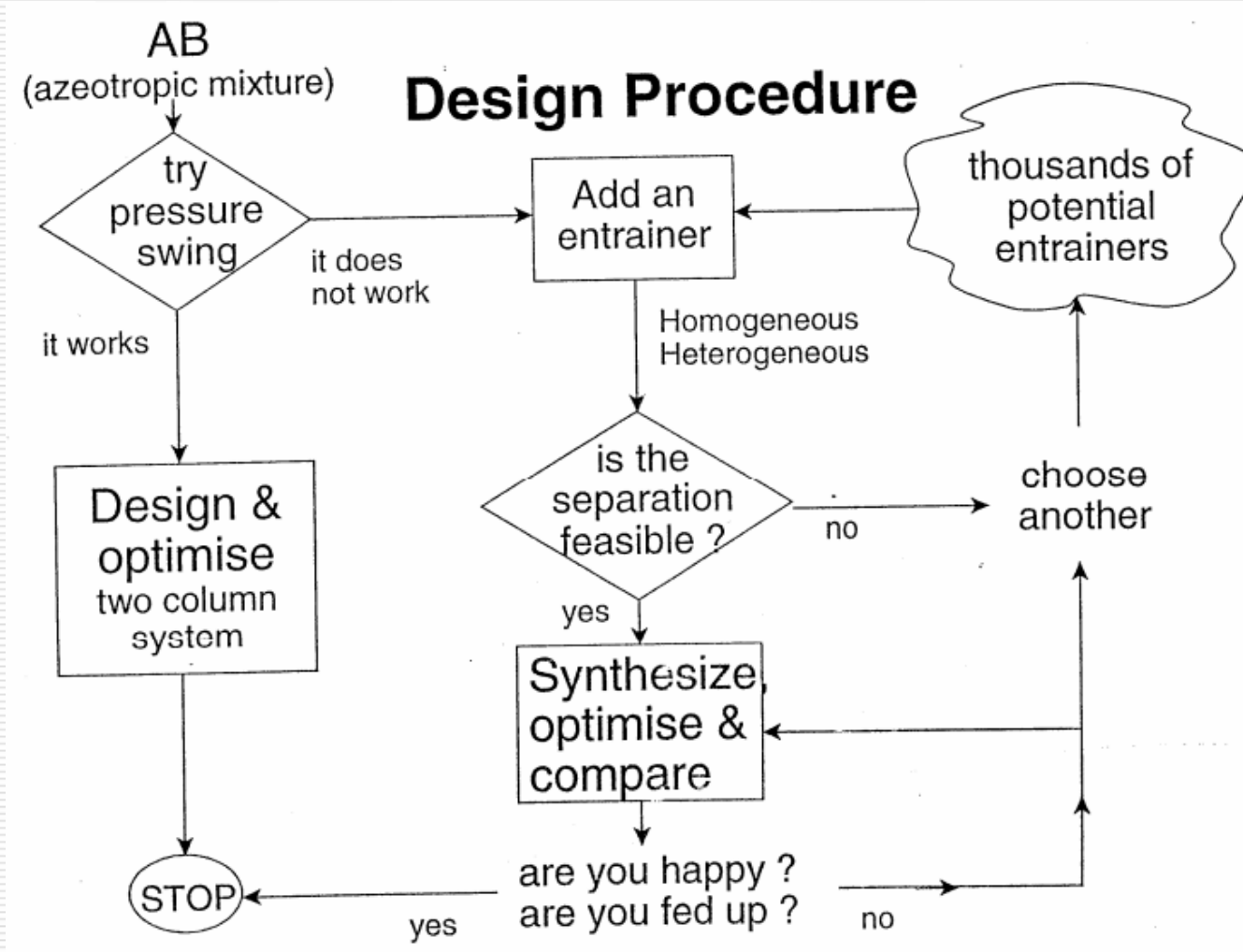
$$f^V = \phi_i \cdot y_i \cdot P$$
$$f^L = \gamma_i \cdot x_i \cdot P_i^s$$

Level 4: Liquid separation system



- Remark:
1. Must change gamma or the ratio of them.
 2. Can possibly induce phase splitting.
 3. Too many choices!

Level 4: Liquid separation system

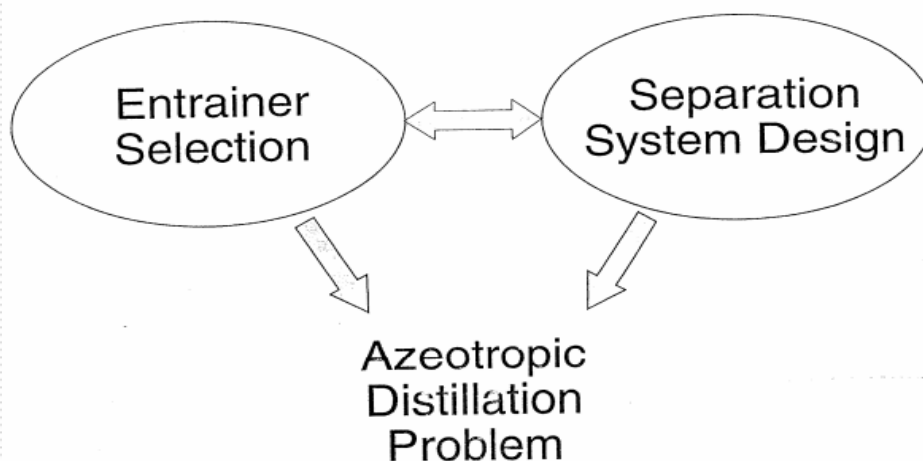


Level 4: Liquid separation system

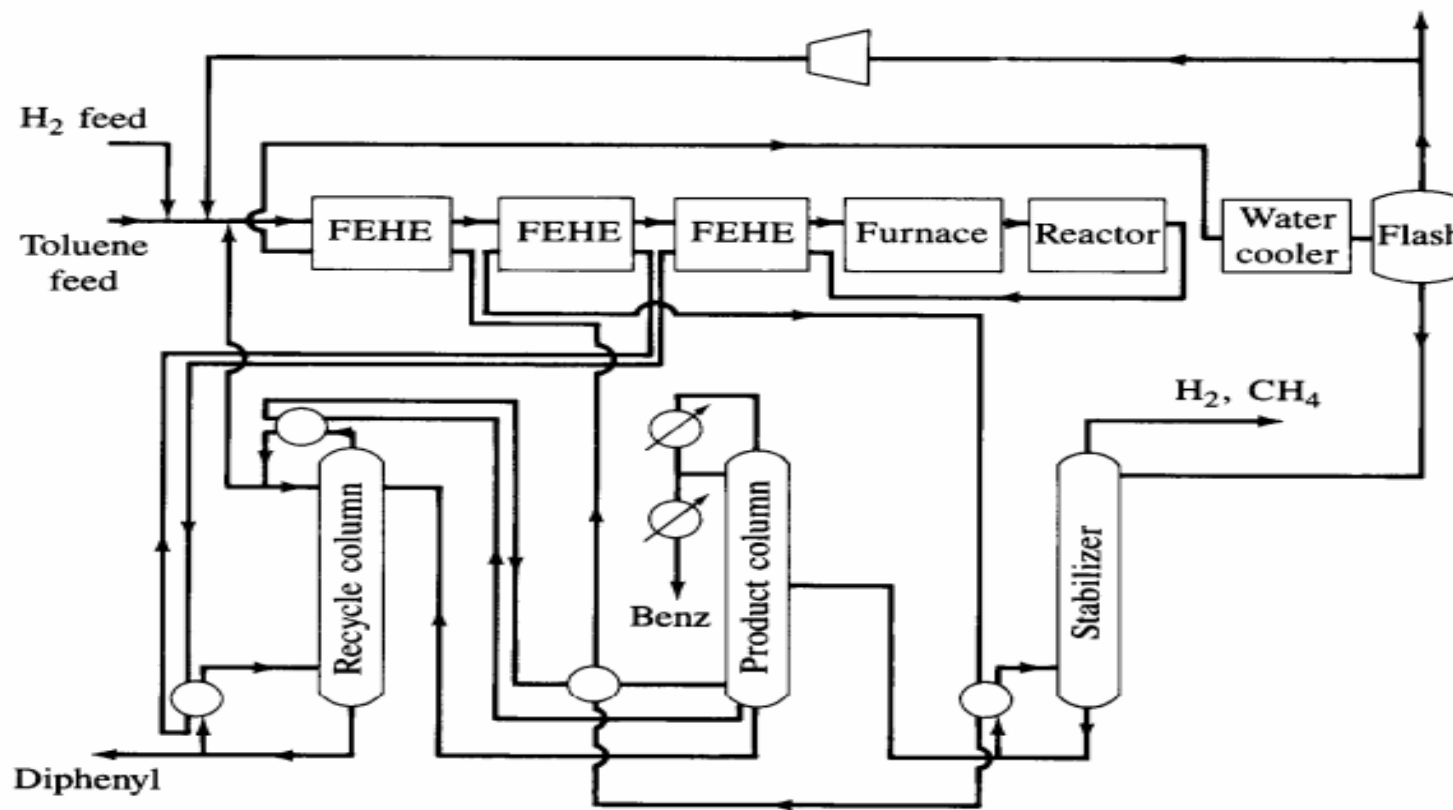
Design Procedure- In practice

- leave azeotropes until last
- try pressure swing
- try extractive distillation using entrainer while not causing phase-split (homogeneous azeotropic distillation)
- try adding a (light) entrainer and induce phase-split (heterogeneous azeotropic distillation)

so the problem has two parts



Level 5: Heat-exchanger network



Alternative 6

FIGURE 8.10-7

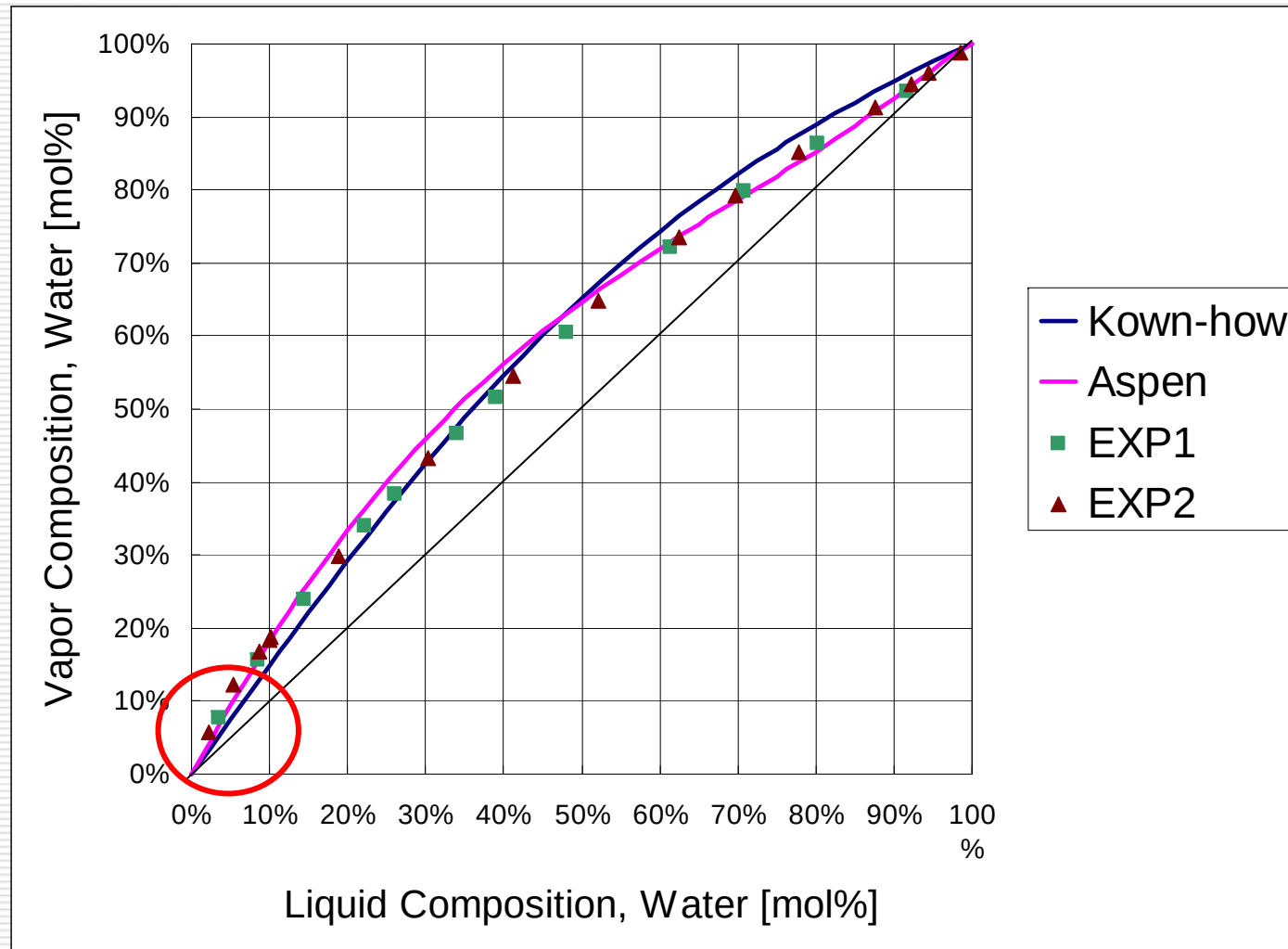
Energy integration alternative 6. [From D. L. Terrill and J. M. Douglas, *I&EC Research*, **26**: 685 (1987), with permission of the American Chemical Society.]

- Attentions for Process Design
 - Flowsheet set up
 - Thermodynamic model selection
 - VLE & VLLE data check
 - Convergence judge
 - Design factor for equipment

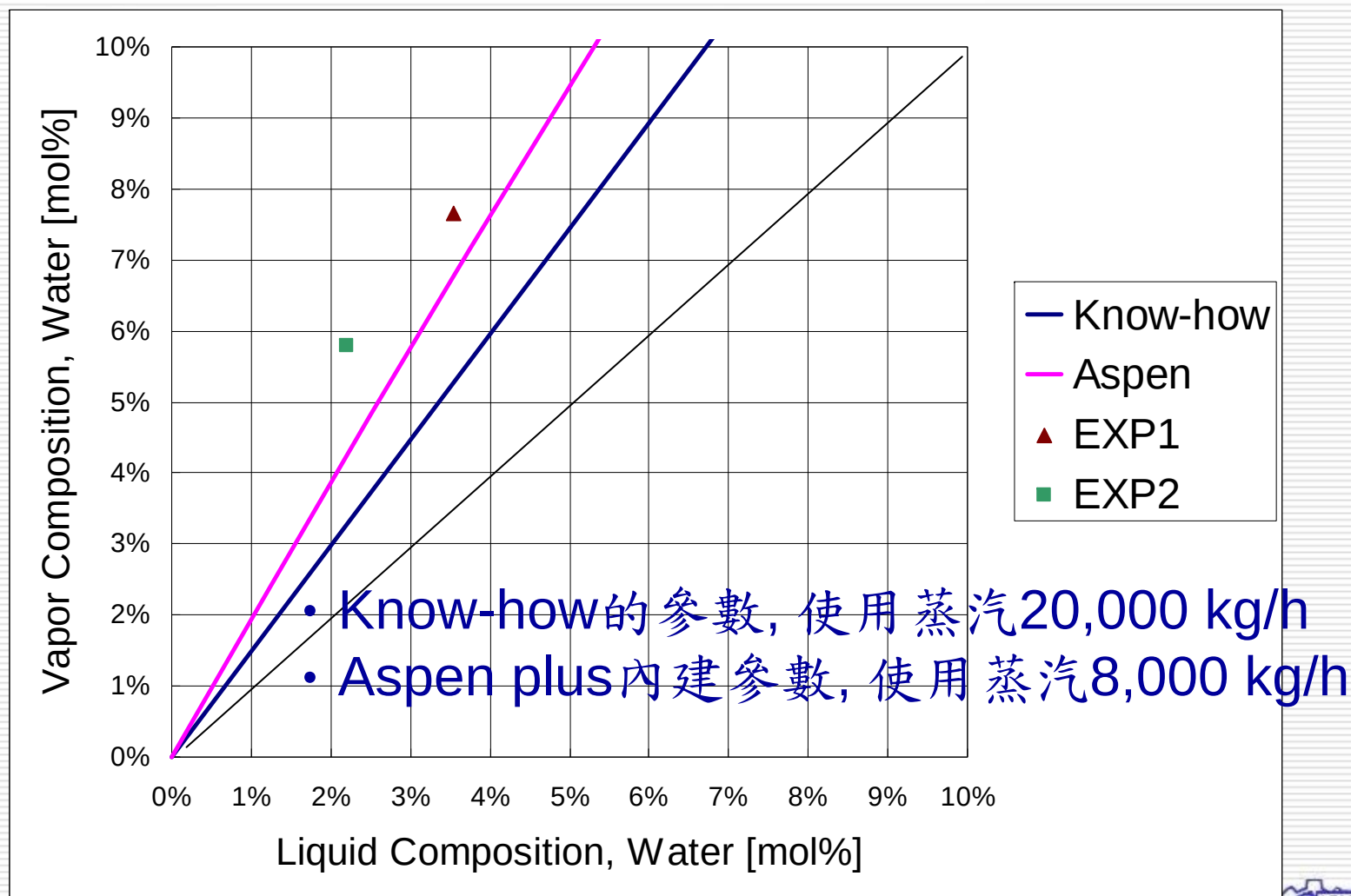
• Thermodynamic Model Parameters Sources

1. Simulator data bank
2. DECHEMA or literatures
3. R & D
4. Predictive model
 - UNIFAC, Regular Solution, Flory-Huggins Models
5. Simplified experimental data
 - Azeotropic data, mutual solubility, infinite dilution activity

VLE Data Check



VLE Data Check (Water Removal Column)



LLE Data Check

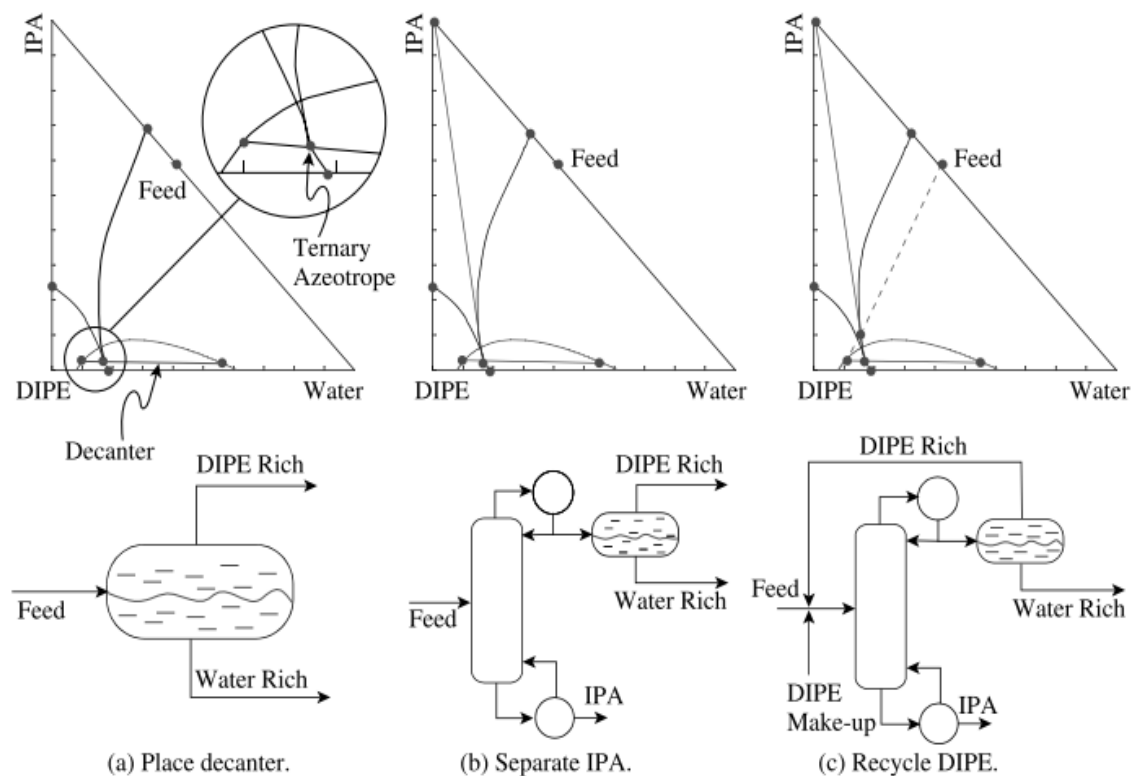


Figure 12.33 Separation of isopropyl alcohol (IPA) and water mixture using di-isopropyl ether (DIPE) as entrainer in heterogeneous azeotropic distillation.

LLE Data Check

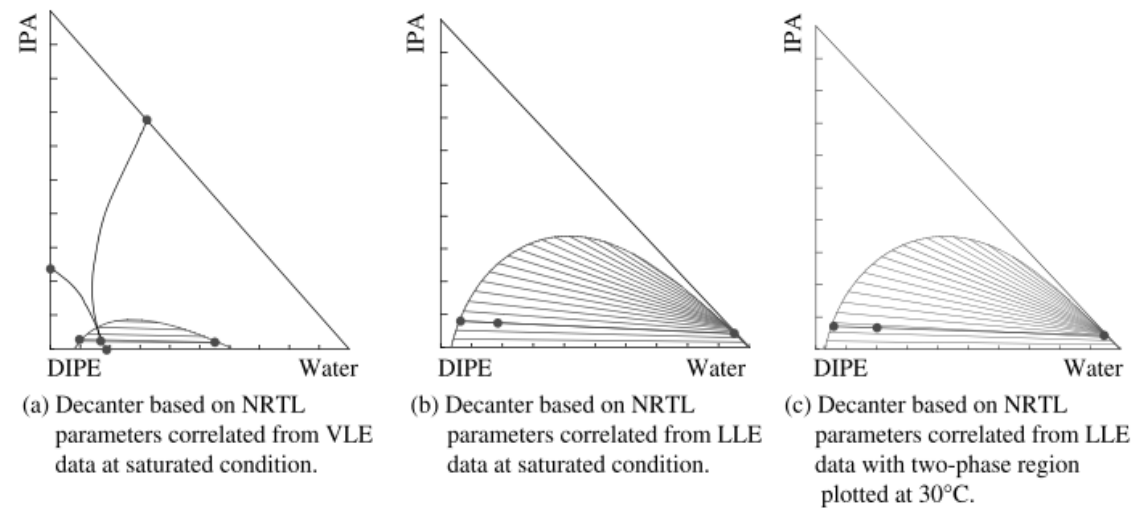


Figure 12.34 The two-phase region can be plotted from correlations fitted to VLE or LLE data, and can be plotted at saturated conditions or at a fixed temperature.

• Optimization

➤ 製程最適化

反應：提高轉化率、減少副產物

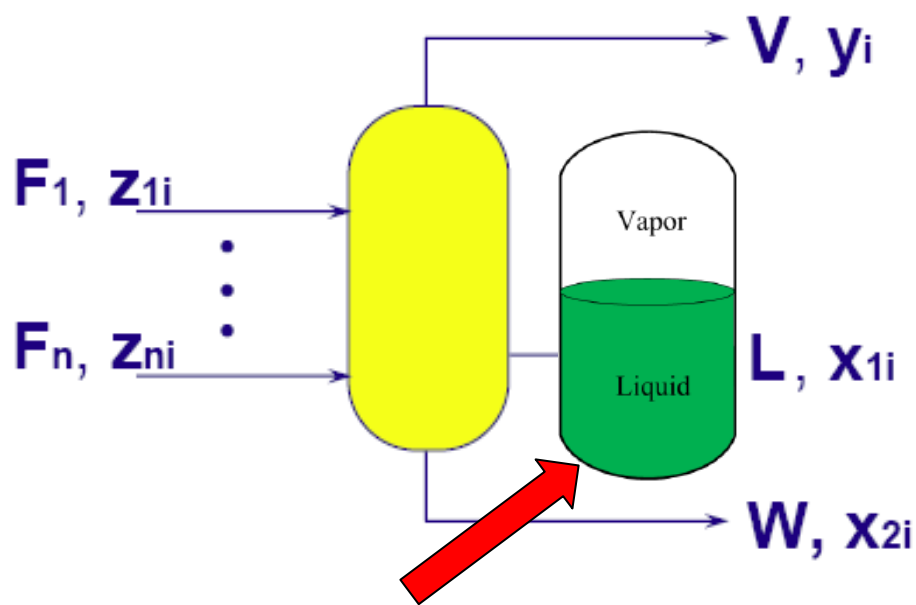
分離：新型分離技術、速率基準計算模式、熱整合
最佳控制策略

➤ 設備最適化

基本設計分析

設備Sizing最適化

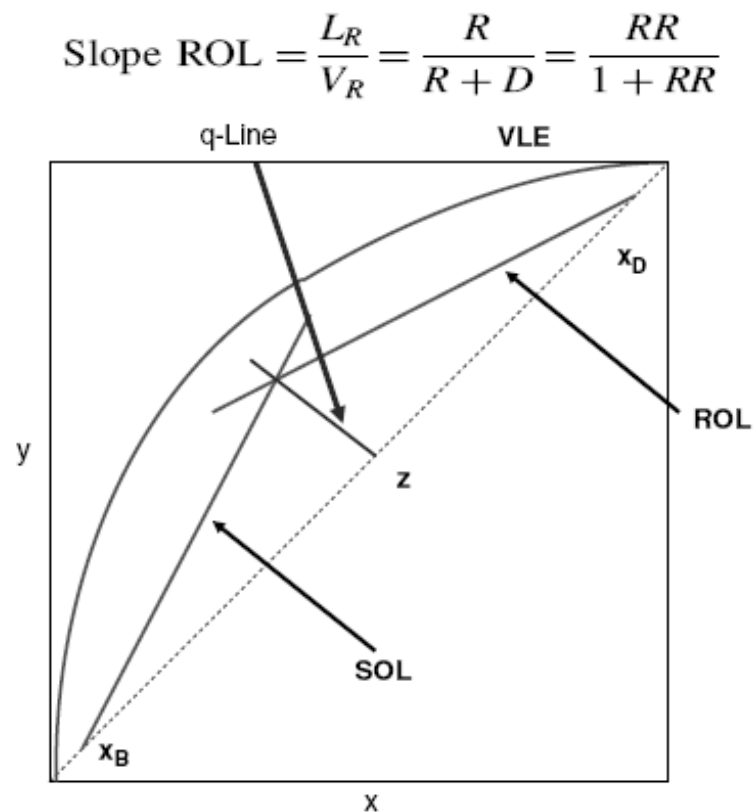
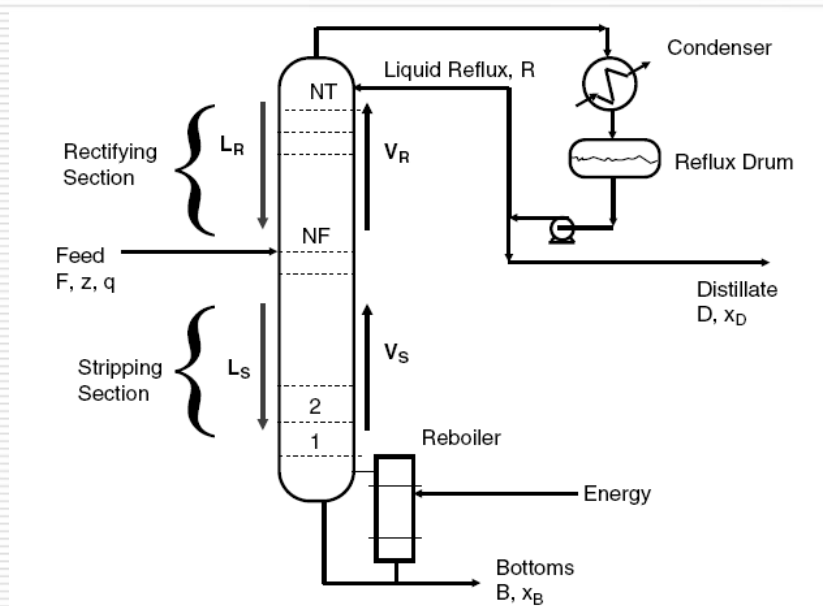
基礎分離單元：氣液分離



Given T, P

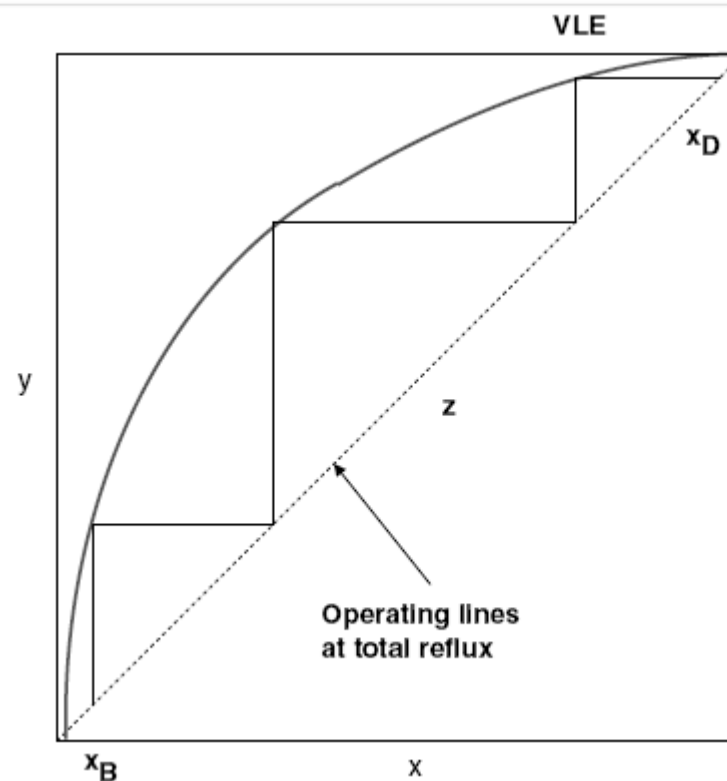
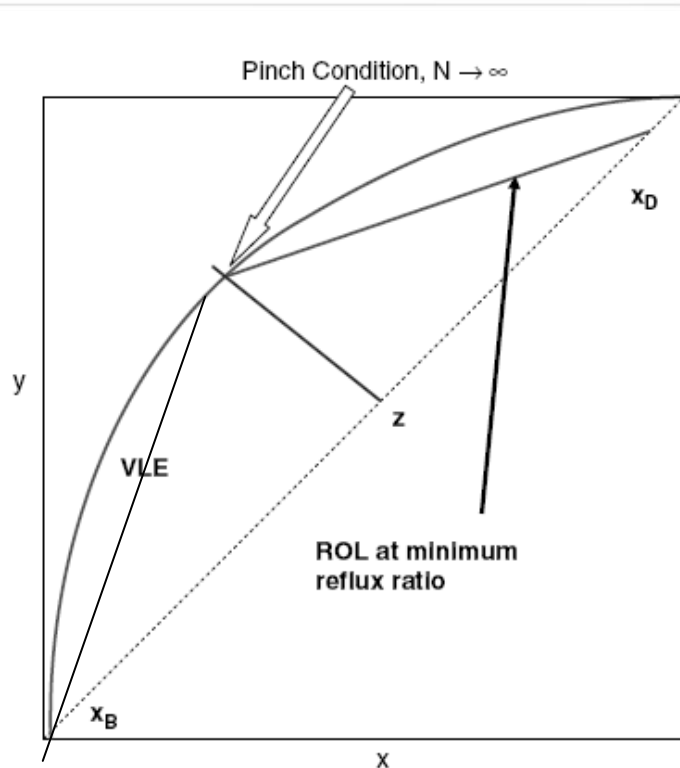
兩成分蒸餾塔理論板數計算

McCabe Thiele method assume Constant molar overflow



McCabe Thiele Method

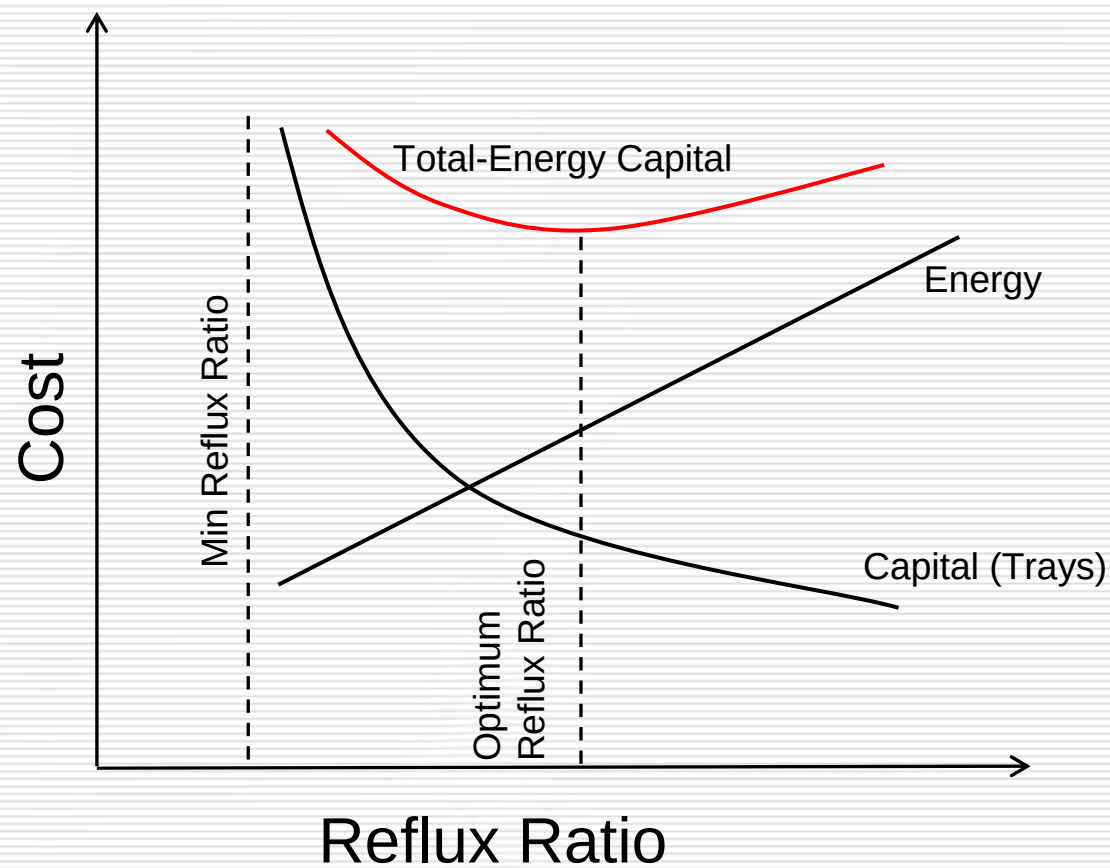
最多理論板數 VS 無限大回流比



雖然McCabe法只能用在兩成分系統，但可以了解蒸餾系統有下列原則：

- a. 在既定的規格下，蒸餾塔有最低迴流比 (RR_{MIN})
最少理論板數 (N_{MIN})
- b. 愈容易分離系統，需要理論板數愈少，回流比愈低
- c. 理論板數(設備成本)與回流比(操作成本)設計上有trade-off 關係

操作成本與設備成本關係



最少理論板數公式解

Fenske Equation:

$$N_{\text{MIN}} = \frac{\log \left[\left(\frac{x_{D,LK}}{x_{D,HK}} \right) \left(\frac{x_{B,HK}}{x_{B,LK}} \right) \right]}{\log(\alpha_{LK/HK})_{\text{avg}}}$$

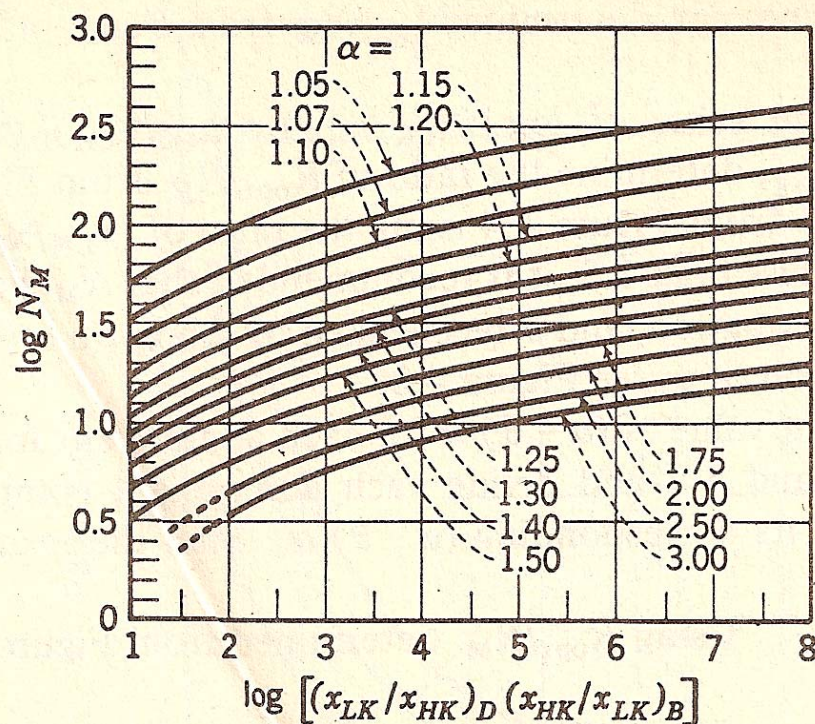


Figure 4.1.⁽¹⁾ Fenske equation for minimum plates expressed in graph form.

Short-cut Method

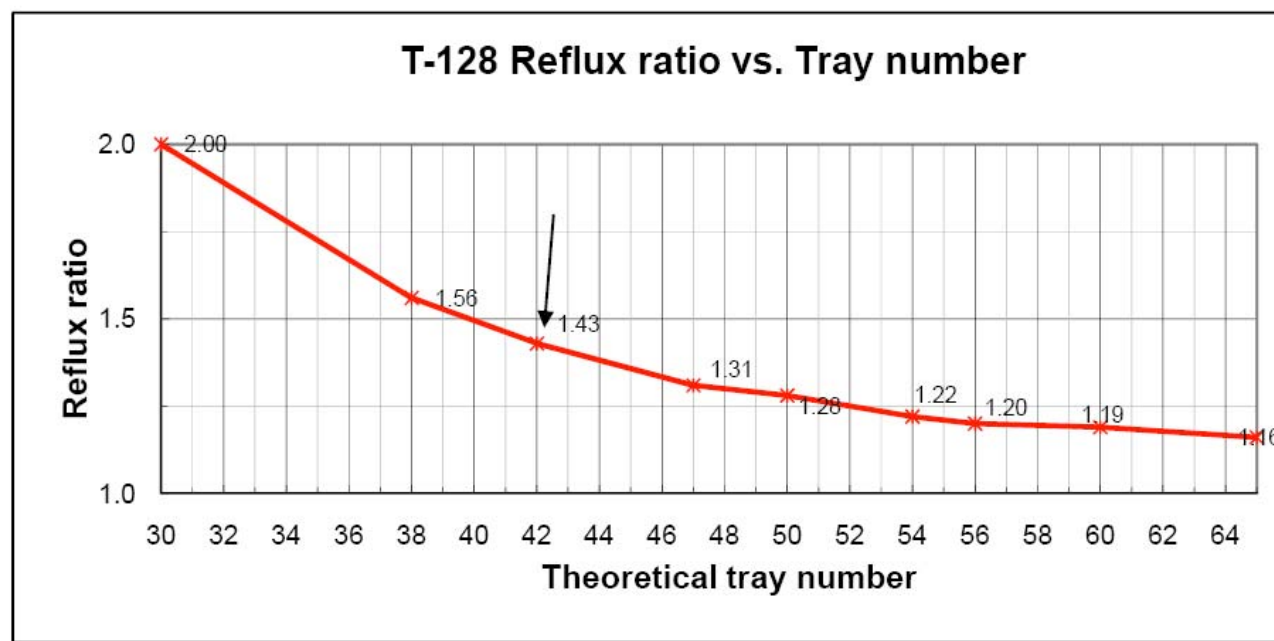
Aspen Plus內的DSTWU模組與Proll Shortcut模組可做Short-cut計算，下圖為Aspen Plus畫面

The screenshot displays the Aspen Plus interface. On the left, the Process Flow diagram shows a distillation column labeled B1. Below it, the Equipment palette is visible, with the DSTWU module highlighted by a red circle and a red arrow pointing to the column B1. The Results Summary window on the right shows the following data:

Results		
Minimum reflux ratio:	5.50557187	
Actual reflux ratio:	99999	
Minimum number of stages:	12.5061604	
Number of actual stages:	12.3948991	
Feed stage:	7.17190032	
Number of actual stages above feed:	6.17190032	
Reboiler heating required:	1.8709E+12	Btu/hr
Condenser cooling required:	1.871E+12	Btu/hr
Distillate temperature:	111.176474	F
Bottom temperature:	183.102183	F
Distillate to feed fraction:	0.399998	
HETP:		

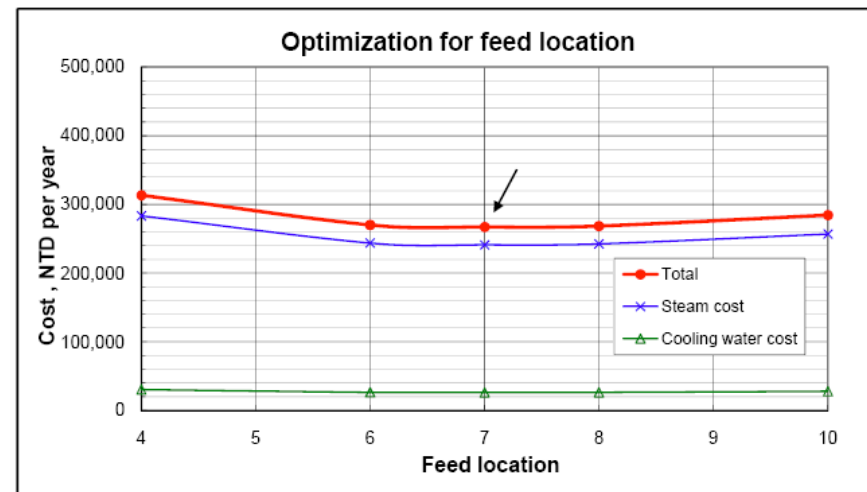
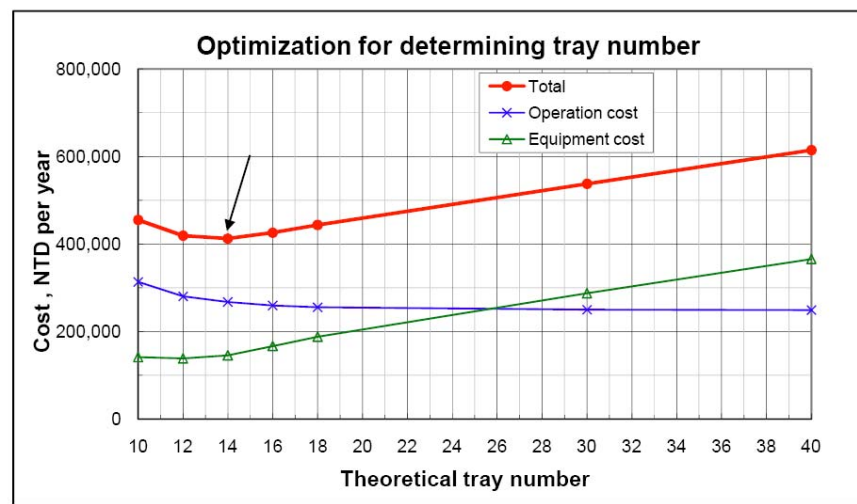
蒸餾塔經驗法設計

一般經驗法則，回流比設計在 R_{MIN} 的1.1 倍，理論板數設計在 N_{MIN} 的2倍



蒸餾塔最佳化設計 - Rigorous Method

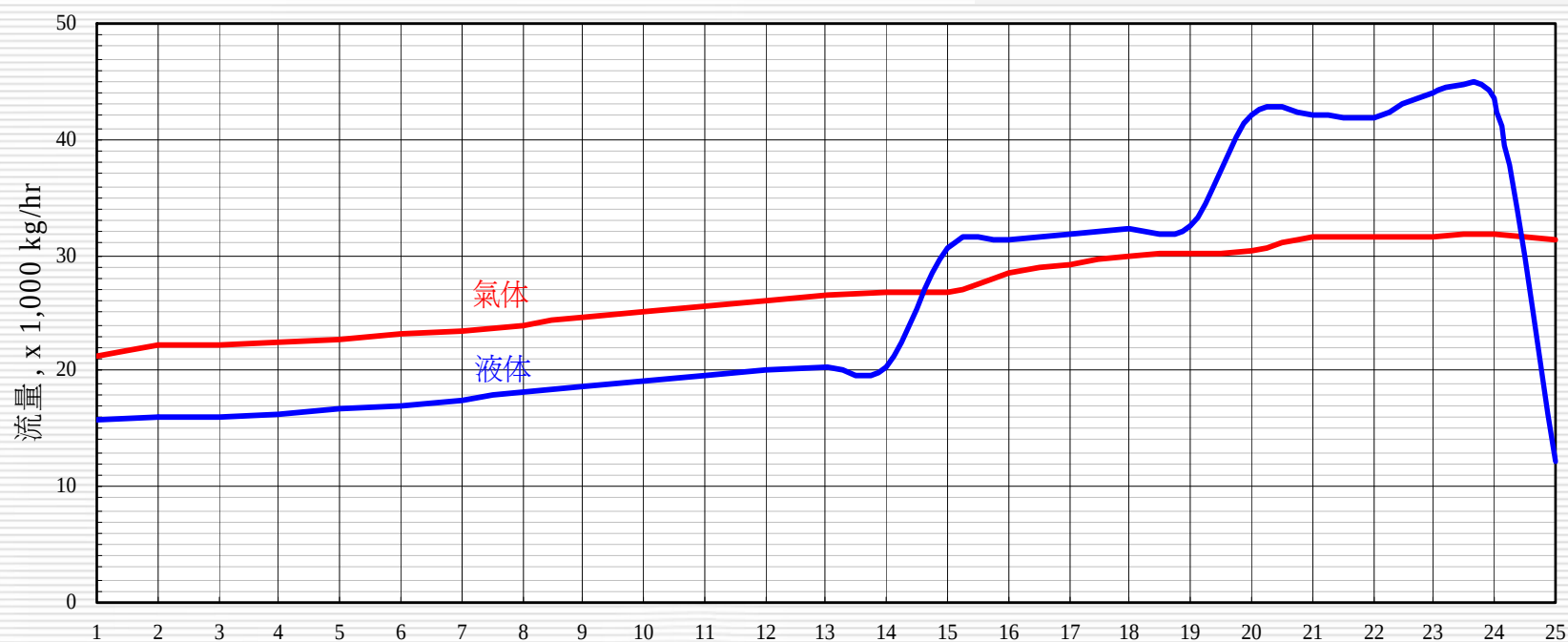
- 設定塔頂與塔底規格進行模擬
- 改變不同理論板、進料板位置。
- 畫出設備成本(塔體及塔內件價格)與操作成本(蒸氣與冷卻水用量價格)關係圖



蒸餾塔模擬結果可知

- a. 理論板數
- b. 進出口流量與組成規格
- c. 再沸器與冷凝器能耗
- d. 塔溫與塔壓力變化 (須搭配實際板數或填料實際高度與塔徑結果)
- e. 塔內水力數據
 - 氣液流量
 - 塔內組成變化
 - 塔內氣液相密度、黏度、分子量、液體表面張力

產生塔內各板氣體、液体流量作為塔徑 Sizing 之用



製程模擬範例：高沸物廢液回收

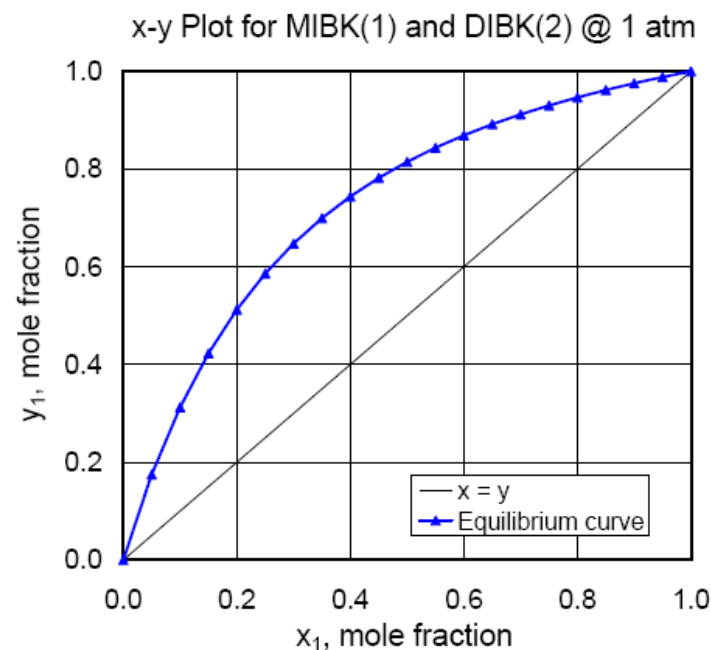
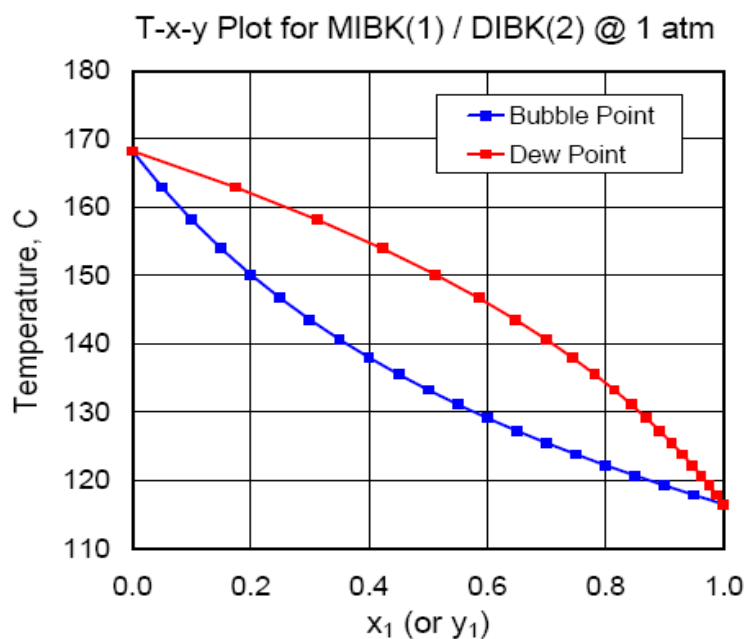
製程高沸物溶液內含MIBK、DIBK及Heavy三種成分，流量為1,000 kg/h，內容物成分有下列兩種情況：

	<u>Case 1</u>	<u>Case 2</u>
MIBK	52.25 wt%	42.75 wt%
DIBK	42.75 wt%	52.25 wt%
Heavy	5.00 wt%	5.00 wt%

以蒸餾分離方式，分別純化MIBK & DIBK兩項產品，產品純度均須為99 wt%，MIBK & DIBK 的產品回收率則分別為99%及90%以上

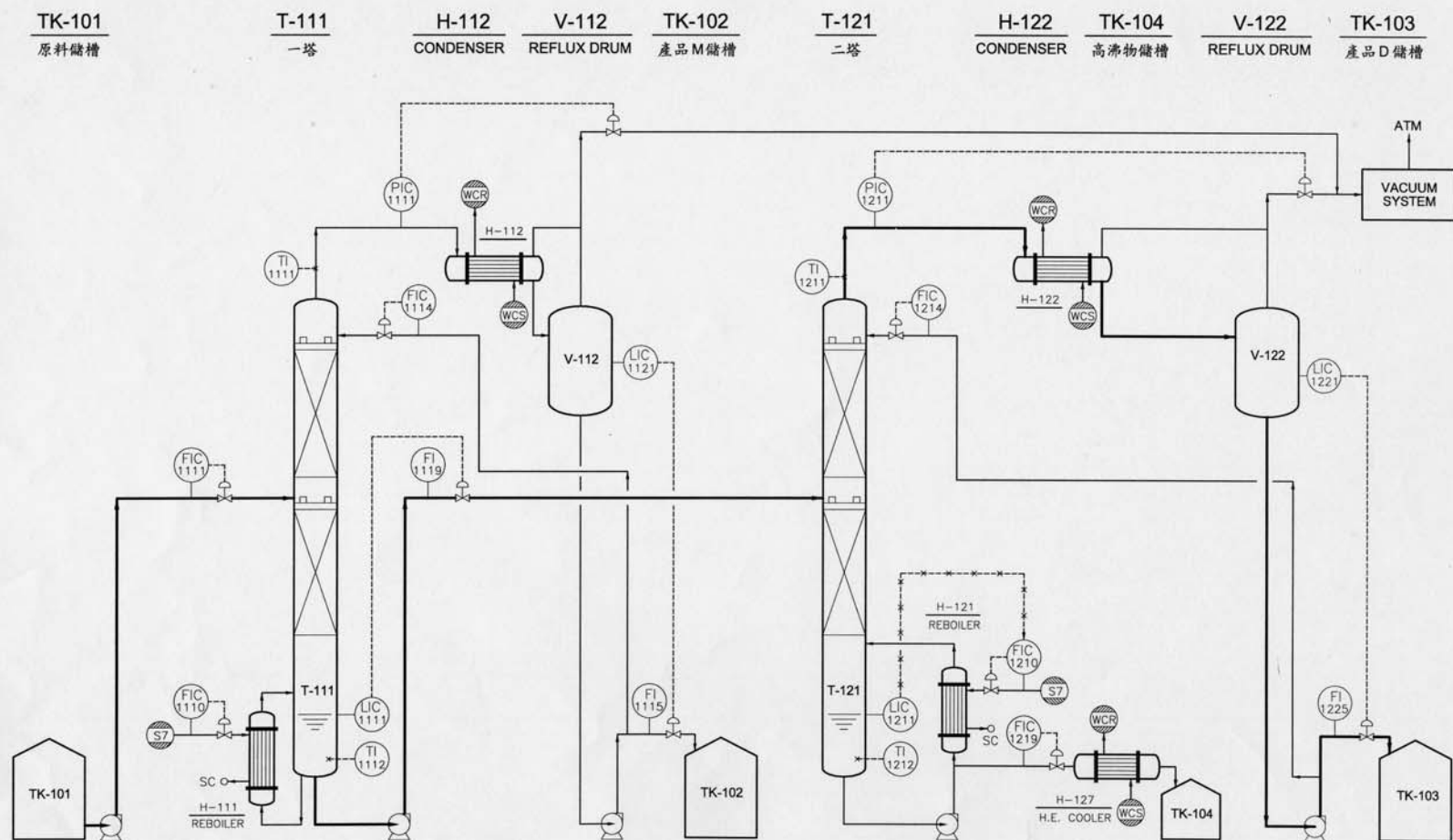
Step 1: 分析 DIBK & MIBK 沸點差異及氣-液相平衡圖

				MW	MP	NBP	Flash point	Auto lg. Temp.	LEL	UEL	Density @20°C	Azeotrope with Water		Water solubility
					°C	°C	°C	°C	vol%	vol%	kg/m ³	Temp.	mole%	g/100m
1.	MIBK	Methyl Isobutyl Ketone	C ₆ H ₁₂ O	100.16	-80.0	117.4	14	448	1.2 %	8.0 %	797.8	87.9	90.85%	1.90
2.	DIBK	Diisobutyl Ketone	C ₉ H ₁₈ O	142.24	-46.4	168.0	49	396	0.8 %	6.2 %	847.0			0.05
3.	Heavy	Unknown												



Step 2：初步畫出流程圖

PROCESS CONTROL FLOW DIAGRAM



Step 3：訂定模擬基準

a：進料條件

	<u>Case 1</u>	<u>Case 2</u>
MIBK	52.25 wt%	42.75 wt%
DIBK	42.75 wt%	52.25 wt%
Heavy	5.00 wt%	5.00 wt%
流量	1,000 kg/h	
溫度	25 °C	

b：蒸餾塔規格

(i) T1

塔頂	MIBK conc. = 99.0 wt%
塔底	MIBK recovery = 1%

Step 3：訂定模擬基準 (續)

(ii) T2

塔頂 DIBK conc. = 99.0 wt%

塔底 DIBK recovery = 10%

Step 4：選擇適合的模擬軟體，決定操作壓力，並依照 Step 2 & 3 資料進行模擬

Step 5：塔板數目及進料位置最適化

Step 6：Sizing 蒸餾塔及相關附屬設備 (以操作負荷較重的Case為 Sizing base)

Step 7：根據模擬結果擬訂控制策略(含溫控點決定)

Step 8：繪製 PCF & PID

高沸物廢液回收質量平衡 - Case 1

Material Balance for Case 1 -- MIBK 55%

Stream Name	S1	S2	S2r	S3	S4	S4r	S5
Stream Description	一塔 進料	一塔 餾出物	一塔 Reflux	一塔 塔頂氣 塔底物	二塔 餾出物	二塔 Reflux 塔頂氣	二塔 塔底物
Phase	液相	液相	液相	氣相	液相	液相	氣相
Total Mass Rate kg/h	1,000	524	321	845	476	384	1,707
Temperature °C	25	72.5		144.1	97.4		144.8
Pressure torr				250			120
kg/cm ² A				0.34	0.52		0.163
Total Weight Comp. Rate kg/h							
MIBK	522.50	518.66		3.84	3.84		0.00
DIBK	427.50	5.24		422.26	380.03		42.23
Heavy	50.00	0.01		49.99	0.04		49.95
Total Weight Comp. Percents wt. %							
MIBK	52.25%	99.00%		0.81%	1.00%		0.00%
DIBK	42.75%	1.00%		88.69%	98.99%		45.81%
Heavy	5.00%	0.00%		10.50%	0.01%		54.19%
Density	799.9	752.5		0.845	697.2		739.1
Viscosity	0.68	0.33		0.008	0.23		0.34
Pipeline Size	1"	3/4"	3/4"	6"	3/4"	3/4"	1"
							8"
							3/4"

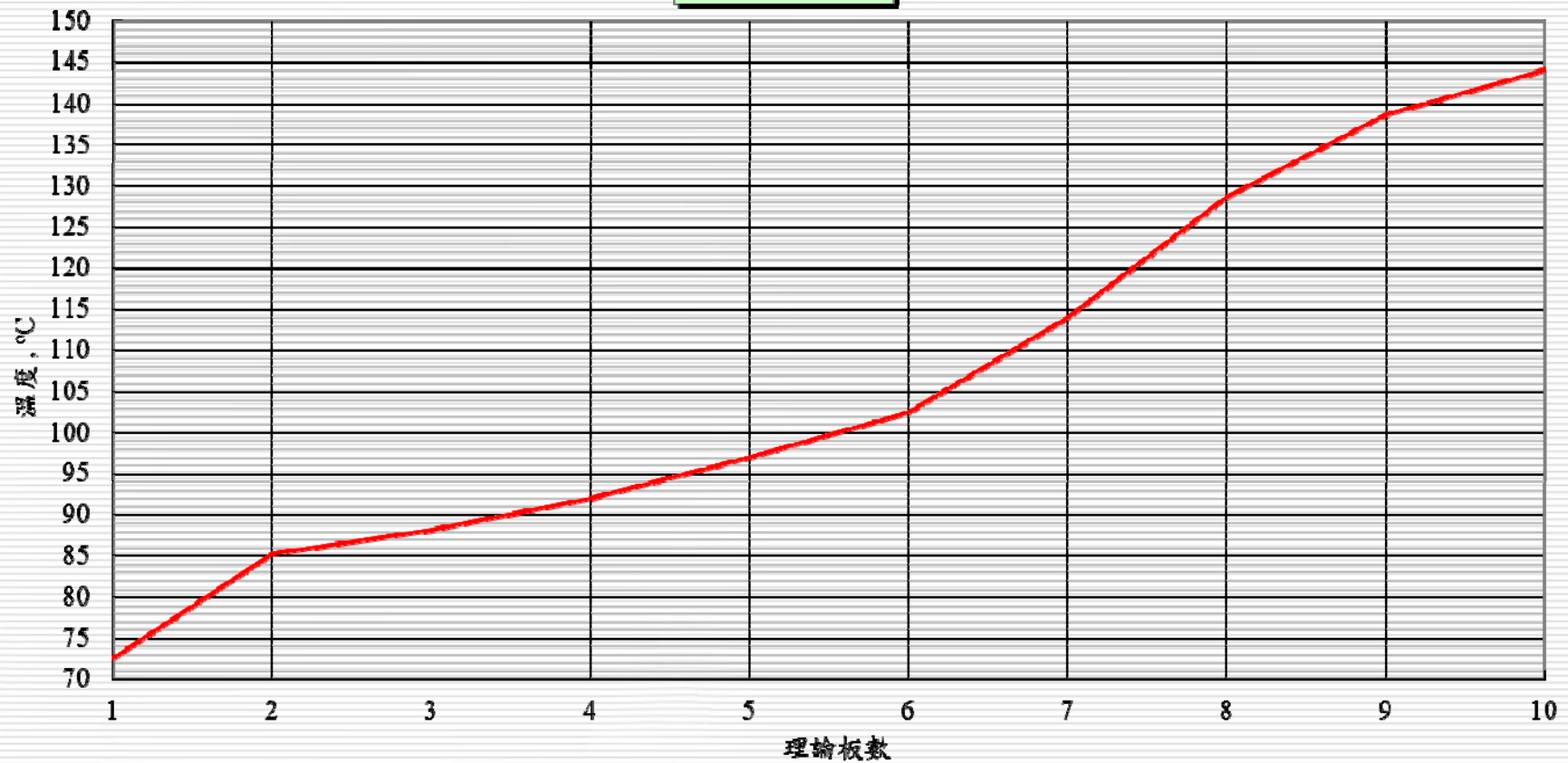
高沸物廢液回收質量平衡 - Case 2

Material Balance for Case 2 -- DIBK 55%

Stream Name	S1	S2	S2r	S3	S4	S4r	S5
Stream Description	一塔 進料	一塔 餾出物	一塔 Reflux	一塔 塔頂氣 塔底物	二塔 餾出物	二塔 Reflux	二塔 塔頂氣 塔底物
Phase	液相	液相	液相	氣相	液相	液相	氣相
Total Mass Rate	kg/h	427	326	753	471	1,856	2,327
Temperature	°C	25	72.5	143.7	97.4		143.6
Pressure	torr			250			120
	kg/cm ² A			0.34	0.52		0.163
Total Weight Comp. Rate	kg/h						
MIBK	427.50	422.75		4.75	4.74		0.00
DIBK	522.50	4.27		518.23	466.41		51.82
Heavy	50.00	0.01		49.99	0.05		49.95
Total Weight Comp. Percents	wt. %						
MIBK	42.75%	99.00%		0.83%	1.01%		0.00%
DIBK	52.25%	1.00%		90.45%	98.98%		50.92%
Heavy	5.00%	0.00%		8.73%	0.01%		49.08%
Density		799.9	752.5	→ 1.23	697.2	739.1	0.805
Viscosity			0.326	0.008	0.23	0.338	0.007

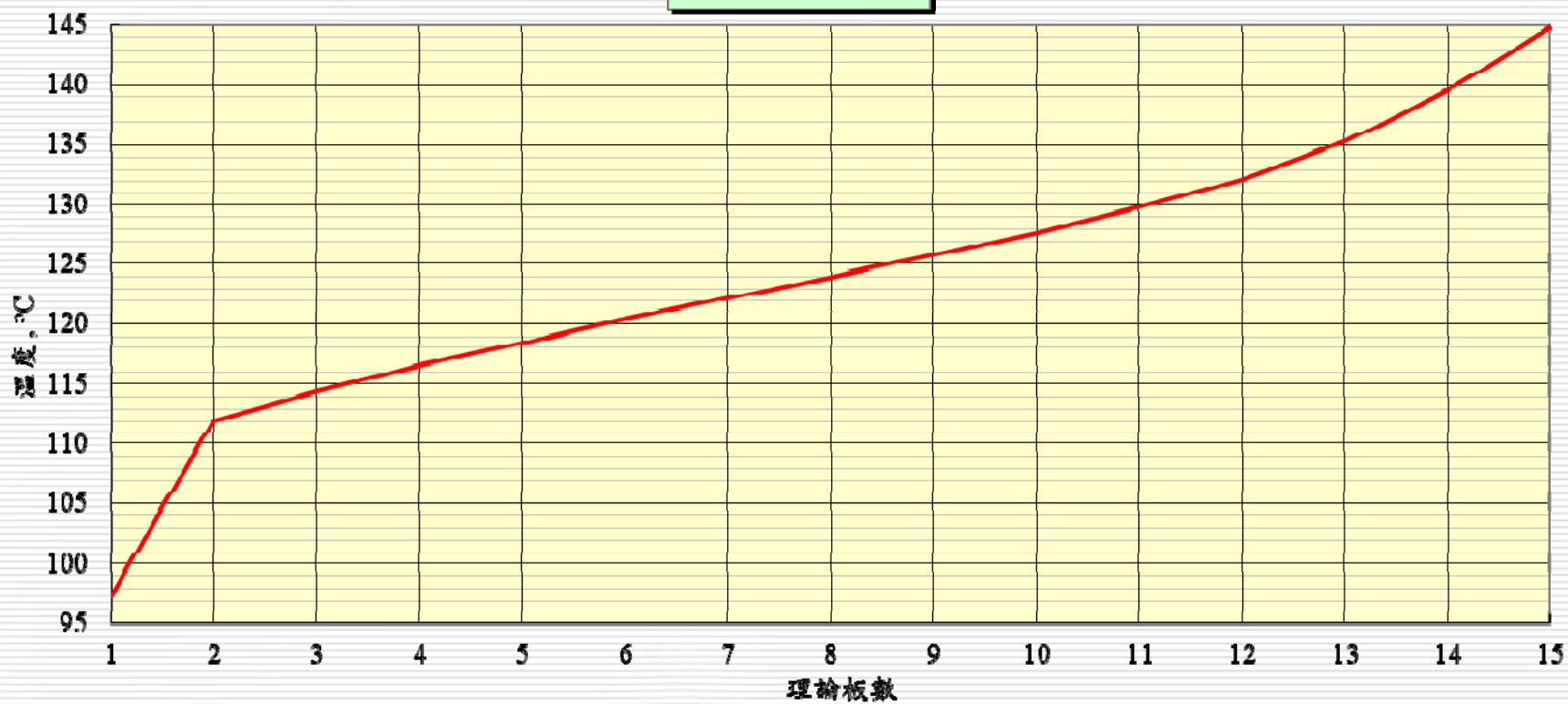
高沸物廢液回收 Case 1

一塔溫度分佈

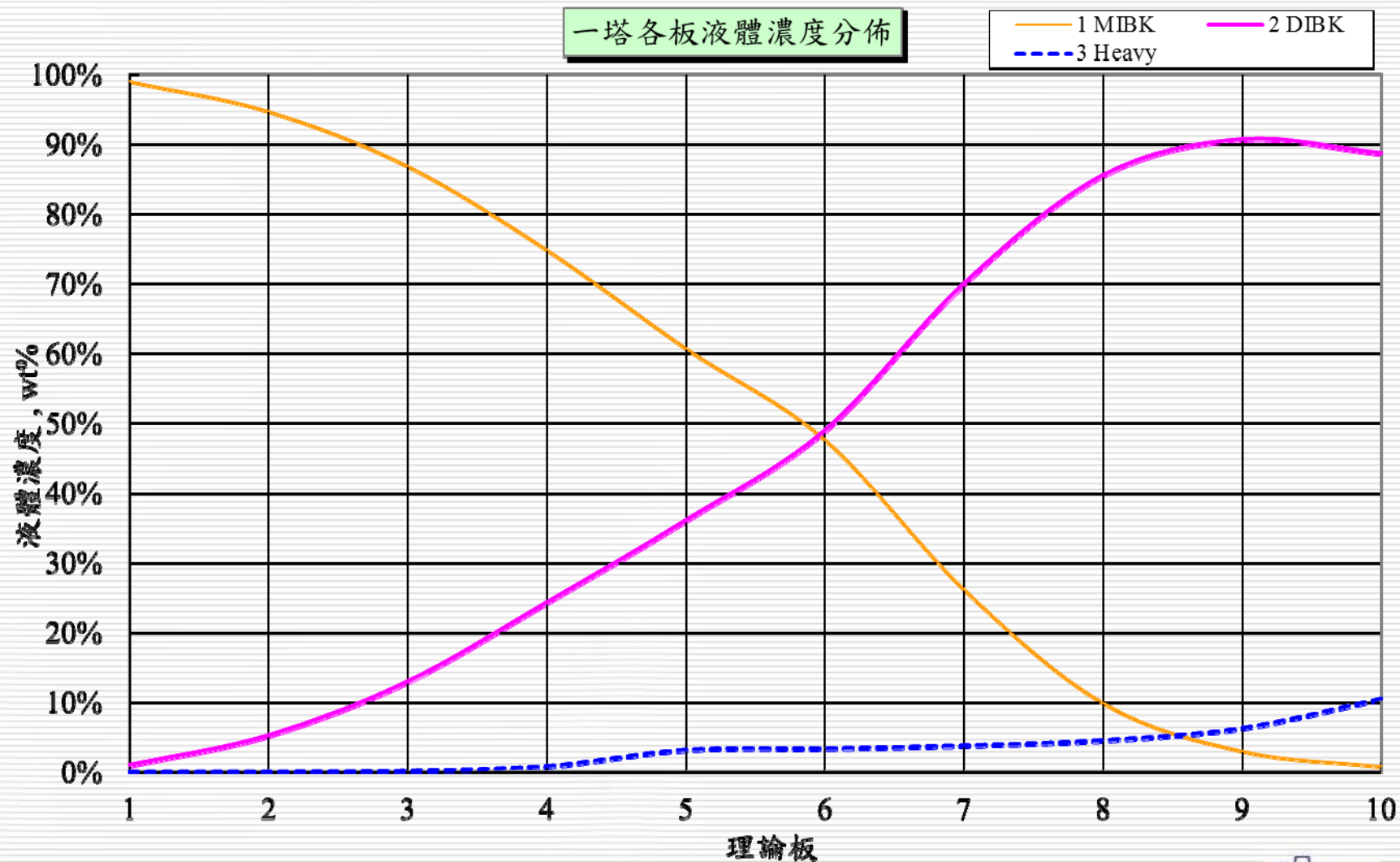


高沸物廢液回收 Case 1

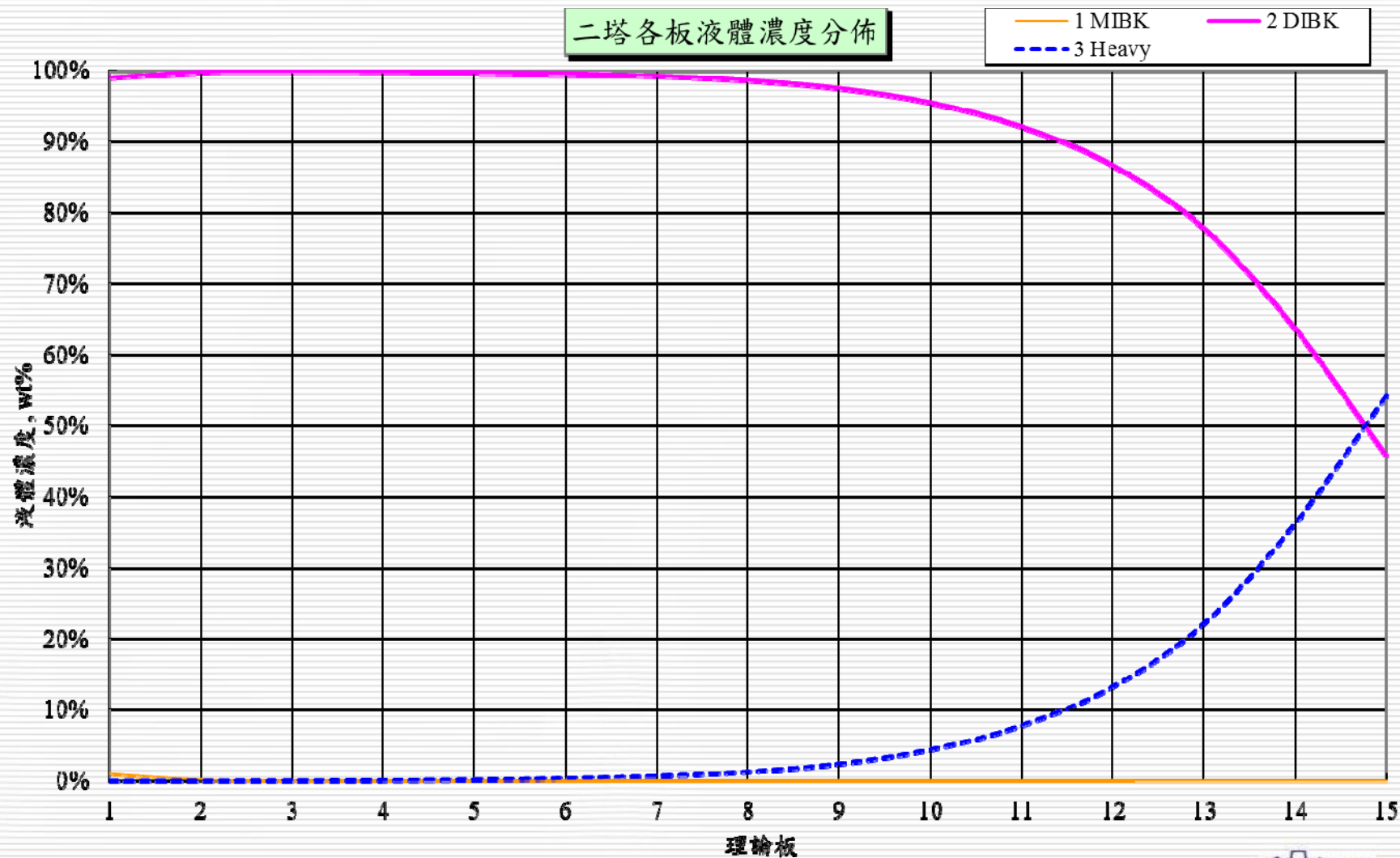
二塔溫度分佈圖



高沸物廢液回收 Case 1



高沸物廢液回收 Case 1



		<u>Case1</u>		<u>Case2</u>	
MIBK	wt%	55 wt%		45 wt%	
DIBK	wt%	45 wt%		55 wt%	
		T-1	T-2	T-1	T-2
Tower Pressure		250 torr	120 torr	250 torr	120 torr
Theo. Plates		10	15	10	15
Feed Location		5th	12th	5th	12th
Feed Rate	kg/h	1,000	476	1,000	573
Overhead		MIBK	DIBK	MIBK	DIBK
		99%	99%	99%	99%
	kg/h	524	384	427	471
Bottoms		DIBK(89wt%)	DIBK(46wt%)	DIBK(90wt%)	DIBK(51wt%)
		+HE(10.5%)	+HE(54%)	+HE(9%)	+HE(49%)
	kg/h	476	92	476	102
Reflux	kg/h	321	1,707	326	1,856
Reflux Ratio		0.6	4.4	0.8	3.9
Condenser Duty	kcal/h	83,100	172,900	74,100	192,400
Cooling Water	m ³ /h	8.3	17.3	7.4	19.2
Reboiler Duty	kcal/h	126,500	162,700	121,400	179,900
Steam	kg/h	253	325	243	360
BTM Vapor	kg/h	1,660	1,947	1,596	2,187
Vaporization Ratio		3.5	21.2	3.4	21.4
Temp.--Top	°C	85	112	85	112
--Bottom	°C	144	145	144	144

		<u>Case1</u>		<u>Case2</u>	
MIBK	wt%	55 wt%		45 wt%	
DIBK	wt%	45 wt%		55 wt%	
Steam cost	NTD/h	126.5	162.5	121.5	180
Cooling Water Cost	NTD/h	5.8	12.1	5.2	13.5
電力成本	NTD/h				
操作總成本	NTD/h	132.3	174.6	126.7	193.5
		307		320	
Tower I.D.	mm	600	800		
Packing Height	mm	5,000	10,000		
Packing Type		IMTP #25	IMTP #25		
Packing Material		SS304	SS304		

蒸餾塔塔徑與塔內件計算

- 將模擬得到的塔內部水力數據輸入塔內件廠商設計軟體，進行塔徑計算與塔內件形式的選擇
- 下列位置做分段計算
 - a. 塔進料位置
 - b. 塔頂塔底出料
 - c. 使用不同形式內件Packing或Tray分界
 - d. 塔側邊氣相或液相取出位置(Chimney tray)
 - e. Recycle 或 Pumping around回塔位置
 - f. 氣液量負荷最大位置

蒸餾塔塔內水力數據 – Vapor phase

----- Vapor to Tray -----						
Tray	Temp deg C	Pressure KG/CM2	Molec Wt	Rate K*KG/HR	Density at Cond KG/M3	Viscosity at Cond CP
1	100.06	0.2240000	117.5539	1.9544204	0.841737	0.0076859
2	109.88	0.2393846	126.8317	2.1392858	0.946963	0.0075239
3	113.49	0.2547692	129.3747	2.2049863	1.019348	0.0074984
4	115.51	0.2701538	129.8736	2.2191758	1.080197	0.0075186
5	117.15	0.2855385	129.9423	2.2190922	1.138246	0.0075476
6	118.69	0.3009231	129.9258	2.2157421	1.195497	0.0075774
7	120.17	0.3163077	129.8914	2.2111984	1.252336	0.0076069
8	121.64	0.3316923	129.8509	2.2054794	1.308782	0.0076360
9	123.12	0.3470769	129.8045	2.1979808	1.364728	0.0076652
10	124.65	0.3624615	129.7491	2.1876947	1.419980	0.0076951
11	126.29	0.3778462	129.6799	2.1732510	1.474245	0.0077268
12	134.63	0.3932308	138.8012	2.8083736	1.611713	0.0075326
13	139.12	0.4086154	141.4593	2.8876626	1.690032	0.0075037
14	143.54	0.4240000	142.0766	2.8257270	1.743998	0.0075358
15	Reboiler					

蒸餾塔塔內水力數據 – Liquid phase

Tray	Temp deg C	Molec Wt	Liquid from tray			
			Rate K*KG/HR	Density at Cond KG/M3	Viscosity at Cond CP	Surface Tension DYNE/CM
1	69.54	117.5539	1.9544204	759.66583	0.3826704	19.418010
2	100.06	133.8660	1.2842424	734.97946	0.3110546	17.065127
3	109.88	138.1754	1.3499429	726.82688	0.2926553	16.336972
4	113.49	139.0048	1.3641323	723.65332	0.2850212	16.051952
5	115.51	139.1334	1.3640488	721.79938	0.2802450	15.883773
6	117.15	139.1269	1.3606987	720.26850	0.2762730	15.745207
7	118.69	139.0956	1.3561550	718.85208	0.2726947	15.618423
8	120.17	139.0615	1.3504360	717.50719	0.2694421	15.500293
9	121.64	139.0294	1.3429374	716.22549	0.2665335	15.391096
10	123.12	139.0012	1.3326513	715.00777	0.2640311	15.292449
11	124.65	138.9788	1.3182076	713.85889	0.2620383	15.206988
12	126.29	138.9646	2.9533310	712.78555	0.2606925	15.138042
13	134.63	141.4950	3.0326200	706.46778	0.2526228	14.651624
14	139.12	142.0831	2.9706843	704.08244	0.2524216	14.530793
15	143.54	142.2098	0.1449574	702.76288	0.2560455	14.537027

蒸餾塔徑與塔內件計算 (by SULCOL software)

SULCOL 2.0.8
File Edit Project Window Help
UnitType: SI

Loadings - D:\Work\ECH PJ\Tower\Calculation\T-601\T-601.sulcol

Sec. 1

Packing

Diam [mm] Fluid Data -> Packing Design -> Packing-Type Height [m] NTS req. HETP [mm]

1050 Fluid_1 Packing1 IR40 9.000 6.0 1500

Flows

	G [kg/h]	L [kg/h]	ρ_G [kg/m ³]	ρ_L [kg/m ³]	σ [mN/m]	η_L [cP]	η_G [cP]	Cap [%]	F-F [Pa ^{0.5}]	Liq. load [m ³ /m ² h]	$\Delta p/\Delta z$ [mbar/m]	hl [%]	dp [mbar]
Top	6248.0	5525.0	1.836	1081.20	30.00	0.480	0.0110	42.6	1.48	5.90	1.23		
Btm	6873.0	6150.0	1.976	1076.10	26.40	0.459	0.0110	45.9	1.57	6.60	1.40	1.9	11.82

Text System factor 1.00 Geom. Details

Sec. 2

Tray

Diam [mm] Fluid Data -> Tray Design -> Tray Type # Pas. Spac. [mm] # Trays Height [m] NTS required Effic. [%]

1450 Fluid_2 Tray1 SVG 1 450 25 11.250 15 60

Flows

	G [kg/h]	L [kg/h]	ρ_G [kg/m ³]	ρ_L [kg/m ³]	σ [mN/m]	η_L [cP]	η_G [cP]	Jet Fl. [%]	Weir load [m ³ /mh]	Δp /Tray dry [mbar]	Δp DC vel [mmH ₂ O]	DC bkp [%]
#1_Min	10342.0	22019.0	2.178	1081.00	26.40	0.467	0.0110	24	12.22	3.96	8.12	0.025
#25_Min	11741.0	23418.0	2.672	1068.00	24.20	0.433	0.0112	25	13.16	4.01	8.53	0.027

Text System factor 1.00 Geom. Details

蒸餾塔控制策略

- 蒸餾塔基本控制：
 - a. 質量平衡控制
 - b. 組成(濃度)控制

質量平衡控制

➤ Rule 1

蒸餾塔出口不可全都流量(FC)控制，至少設定一股為Free或用液位(LC)控制

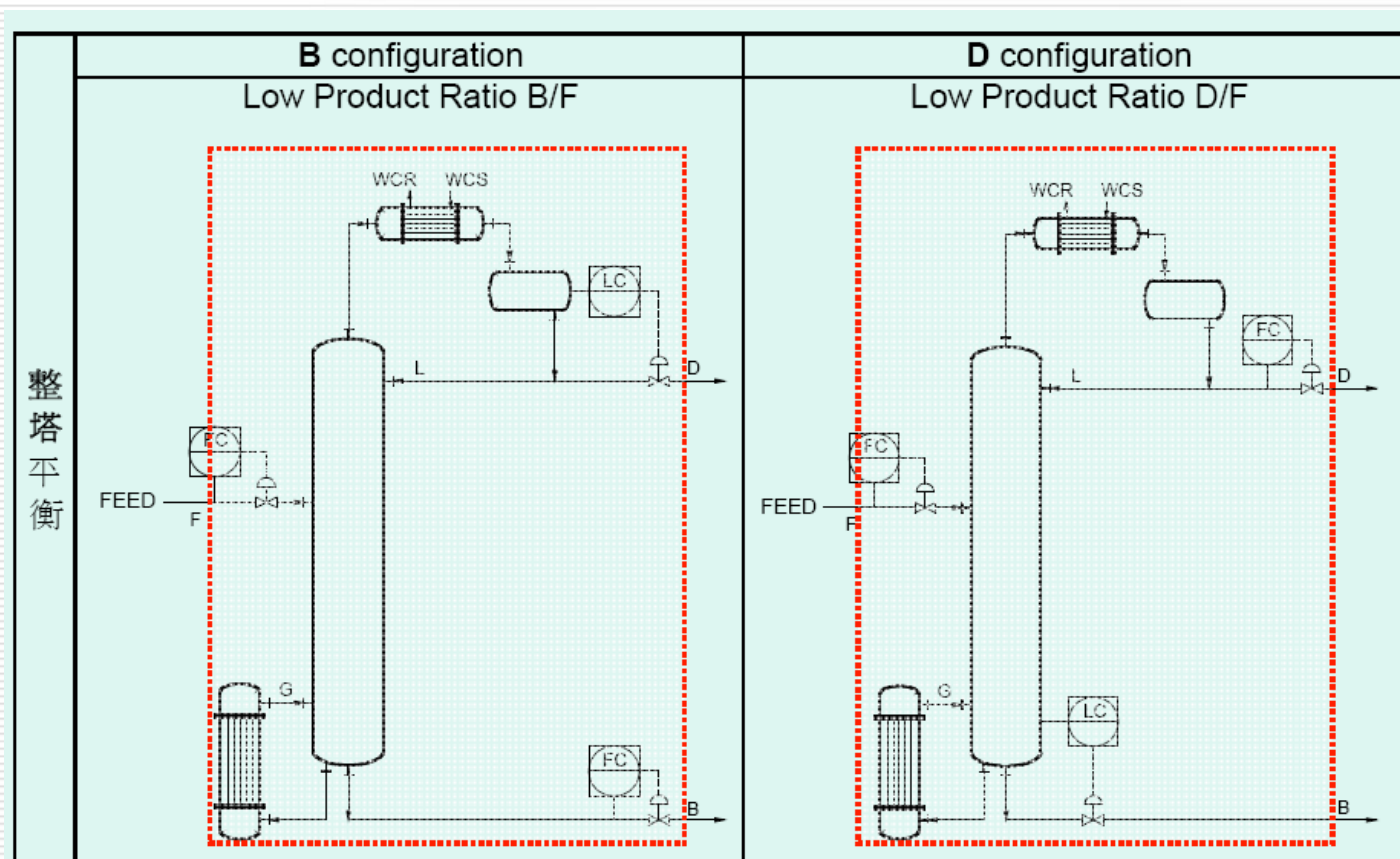
➤ Rule 2

對於流量較小採用FC，而流量較大採用LC

➤ Rule 1為質量平衡考量必須遵守，當1, 2項條件無法同時滿足時，須以第1項為優先

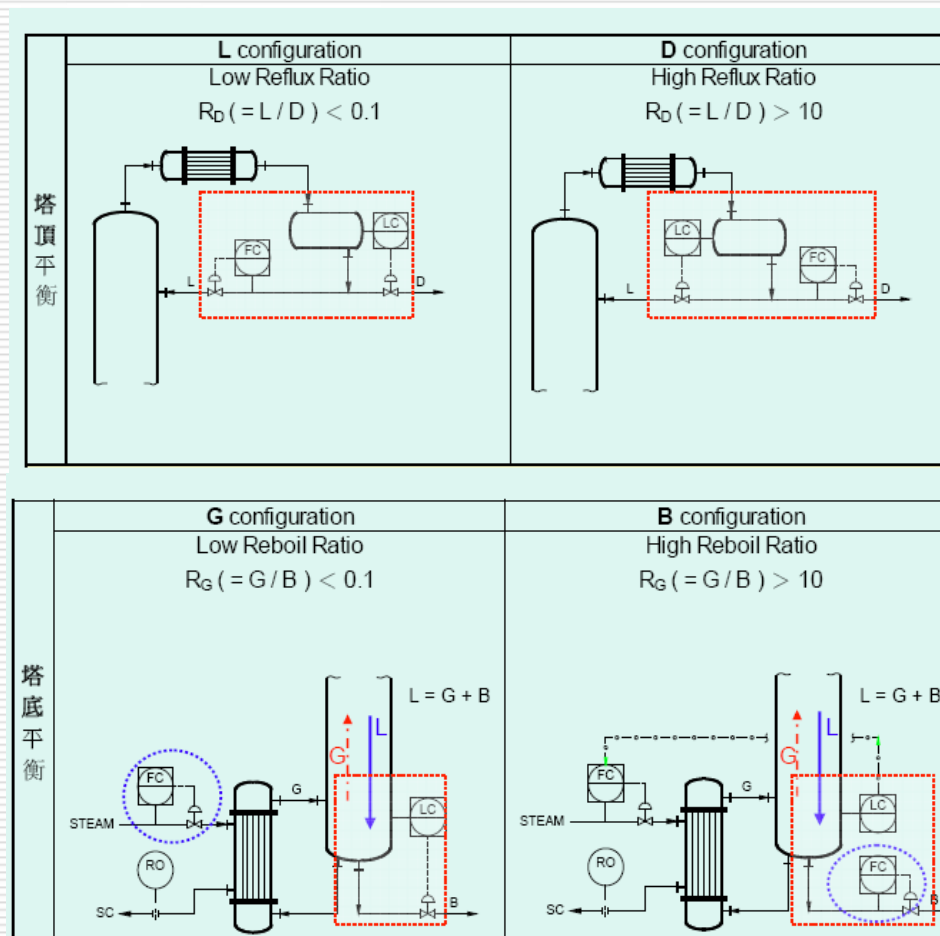
質量平衡控制

➤ 整塔平衡控制架構： $F=D+B$



質量平衡控制

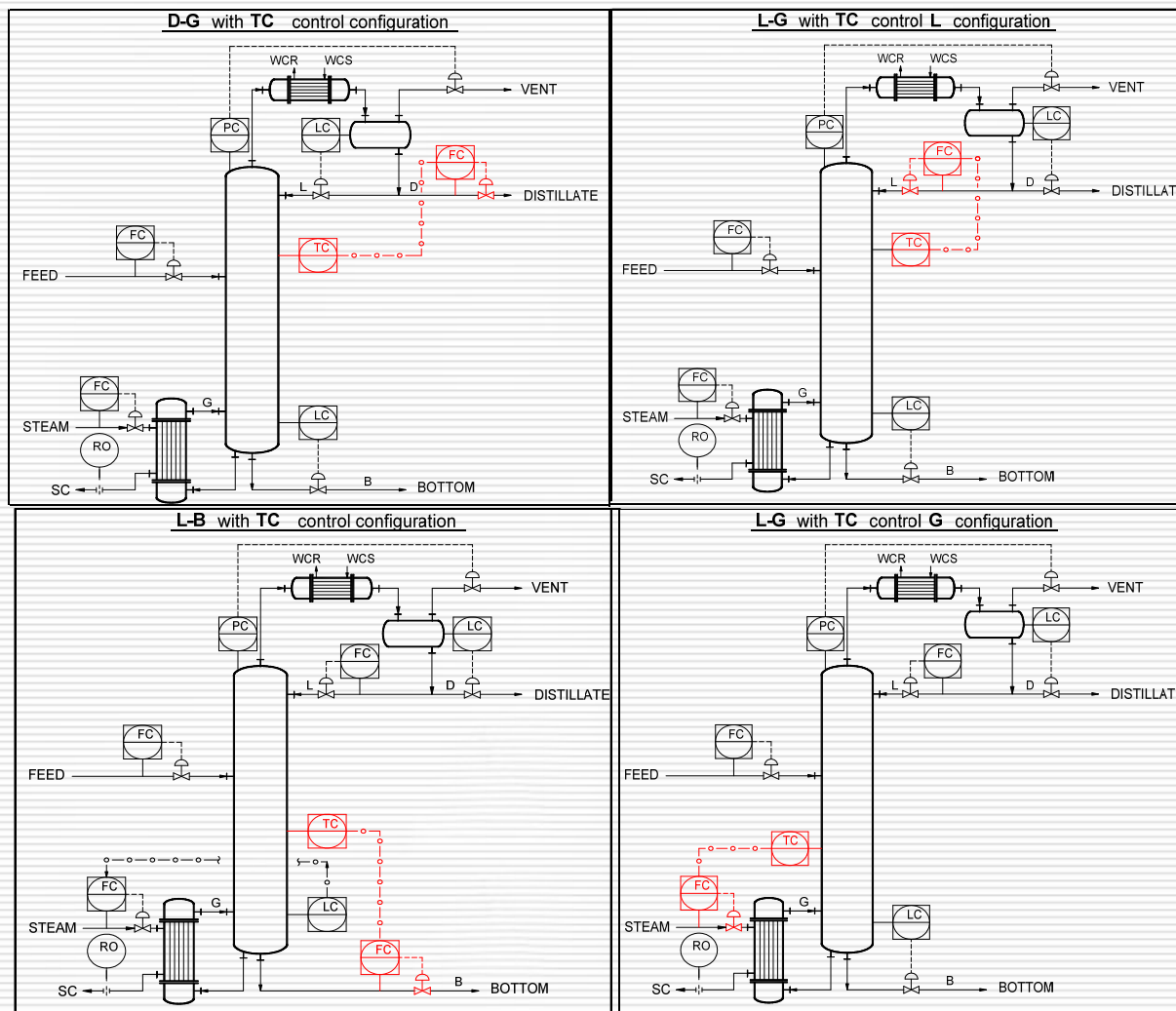
- 塔頂與塔底控制：由L、D、G、B值的大小來做判斷



組成(濃度)控制

- 使用線上組成分析器較昂貴，且準確性與可靠性較差，除非萬不得已才使用
- 溫度與組成有對應關係，溫度控制是最便宜且有效的組成控制方法
- 溫度控制的位置，以及採用那個操作變數來控制，須利用模擬進行溫度靈敏點分析決定

蒸餾塔常用溫度控制架構



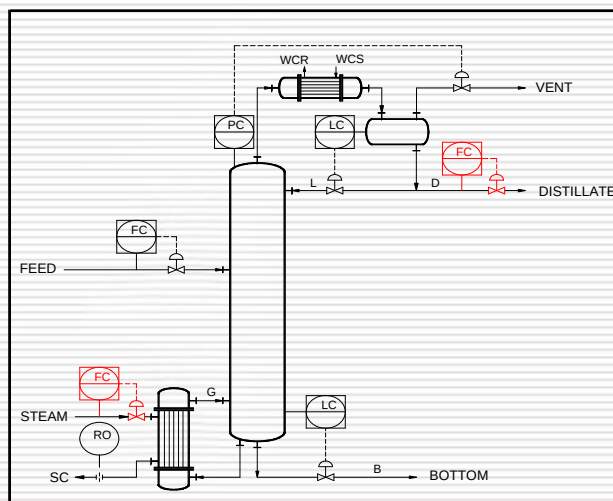
蒸餾塔控制策略擬定範例

Tower No.			281	341	361	371	421	441	551	561	581
Feed	F	kg-mol/h	1,288.2	20.9	1,071.0	984.0	211.8	550.9	54.6	54.9	95.3
Feed 2, recycle	F2	"	82.0	3.2						1.8	7.8
Reflux	L	"	838.8	42.7	297.4	487.1	685.2	118.3	0.0	11.3	3.9
Distillate	D	"	357.5	21.2	87.0	974.1	5.6	19.0	3.7	4.4	2.5
Bottom Liquid	B	"	1,012.8	2.9	984.0	9.8	206.1	531.7	50.9	50.8	100.6
Bottom Vapor	G	"	1,017.1	59.9	642.7	1,419.3	730.5	400.7	10.6	21.6	11.4
Partition of the feed	D / B		0.35	7.32	0.09	99.08	0.03	0.04	0.07	0.09	0.02
	判定			可 B	必 D	必 B	必 D	必 D	必 D	必 D	必 D
Partition of the overhead liquid	$R_L = L / D$		2.35	2.02	3.42	0.50	122.11	6.22	0.00	2.55	1.58
	判定						必 D	可 D	必 L		
Partition of the bottom liquid	$R_G = G / B$		1.00	20.71	0.65	144.37	3.54	0.75	0.21	0.42	0.11
	判定			必 B		必 B					可 G
Factor = 4.0	可能之策略		D-G	L-B	D-G	L-B	D-G	D-G	L-G	D-G	D-G
Control Configuration	Eastman(K/H)		D-G	L-G	D-G	L-B	D-G	D-G	L-G	D-G	L-G
D 表示: D 控制 (FC), L free (LC)			● D-G	○ D-G	● D-G	○ D-G	● D-G	● D-G	○ D-G	● D-G	● D-G
L 表示: L 控制 (FC), D free (LC)											
B 表示: B 控制 (FC), G free (LC)			○ L-B	● L-B	○ L-B	● L-B	○ L-B	○ L-B	○ L-B	○ L-B	○ L-B
G 表示: G 控制 (FC), B free (LC)			○ L-G	○ L-G	○ L-G	○ L-G	○ L-G	○ L-G	● L-G	○ L-G	○ L-G

溫度控制點敏感度分析

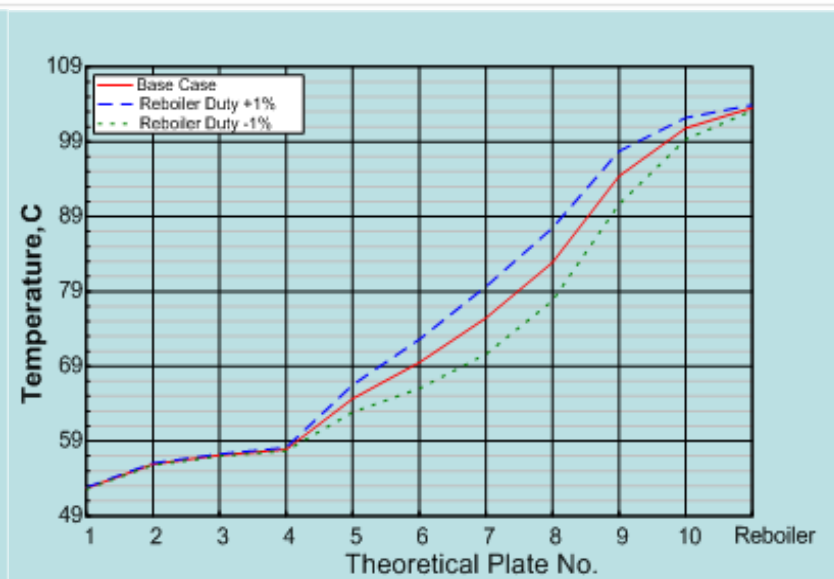
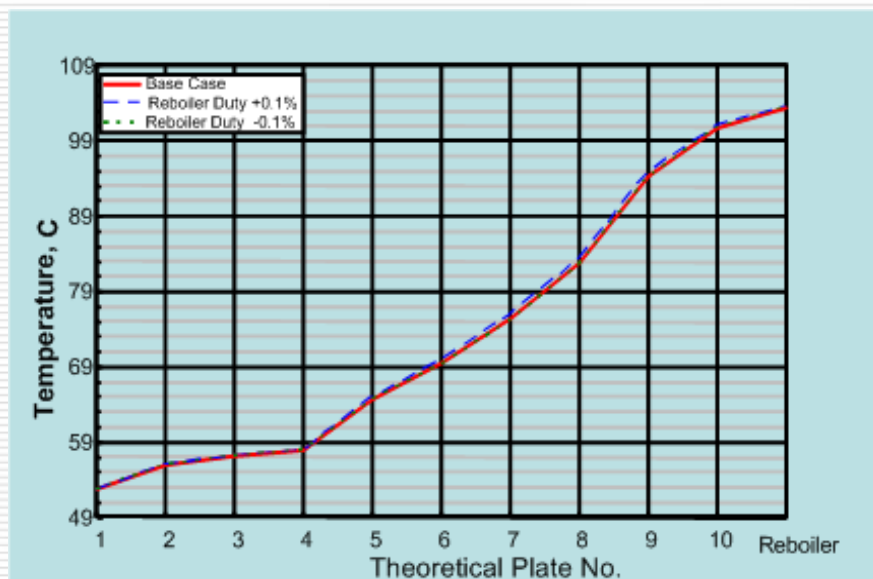
- 選定蒸餾塔控制架構(D-G、L-G or L-B)
 - 固定一個操作變數，對另一個操作變數正負方向改變量，進行穩態模擬
 - 記錄塔內溫度，選擇對塔板溫度變化大者
- 例 D-G control:

- ➔ 固定Reboiler Duty 改變Distillate Rates
- ➔ 固定Distillate Rates 改變 Reboiler Duty



觀察塔內溫度分佈

- 建議由小幅度改變逐步增加，但有時改變量太小或太大，不易判別溫度最靈敏的位置



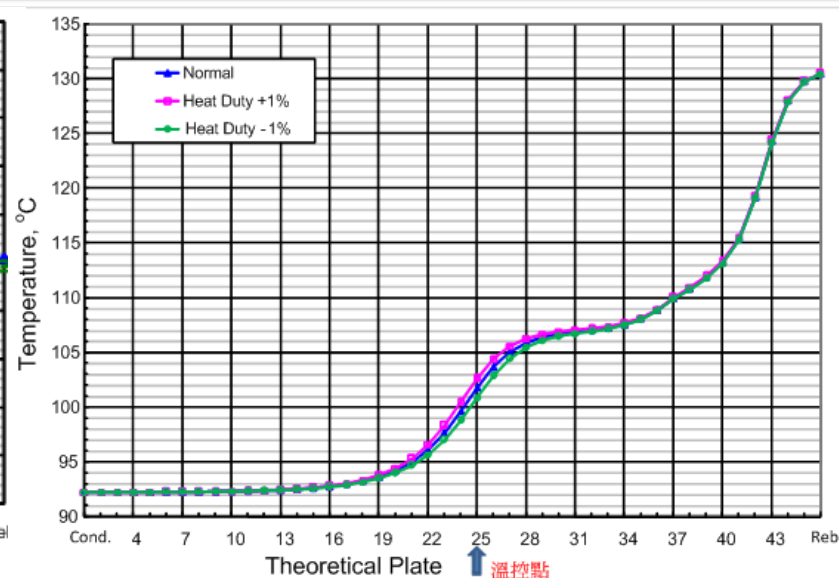
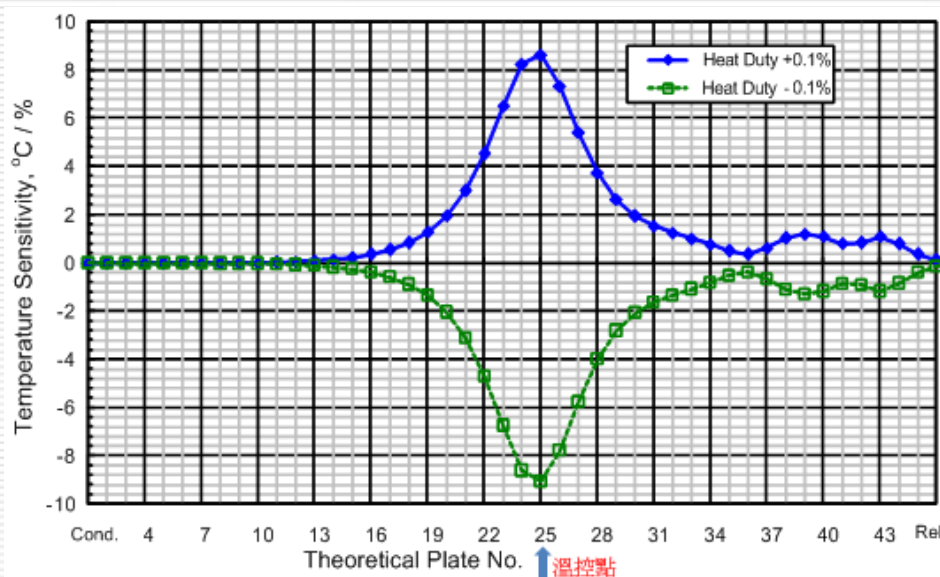
觀察溫度偏差程度

➤ 溫度偏差量/操作改變量

$\Delta T / |\Delta u|$: ΔT 塔板溫度偏差量

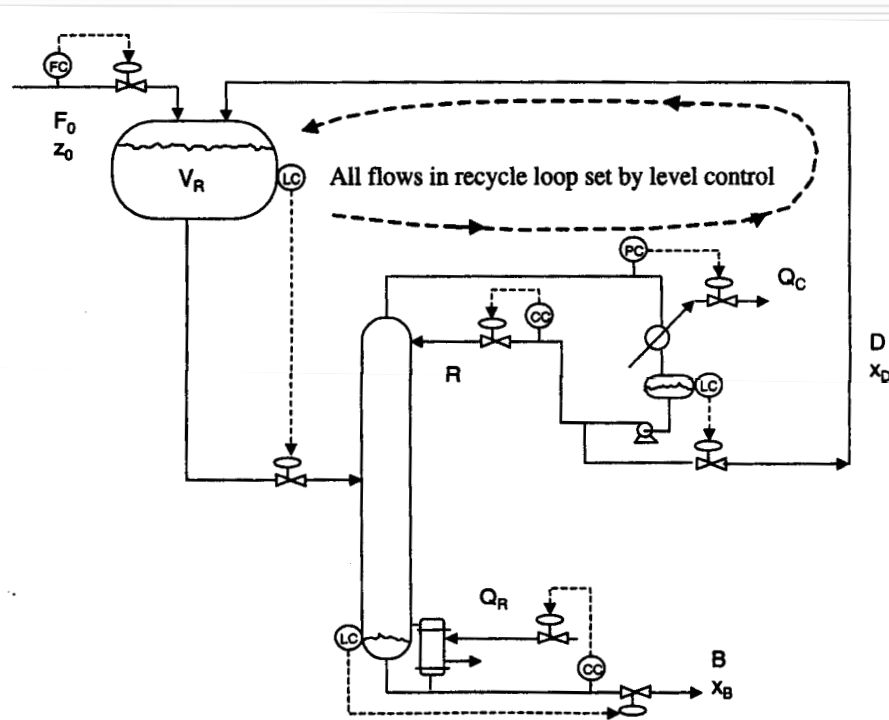
Δu 操作變數變化量

可以在小幅度操作變動下觀察溫度靈敏板



整廠控制

- 整廠控制需注意質量平衡，避免雪球效應(Snow Ball Effect)
- 如下圖 F_0 增加 $\Rightarrow V_R$ 增加 $\Rightarrow D$ 增加 (控制失效)

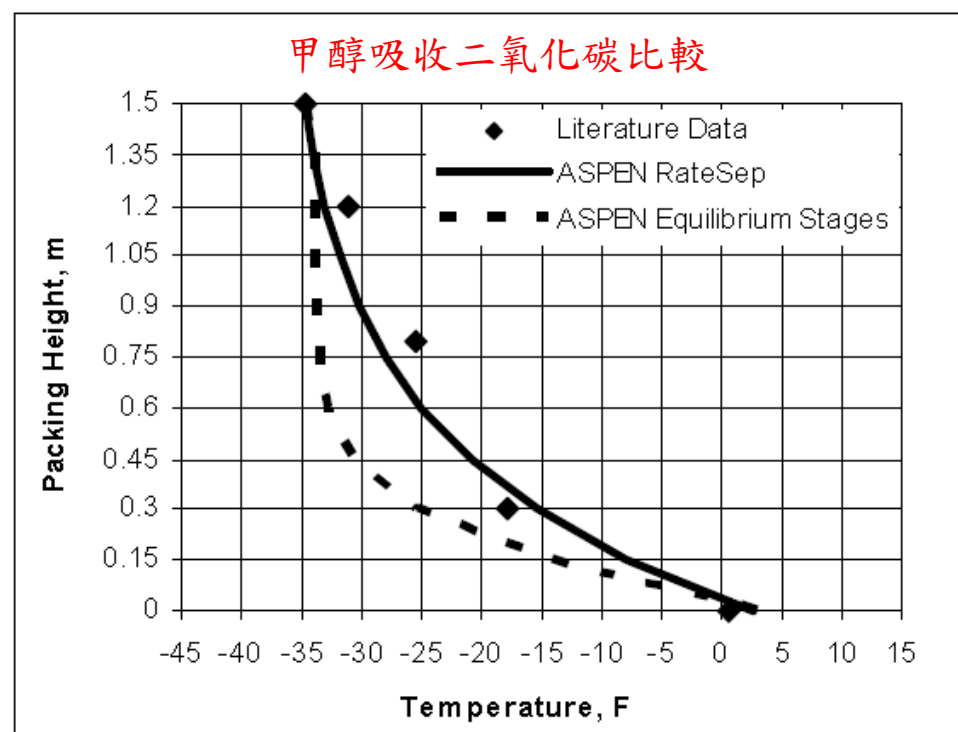


非平衡計算模式 (速率基準模式)

- 一般蒸餾塔模擬大多採用每個理論板均達到氣液相平衡，實際上會因選用的塔板或填充物的不同，質(熱)傳的效率也會有所差異
- 現行模擬軟體可提供非平衡模擬模式，使模擬的塔內組成更符合現實狀況，如
 - Aspen Plus : **Rated based model**
 - Pro II : **RadFrac model**
- 特別適用於洗滌塔、吸收塔

範例：CO₂ Capture

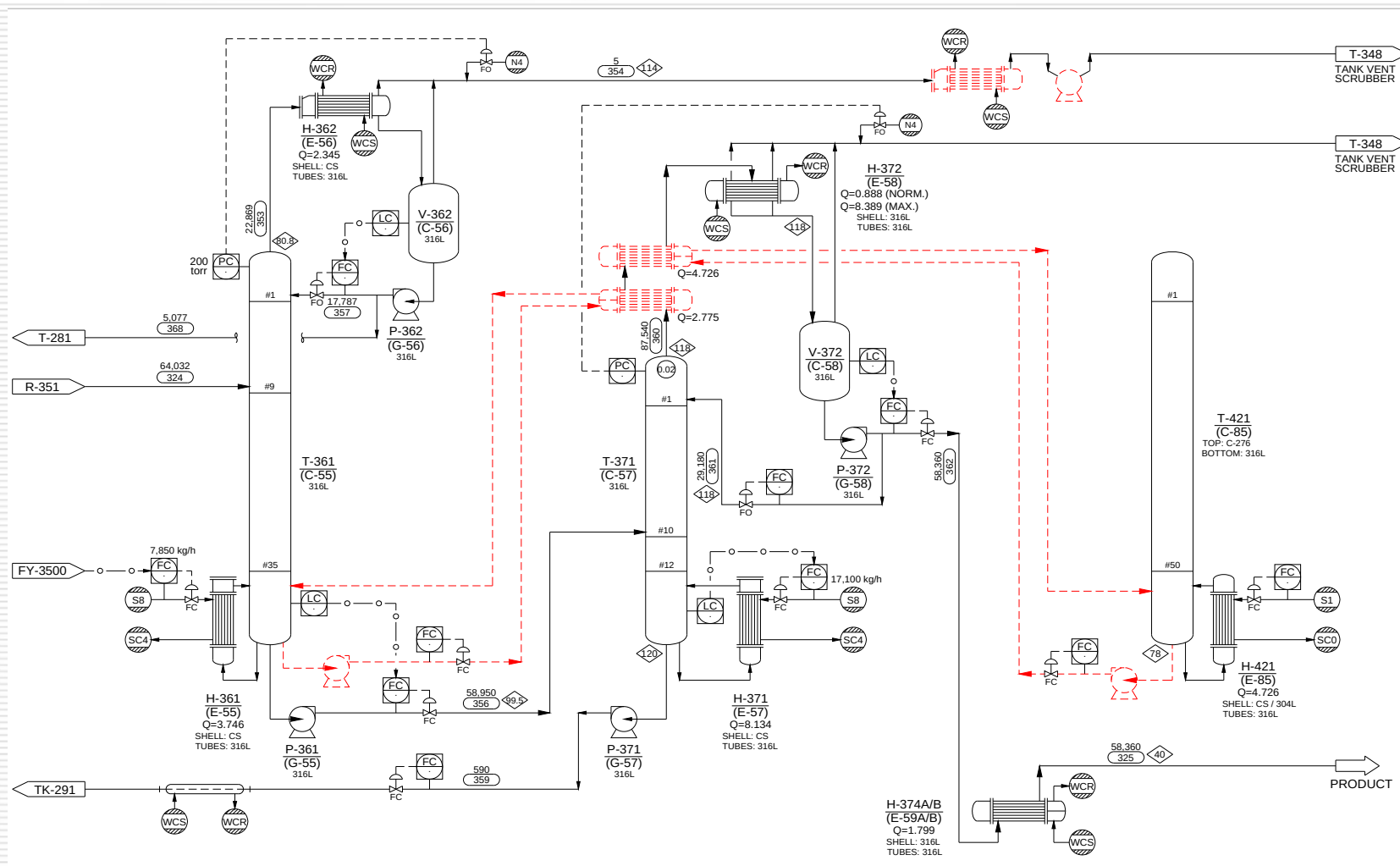
- 目前發展中的二氧化碳捕捉吸收技術
化學溶劑：MEA, MDEA, DEA, DGA, K₂CO₃, NH₃...
物理吸收：Methanol, Selexol, NMP...



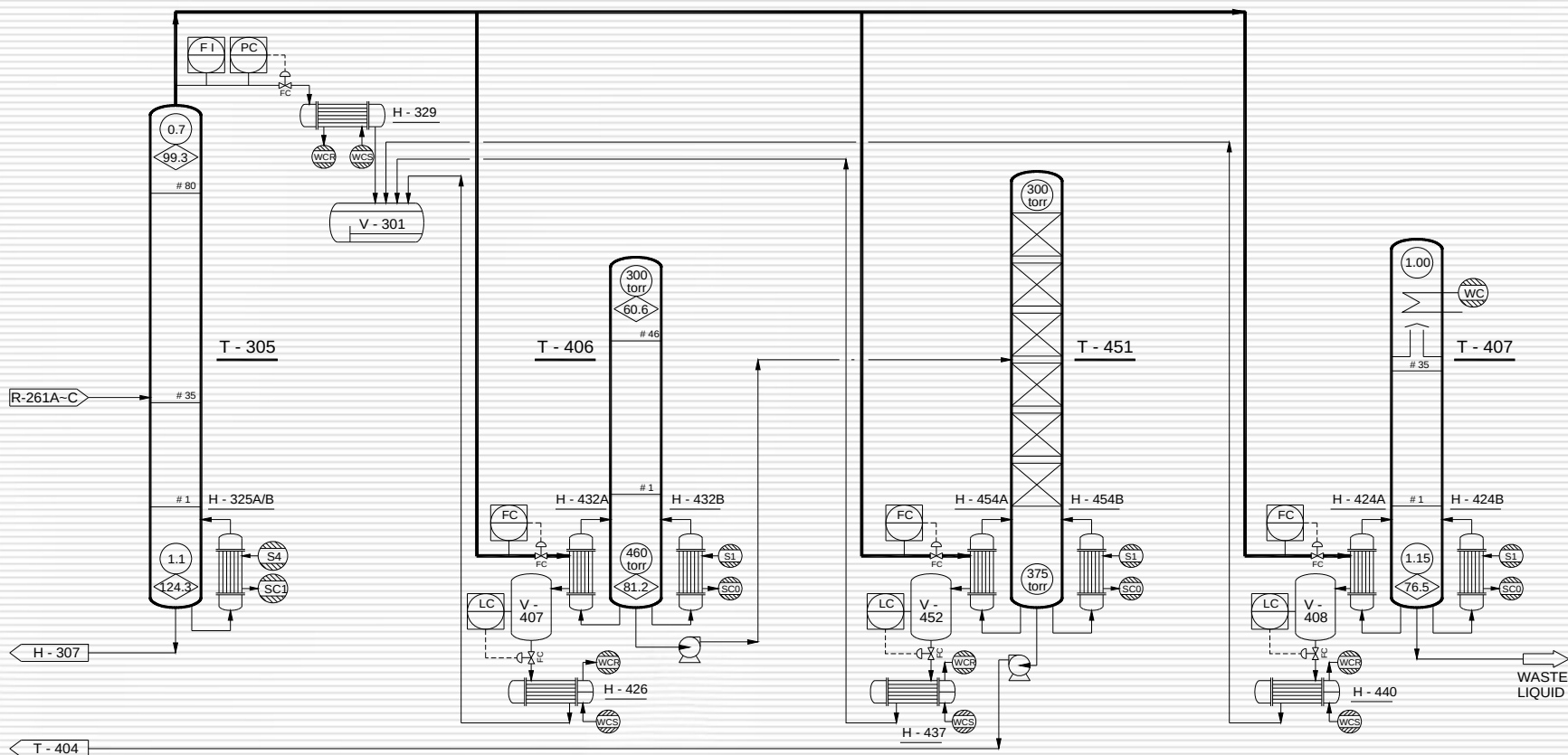
熱整合設計

- 製程廢熱回收
 - 直接靠熱交換器換熱
 - 產生蒸汽再利用

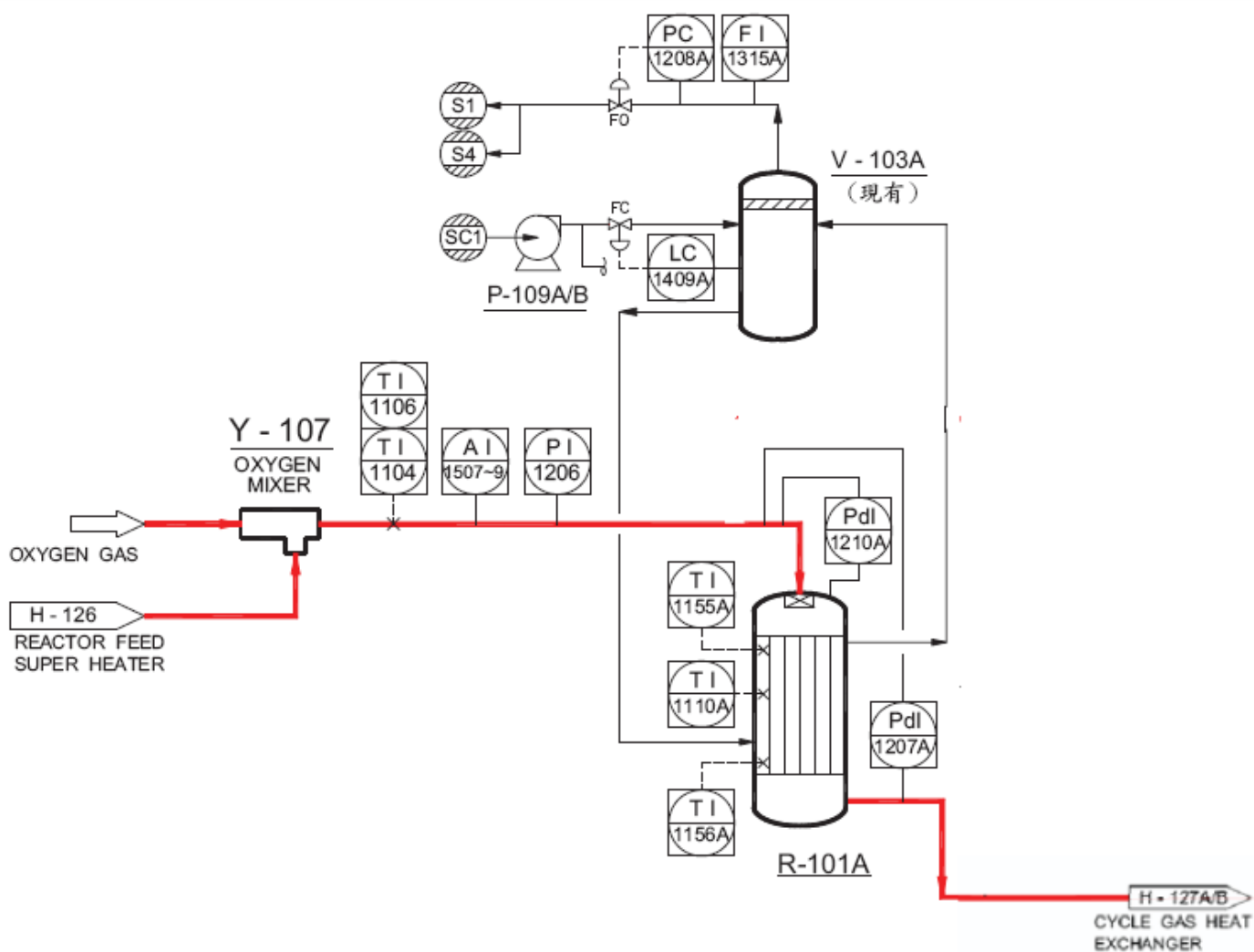
範例 1:A塔塔底液(低溫)泵送到B塔冷凝器(高溫)



範例 2: 塔頂熱氣體靠壓差送到再沸器當加熱熱源



範例 3: 利用反應熱產生蒸汽再利用



換熱器設計

➤ 設計目標

在合理的設計下，提供足夠換熱面積，且壓降及流速均在合理範圍內，避免振動或結垢

➤ 最適化設計

設備尺寸愈小價格愈低，但需考慮操作成本、安裝與維護等需求

應用模擬軟體設計換熱器

➤ 製程模擬軟體搭配換熱器設計程式

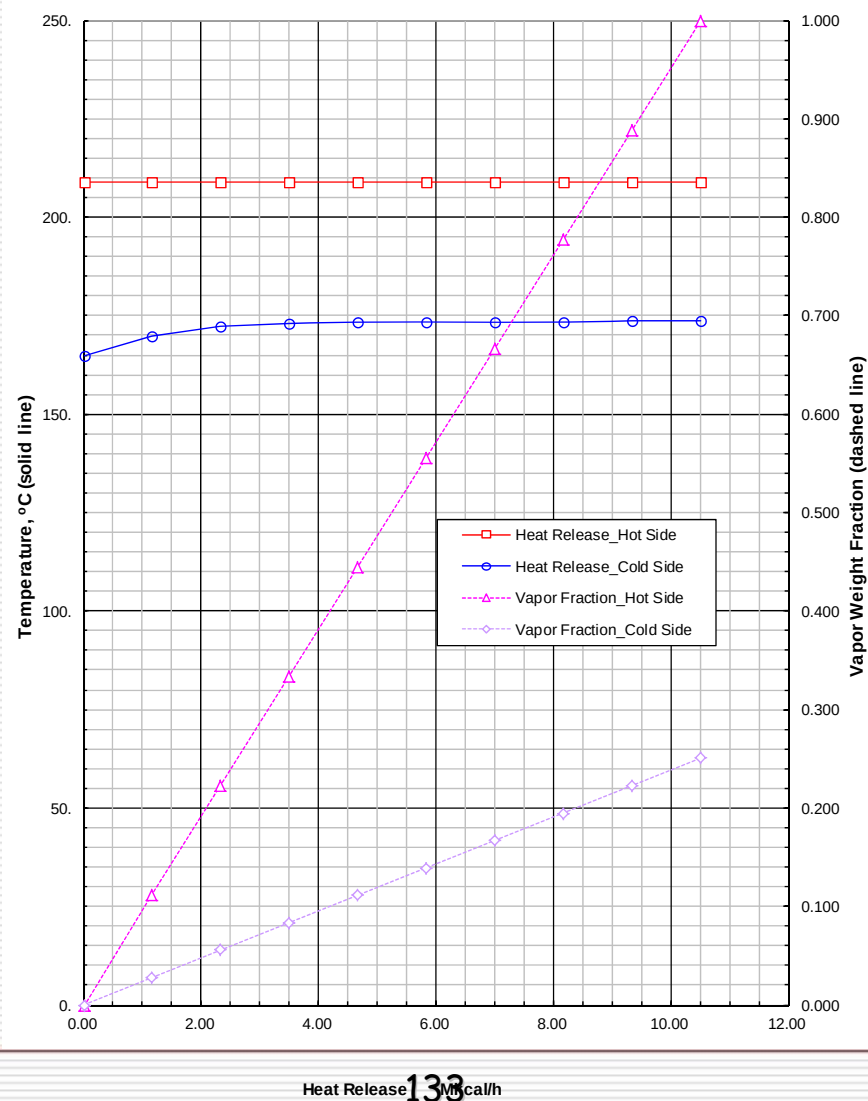
製程模擬軟體

Aspen Plus or Pro II產生Condensing Curve

換熱器設計軟體

HTFS+ (Aspen Tech)、HTRI

應用模擬軟體設計換熱器 (產生 Condensing curve)



流體走管側或殼側之選擇 (殼管式換熱器)

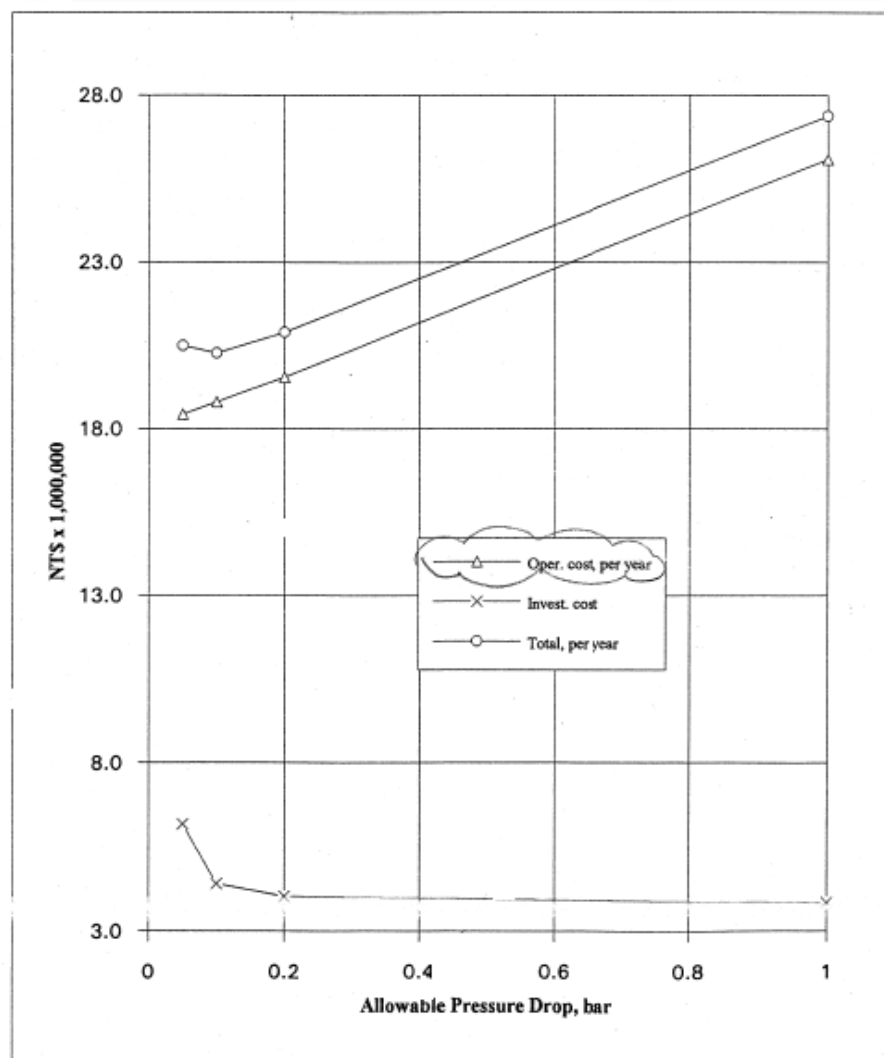
流體特性	流路選擇		說明
	殼側	管側	
容易結垢		☐	容易清除 (U-Tube除外)
含有固體粒狀物		☐	
高壓		☐	成本較低
高溫		☐	可減少熱損失及保溫費
具腐蝕性		☐	成本較低
黏度高	☐		
流量低	☐		
熱傳係數低	☐		

換熱器最適設計範例

Example: H-123A (T-110 overhead 1st condenser), VAM plant

Allowable Pressure Drop bar	Shell ID mm	Pressure Drop		Baffle		Tube Number (2 pass)	Cost USD
		Shell Side bar	Tube Side bar	Spacing mm	No.		
1	1150	0.615	0.415	248	19	1204	144,880
0.2	1200	0.187	0.344	464	11	1341	152,030
0.1	1250	0.094	0.288	654	7	1488	166,070
0.05	1500	0.048	0.156	692	7	2142	232,710

換熱器最適設計範例 (cont'd)



容許壓降愈大，設備尺寸可愈小，設備成本可較低，但壓降大，電力成本變高，操作成本提高，兩者之間有一最適化之設計

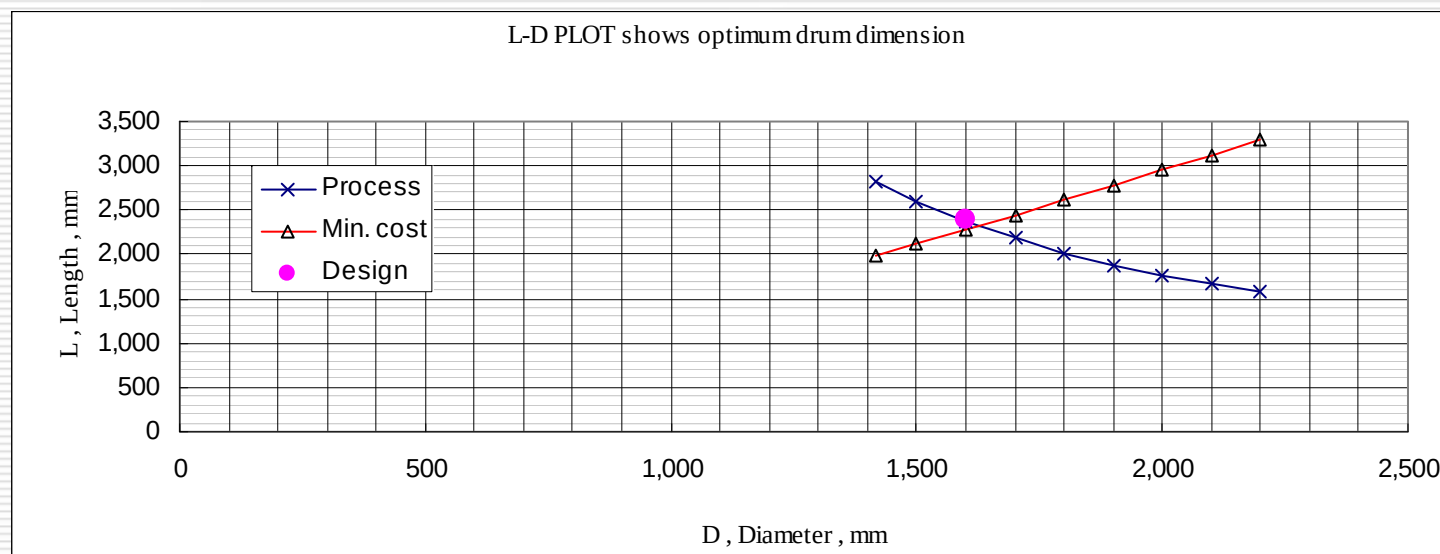
貯槽設計

➤ 設計目標

設備有足夠的緩衝(或滯留)時間，方便製程操作

➤ 最適化設計

設備內徑及高度可調整，但需考慮場地限制及設備成本等因素



結論

- 化工設計通常會有多種組合解，最佳的設計是在合適的操作條件下，設計出操作簡單、價格低廉、操作彈性大的設備
- 藉助於化工設計軟體，可大大節省計算時間，設計出最適化的規格

References

1. J. M. Coulson and J. F. Richardson, Chemical Engineering, Headington Hill, 1983.
2. E. E. Ludwig, Applied Process Design for Chemical and petrochemical Plants, 3rd Ed., Gulf Publishing Co., Vol.1, 1995, Vol.2, 2000, Vol.3, 2001.
3. M. F. Doherty and M. F. Malone, Conceptual Design of Distillation Systems, McGraw-Hill, 2001.
4. J. M. Douglas, Conceptual Design of Chemical Processes, McGraw-Hill, 1988.
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問題及討論